# Increasing pollution, increasing heat, and rapid rise of compound environmental risks for Latin American and Caribbean children

Preliminary draft

The latest version of this paper is available at this link.

Angelo dos Santos, Jere R. Behrman, Emily Hannum, Oscar Morales, Fan Wang*

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**Abstract**

Globally, air pollution and extreme heat pose important risks to human health and well-being, and children are particularly vulnerable to these hazards. We analyze how ambient exposures to these hazards are distributed across, and overlap with, the young child population of Latin America and the Caribbean. We then compare ambient exposure across income, human development, remoteness, urbanicity, and Indigenous land, and track how it changed between 2000 and 2020. Our findings highlight substantial increases in the mean and in the dispersion of ambient exposures from 2000 to 2020. Compared with 2000, in 2020 an additional 7.9 million children were exposed to air pollution above $10\,\mu g/m^3$, the World Health Organization limit for safe exposure until 2021. For heat stress, 3.5 million additional children were exposed to a Universal Thermal Climate Index (UTCI) above 32°C for at least 3 months of the year. Beyond increases in single-hazard ambient exposures, we find growing compound exposures. In 2020, 4.3 million children (9.2% of the child population) were exposed to both $PM_{2.5}$ above $15\,\mu g/m^3$ and at least 3 months of extreme heat, 3.2 million more than in 2000. Moreover, poorer, less developed, more remote, and more rural areas face both the highest compound exposure and the largest increases since 2000. These findings show that ambient exposure compounds socioeconomic and geographic disadvantages and threatens young children's health and wellbeing.

**Keywords:** Climate; Air Pollution; Extreme Heat; Young Children; Latin America and the Caribbean

*Angelo dos Santos: Population Studies Center, University of Pennsylvania, PA, USA; Jere R. Behrman: Department of Economics and Sociology and Population Studies Center, University of Pennsylvania, Philadelphia, PA 19104, USA; Emily Hannum: Department of Sociology and Population Studies Center, University of Pennsylvania, PA, USA; Oscar Morales: Population Studies Center, University of Pennsylvania, Philadelphia, PA 19104, USA; Fan Wang: Department of Economics, University of Houston, Houston, Texas, USA. This paper is part of the project "Climate Risk, Pollution, and Childhood Inequalities in Low- and Middle-income Countries," which is supported by National Science Foundation Grant 2230615 (PI: Hannum)

## 1 Introduction

Environmental hazards such as air pollution and extreme heat are unevenly distributed across and within countries (Sager 2025; Rentschler and Leonova 2023; Tuholske et al. 2021). Young children are especially vulnerable to them, particularly from conception to age five, when development is most sensitive. In this life-cycle stage, children are less able to regulate their body temperature during heat waves and are physiologically more sensitive to air pollution than adults (Rees et al. 2021; Landrigan et al. 2019; UNICEF 2026). Latin America and the Caribbean (LAC) is among the most unequal regions in the world, with stark differences in income, access to health care, and the provision of public services (UNECLAC 2023). These inequalities shape how much exposure children face and limit households' ability to adapt to it (Mullins and White 2020).

We develop a framework to document and decompose the child population distribution of ambient hazards exposure, focusing on air pollution and extreme heat in LAC between 2000 and 2020. Our analysis combines continental-scale geographically-granular ambient exposure data with the distribution of child population across several decades, and provides a full accounting of the number and share of children exposed separately and jointly to compounding exposure types. Furthermore, to understand the roles of socio-economic conditioning and population dynamics in the variations and changes in child population ambient hazard distributions, we conduct cross-sectional and intertemporal decompositional analyses. Cross-sectionally, we decompose the child exposure distribution into conditional distributions by income, human development, remoteness, urbanicity, and Indigenous land, and compare their means, percentiles, and dispersion. Over time, we replace later-period child-population weights with 2000 weights, which provides a counterfactual that decomposes changes in ambient conditions from changes in where children live. Jointly, these two decompositions show the relative roles of demographic shifts as well as the distribution and shifts of location-bound socio-economic conditions on the ambient environmental exposure for children. The framework also applies to the analysis of recent ambient exposure for any population group. The unique feature of child population exposure is that recent exposure is equivalent to the child's entire life cycle of exposure, when the focus is on children below the age of five.

Exposure to air pollution and extreme heat is increasingly well documented, and recent studies map their co-occurrence at global and regional scales (Huang et al. 2026; Stalhandske et al. 2025; Chen et al. 2025). However, this work mainly describes the exposure of the general population, rather than how joint exposure is distributed among young children or how it differs across socioeconomic and geographic groups. A few studies separate population growth from changes in hazard exposure (Stalhandske et al. 2025; Feng et al. 2026), but they are limited to one or a few hazards or the general population, and to our knowledge, none combines this with socioeconomic conditioning of joint exposures. For the child population analysis, global assessments map overlapping climate hazards at fine resolution but are limited to a single point in time, and treat air pollution separately from heat hazards (UNICEF 2026) or analyze only heat hazards (Feng et al. 2026). In LAC, regional monitoring reports track heat and air pollution as separate indicators (Hartinger et al. 2024; Hartinger et al. 2025). Studies of exposure inequalities examine a single hazard, mainly in one country or city (Coelho and

Requia 2025; Aguilar-Gomez, Cardenas, and Salas Diaz 2025; Chakraborti and Voorheis 2025), and global studies on exposure inequality are restricted to one hazard and do not focus on child populations (Rentschler and Leonova 2023; Sager 2025; Tuholske et al. 2021). Consequently, little is known about how many young children in LAC are exposed to both hazards, how that joint exposure has changed since 2000, and which groups bear the highest burden.

We apply our framework to 2,180 grid cells ($1° \times 1°$ resolution) covering the countries and subregions of LAC (South America, the Caribbean, and Central America) in 2000, 2010, and 2020. Estimates of the population of children under five (ages 0–4) come from WorldPop, aggregated from their native 30 arc-second resolution to the cell level. The total number of children in the region was 45 million in 2000 and 47.0 million in 2020. Air pollution is measured by annual mean $PM_{2.5}$ concentrations from satellite-based estimates (Van Donkelaar et al. 2021). We identify the severity of exposure using the World Health Organization (WHO) guideline of $10\,\mu g/m^3$ and its interim targets (ITs) of 15 (IT3), 25 (IT2), and 35 $\mu g/m^3$ (IT1).[1] Extreme heat is measured by the hourly Universal Thermal Climate Index (UTCI), a composite thermal-stress metric that integrates air temperature, radiation, wind, and humidity, from the ERA5-HEAT reanalysis (Napoli 2020), which we summarize as the share of a given year above 32°C (a share of 0.25 corresponds to three months). We define compound exposure as a child living in a cell above both a given $PM_{2.5}$ threshold and a heat-duration threshold. All statistics are weighted by each cell's child population, which gives a more accurate picture of the exposure children actually experience.

We find substantial increases in exposure between 2000 and 2020. Compared with 2000, nearly 8 million additional children in 2020 were exposed to $PM_{2.5}$ above $10\,\mu g/m^3$, the WHO's safety threshold until 2021 (*WHO global air quality guidelines* 2021), and 3.5 million additional children were exposed to a UTCI above 32°C for at least three months a year. We show that these increases reflect changes in the ambient exposure distribution, not changes in the population distribution of children. We also document increases in compound exposure. In 2020, 4.3 million children (9.2% of young children) lived where $PM_{2.5}$ exceeded $15\,\mu g/m^3$ and the UTCI exceeded 32°C for at least three months, 3.2 million more than in 2000. These exposures are unequally distributed across socioeconomic and geographic groups. Poorer, less developed, more remote, and more rural areas face both the highest compound exposure and the largest increases since 2000. In 2020, the share of children facing compound exposure was 13% in the lowest income quartile, compared with 5% in the highest, and 16% in the lowest HDI quartile, compared with none in the highest. Among geographic groups, the share was 51% in the most remote quartile of cells, compared with 6% in the most accessible, and 40% in the most rural quartile, compared with 7% in the most urban. In 2020, children in Indigenous territories also faced higher compound exposure (13%) than children in non-Indigenous territories (9%). These findings identify ambient exposure as an important dimension of vulnerability that compounds socioeconomic disadvantage and threatens the health and wellbeing of young children.

1. More information on the WHO interim targets can be found in the WHO global air quality guidelines (*WHO global air quality guidelines* 2021).

## 2 Materials and Methods

### 2.1 Children and exposure data

Section A.2 provides the data sources and aggregation details.

We combine gridded data on children, air pollution, and heat on a common $1° \times 1°$ grid for 2000, 2010, and 2020. Child population comes from WorldPop.org (Tatem 2017), which provides population by various age groups at roughly 1 $km^2$ resolution. To capture the young children population, we aggregate counts for those aged 0–4 (under five). Air pollution is the annual mean $PM_{2.5}$ concentration from Van Donkelaar et al. (2021) at $0.01° \times 0.01°$, averaged within each $1° \times 1°$ cell. We classify exposure severity by using the World Health Organization (WHO) guideline and interim targets (IT4 to IT1), which correspond to 10, 15, 25, and 35 $\mu g/m^3$ (*WHO global air quality guidelines* 2021).

Extreme heat exposure is measured with the Universal Thermal Climate Index (UTCI) from the ERA5-HEAT dataset (Napoli 2020), which provides hourly data at $0.25° \times 0.25°$. UTCI is an index of the human-perceived equivalent temperature that accounts for air temperature, humidity, wind, and radiant heat (Bröde et al. 2012). Following Feng et al. (2026), we compute for each cell the share of hours in a year above a UTCI threshold. Weighted by children, this is the Share of Time for the Average Child (STAC). As a severity benchmark, we express it in months per year. For example, a share of 0.25 corresponds to three months. Our main threshold is 32°C, the lower bound of strong heat stress (Bröde et al. 2012), we add results at 26°C and 38°C to our Supplementary Materials.

### 2.2 Socioeconomic and geographic groups

We assign each cell to groups using five layers and hold the categories fixed to the most recent year available for each layer, so that changes within a group do not reflect cells moving between groups. For socioeconomics layers, we use as income the GDP per capita in constant 2021 PPP dollars from the Becker Friedman Institute (Rossi-Hansberg and Zhang 2026), and human development is the Subnational Human Development Index from the Global Data Lab (Smits and Permanyer 2019). We divide cells into quartiles using 2020 values, assigning Q1 as the lowest chunks of the distribution. For geographic layers, accessibility is the travel time to the nearest city of at least 50,000 inhabitants in 2015 (Weiss et al. 2018), averaged within each cell and divided into quartiles (Q1 is the most remote). Urbanicity is defined by the share of each cell classified as urban in the Global Human Settlement Layer (Pesaresi et al. 2024). Because about 40% of cells contain no urban land, Q1 holds all cells with no urban land (the most rural) and Q2 to Q4 split the remaining cells into tertiles. Land type separates cells whose centroid lies within an Indigenous territory in the LandMark database (LandMark 2026) from all other cells. Section A.2 describes each layer, and Section A.1 formalizes conditioning on these fixed spatial groups.

### 2.3 Statistics

To capture overlapping exposures, we merged cell-level data on air pollution, heat, and children, and count of children exposed to both poor air quality and months of extreme heat. For example, $PM_{2.5}$ above IT3 ($15\,\mu g/m^3$) and at least three months above 32°C, for the region, its subregions, countries, and groups. For each group and year, we report the child-weighted mean and the 20th and 80th percentiles of exposure, and the difference between the two percentiles (P80−P20) as a measure of dispersion. To separate environmental change from changes in where children live, we recompute each statistic holding the 2000 distribution of children fixed. All statistics are child-population-weighted. Section A.1 presents the framework: marginal statistics in Section A.1, joint exposure in Section A.1, cross-sectional conditioning in Section A.1, and intertemporal counterfactuals in Section A.1.

## 3 Results

### 3.1 Child-population distributions of pollution and heat exposure

Our results show that between 2000 and 2020, the number of children in the region facing lower air quality and rising extreme heat exposure increased sharply. Our distributional results in Figure 1 present the distributions of $PM_{2.5}$ and shares of hours of the year above 32°C for children in years 2000, 2010, and 2020, using cell-level statistics. These distributions are computed using child population, pollution, and temperature data aggregated at the cell level. To have benchmarks for severity of exposure, we add vertical lines to indicate WHO's Interim Targets (ITs) and the share of hours of exposure summarized in months thresholds. We complement the visualization with shaded areas to make clear the range of values associated with each severity category. Figure 2 reports the corresponding number of children (in millions) in each severity category for $PM_{2.5}$ (panel a), months above 32°C (panel b), and both hazards combined (panel c).

**Air pollution.** The mean child air pollution exposure across the region increased from $13.05\,\mu g/m^3$ in 2000 to $15.96\,\mu g/m^3$ in 2020 (Table 1).[2] Figure 2a shows the shift across severity categories. In 2020, 40.6 million children (87% of the child population) were exposed to air quality that did not meet the WHO limit of $10\,\mu g/m^3$, an increase of 7.9 million relative to 2000. Children also moved up the severity scale. The number in the 15–$25\,\mu g/m^3$ category (IT3) rose from 11.2 to 20.7 million, and the number above $25\,\mu g/m^3$ (IT2 and above) rose from 0.4 to 3.5 million. In 2020, 0.2 million children were exposed to levels above $35\,\mu g/m^3$ (IT1), the most severe category, to which no children were exposed to in 2000.

Subregional trajectories differ markedly. Mean $PM_{2.5}$ rose most in South America, from 11.5 to $15.2\,\mu g/m^3$, and by about $1.2\,\mu g/m^3$ in the Caribbean (from 10.7 to $11.9\,\mu g/m^3$) and Central America (from 16.9 to $18.1\,\mu g/m^3$) between 2000 and 2020. Central America remains the most polluted subregion ($18.1\,\mu g/m^3$ in 2020) and is the only one with children above the highest WHO threshold, $35\,\mu g/m^3$. Moreover, 16% of its children live above $25\,\mu g/m^3$. In the

2. Appendix Table C.1 reports regional means. Appendix Tables C.2 and C.4 report marginal category counts; Appendix Tables C.3 and C.5 report category shares.

Caribbean, all children lived below 15 $\mu g/m^3$ in 2000, but 38% were above that level by 2020. Appendix Table C.3 isolates the subregional child exposure shares, and Appendix Table C.7 provides the country count disaggregation.

**Heat stress.** Focusing on extreme heat, children were more exposed in 2020 than in 2000. The average child spent about 9.6% of the year (around one month) above a UTCI of 32°C in 2000, rising to 12.5% in 2020.[3] Figure 2b shows the number of children that experienced differential shares of the year above 32°C. The number of children exposed to at least three months a year above 32°C nearly doubled, from 4 to 7.5 million, an increase of 3.5 million that represents 16% of the child population in 2020. Exposure to the longest durations grew fastest. The number of children exposed to four to five months rose from 0.5 to 2.5 million, and none were exposed to five or more months.

The Caribbean has the largest increase in the share of the year above 32°C, from 14% to 18%, about 15 additional days, followed by South America and Central America. Long-duration exposure grew fastest in South America, where the number of children facing four to five months a year above 32°C rose from 0.47 to 2.32 million (7.6% of its children in 2020). In Central America, the share facing three to four months rose from 4% to 10%. Appendix Table C.5 isolates the subregional shares. Country means and category counts are in Appendix Tables C.10 and C.11; Appendix Figures D.1 and D.2 visualize country means and dispersion.

### 3.2 Joint distribution of pollution and heat exposure

Using the joint distribution and compound-exposure statistics defined in Section A.1, we find that overlapping exposure to air pollution and heat rose sharply as well (Figure 2c).

In 2020, 4.3 million children (9.2%) lived in areas where $PM_{2.5}$ exceeded 15 $\mu g/m^3$ and temperatures exceeded 32°C for at least three months a year, up from 1.1 million (2.4%) in 2000. Exposure to both $PM_{2.5}$ above 15 $\mu g/m^3$ and at least four months of heat rose from 0.1 to 1.7 million children (3.6% in 2020). At the higher pollution threshold of 25 $\mu g/m^3$ (IT2), no children were exposed to both hazards in 2000, but in 2020 0.8 million simultaneously faced at least three months of heat and 0.6 million at least four months. Compound exposure is concentrated in South America and, increasingly, Central America, and is absent in the Caribbean (Table 2). In South America, 10.4% of children faced $PM_{2.5}$ above 15 $\mu g/m^3$ and at least three months of heat in 2020 (2.2% in 2000), and 4.9% faced $PM_{2.5}$ above 15 $\mu g/m^3$ and at least four months (0.2% in 2000). In Central America, the corresponding shares were 7.8% (3.1% in 2000) and 1.2% (none in 2000). Appendix Table C.6 gives regional exceedance child population counts, and Appendix Table C.12 gives counts by country.

### 3.3 Decomposition by socioeconomic and geographic groups

We now apply the cross-sectional decomposition into conditional distributions in Section A.1, comparing within-group dispersion and differences across groups, and examine these differences across time. In addition to overlapping air pollution and extreme heat distributions with the

3. Appendix Tables C.1, C.4, and C.5 report the regional means, heat-duration counts, and shares.

child population, we add socioeconomic and geospatial information layers to investigate if ambient exposures differ across people grouped by common living conditions or socioeconomic status. Figure 3 shows the mean, 20th percentile, and 80th percentile of exposure for each of these groups. Table 1 reports these statistics together with measures of inequality, and Table 2 shows the share of children in each of these groups who were exposed to both extreme heat and high levels of air pollution.[4]

**Socioeconomic layer** To characterize socioeconomic groups, we use cell-level estimates of GDP per capita and the subnational Human Development Index (HDI). Both are important stratifiers because they are strongly correlated with the capacity to mitigate the effects of ambient exposures on children and with other vulnerabilities, such as the availability and quality of public services. To rank groups, we split the distributions of income and HDI into quartiles, where Q1 denotes the lowest values.

The results in Figure 3 show that children in poorer and less developed areas face both the highest ambient exposure and a wider dispersion of exposure than children in richer and more developed areas. For air pollution, mean $PM_{2.5}$ exposure in the lowest income quartile rose from 13.7 to 17.9 $\mu g/m^3$ between 2000 and 2020. Over the same period, the range between the 20th and 80th percentiles widened from 11.0 to 15.5 $\mu g/m^3$. In the highest income quartile, the mean rose only from 11.4 to 12.8 $\mu g/m^3$, and the P20–P80 range stayed narrow (3.5 to 4.0 $\mu g/m^3$). As a result, the gap in mean exposure between the bottom and top income quartiles doubled, from 2.4 to 5.0 $\mu g/m^3$. The lowest HDI quartile shows the highest mean exposure of any group, at 19.3 $\mu g/m^3$ in 2020, up from 15.5, about 4 $\mu g/m^3$ above WHO Interim Target 3. Its dispersion remained roughly stable over the period.

For extreme heat, the lowest income and HDI quartiles both went from about 1.5 to 1.9 months per year above 32°C (shares of 0.125 to 0.161 and 0.126 to 0.162). Their P20–P80 ranges widened from about 2.4 to 3.1 months for income and from 2.7 to 3.2 months for HDI. By contrast, children in the highest HDI quartile spent less than one month above 32°C in 2020 (0.7 months), and the range of their exposure stayed below one month.

Figure 4 makes this pattern even clearer. It shows the share of children exposed to both air pollution and extreme heat in the lowest and highest income and HDI quartiles. In both dimensions, exposure rises most among children in the lowest quartile. In the lowest income quartile, the share of children exposed to $PM_{2.5}$ above 15 $\mu g/m^3$ and at least three months above 32°C rose by 8.6 percentage points between 2000 and 2020. With a four-month heat threshold, it rose by 4.8 points. In the highest income quartile, the corresponding increases were 4.1 and 1.3 points. The contrast is starker for HDI. In the highest HDI quartile, no children face this combined exposure in any year. In the lowest quartile, the share rose by 12.3 points under the three-month threshold and by 4.7 points under the four-month threshold. These differences are computed from the unrounded shares in Table 2, so they can differ by 0.1 from the difference of the rounded values shown in Figure 4.

4. Appendix Tables C.8 and C.9 complement the mean and percentile comparisons with marginal category shares. Appendix Figures D.8 and D.9 show income- and HDI-specific densities; Appendix Figures D.10, D.11, D.13, and D.14 visualize the corresponding marginal shares. These are distinct from the joint exceedances in Figure 4.

**Geographic groups** For the geographic layers, we stratify cells along three dimensions. First, we group cells into remoteness quartiles based on travel time to the nearest city of at least 50,000 inhabitants (Weiss et al. 2018). Q1 contains the most remote cells and Q4 the most accessible. Second, we group cells by urbanization, measured as the share of urban land cover within each $1^\circ \times 1^\circ$ cell in 2020. Because about 40% of cells contain no urban land, the quartile cut-offs are not unique. We assign all cells with zero urban share to Q1 and split the remaining cells into Q2–Q4 by tertiles of urban share. Q1 thus captures entirely rural cells and Q4 the most urbanized. Third, we classify cells by land tenure into two groups, Indigenous and non-Indigenous land. [5]

Exposure differs sharply across geographic groups, and the gaps are largest for compound exposure. Children in the most remote quartile experienced mean $PM_{2.5}$ increase from 12.6 to 18.6 $\mu g/m^3$ between 2000 and 2020, while children in the most accessible quartile moved less from 13.1 to 15.6. The share exposed to both $PM_{2.5}$ above 15 $\mu g/m^3$ and at least three months above 32°C increased from 15.2% to 51.2% among the most remote children (59.3% in 2010), compared to 1.0% to 5.5% among the most accessible (Table 2). The urbanization gradient is similar. Mean $PM_{2.5}$ among children in entirely rural cells rose from 11.7 to 16.8 $\mu g/m^3$, against 13.5 to 16.2 in the most urbanized cells, and compound exposure rose from 9.4% to 40.5% in rural cells but only from 1.7% to 6.9% in the most urbanized. Children on Indigenous lands were exposed to higher mean $PM_{2.5}$ than other children in 2000 (14.4 vs. 13.0 $\mu g/m^3$), but by 2020 the difference had reversed (15.5 vs. 16.0). Compound exposure nevertheless remained higher on Indigenous lands, at 12.6% versus 9.0% in 2020 (3.1% versus 2.4% in 2000).

### 3.4 Intertemporal decomposition of environmental change and population shifts

We apply the intertemporal counterfactual decomposition in Section A.1 to distinguish changes in ambient conditions from changes in where children live. Exposure can rise because pollution or heat worsens within cells, or because a larger share of children lives in more exposed cells. We therefore weight the 2010 and 2020 cell-level distributions of $PM_{2.5}$ and of the share of the year above 32°C by the 2000 spatial distribution of children. The counterfactual preserves each cell's baseline share of the child population while retaining its later-year ambient exposure. Comparing this distribution with the baseline describes environmental change under fixed spatial weights; comparing it with the observed later-year distribution describes the contribution of population redistribution under the same ambient conditions.

The counterfactual distributions are nearly identical to the observed ones (Figure 1, panels b and d, for 2020). The substantial increases in pollution and heat exposure would therefore have occurred even had children's relative spatial distribution remained unchanged. The distributional shift primarily reflects worsening ambient conditions, not a larger share of children moving to, or being born in, more exposed places. We should note that while the comparison isolates spatial redistribution, increases in total child population can still affect absolute exposed counts. The population maps in Appendix Figure D.5 illustrate the spatial input to this comparison; the observed and counterfactual distributions themselves are shown

5. Appendix Figures D.12, D.15, and D.16 visualize marginal category shares for accessibility and land type. Appendix Figures D.17 and D.18 visualize their joint exceedances, tabulated alongside urban-share comparisons in Appendix Table C.13 and main Table 2.

in Figure 1b and 1d.

## 4 Discussion

Exposure to air pollution and extreme heat in early childhood can have lasting consequences for health and development, so knowing where these hazards overlap with young children is a central concern. The capacity to mitigate these effects is also strongly linked to socioeconomic and geographic characteristics. In this study, we combine data on child population counts, ambient exposure levels, and socioeconomic and geographic characteristics to examine how these layers overlap in Latin America and the Caribbean, a region that is unequal along many dimensions.

We show that between 2000 and 2020, exposure to air pollution and extreme heat increased, and the gap between the most and least exposed children (P80 minus P20) widened in absolute terms. The number of children facing both $PM_{2.5}$ above $15\,\mu g/m^3$ and at least three months of extreme heat rose from 1.1 to 4.3 million. Children in the poorest and least developed areas bear the highest burden of both pollution and heat. We also find that geography shapes exposure: children in remote and rural areas face the most heat and the largest increases in air pollution.

Our contribution has two layers: a distributional framework and its empirical application to young children in LAC over two decades. The framework in Section A.1 integrates four parts: child-population distributions of individual ambient exposures; their joint distribution and compound-exposure probabilities; intratemporal decomposition into conditional distributions by socioeconomic and geographic characteristics; and intertemporal counterfactual decomposition separating environmental change from changes in where children live. Joint analysis addresses calls to account for overlapping environmental burdens (Mohai and Lee 2026), while population-weighted distributions extend attention beyond average locations and group means to exposure heterogeneity (Feng et al. 2026).

The empirical application contributes to the environmental justice literature in three ways. First, our LAC-wide coverage identifies patterns beyond the cities emphasized in much of the region's existing evidence (Fernández, Koplow-Villavicencio, and Montoya-Tangarife 2023): rural and remote areas face high burdens, and Indigenous lands have higher compound exposure. Second, focusing on young children reveals how ambient exposure is distributed within this population. Children in the poorest and least developed areas face both high exposure levels and wide within-group dispersion, differences that regional or group means alone do not capture. Third, tracking the child-population exposure distributions from 2000 to 2020 documents substantial increases in pollution, heat, and their overlap, together with widening exposure gaps. These patterns are consistent with broader evidence of disproportionate environmental burdens among vulnerable populations (Rees et al. 2021; Drupp et al. 2025). The counterfactual results further show that worsening ambient conditions, rather than shifts in where children live, account for most of the distributional change, emphasizing the need to reduce pollution and heat stress where children reside.

Our data sources and construction have several limitations. First, aggregating to $1^\circ \times 1^\circ$ cells (about 110 km at the equator) averages over substantial variation within each cell, such

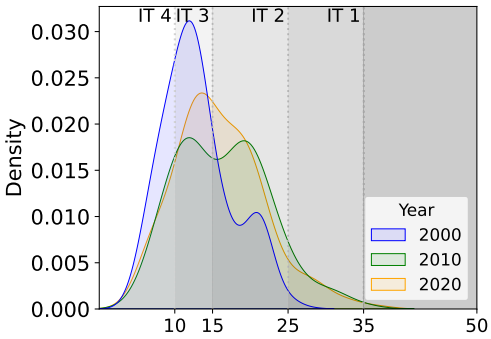
as differences between urban and rural areas or within cities. It can also introduce boundary errors, especially for small countries and islands. Second, our exposure and population measures come from satellite retrievals, reanalysis models, and gridded population estimates, which rely on interpolation for missing periods or locations. Measurement error from these sources carries into our estimates. To minimize this concern, we use published and validated datasets from the scientific literature. Third, because of data availability, we hold the geographic and socioeconomic groups fixed at their 2020 values (2015 for accessibility). Some characteristics, such as land type and remoteness, change little over two decades, and relative rankings by income and development tend to be persistent, but some cells may have moved between quartiles over the period. Fourth, our measures capture ambient outdoor conditions. They do not reflect indoor or household exposure, time spent outdoors, or access to cooling, so actual personal exposure may differ from ambient levels. In addition, the 32°C UTCI threshold was developed for adults, and young children may experience heat stress at lower levels. Finally, we document exposure, not its consequences: linking these exposure patterns to child health and development outcomes is an important next step.

Our study draws attention to the growing environmental pressures faced by young children in Latin America and the Caribbean. These pressures fall most heavily on the children with the least socioeconomic capacity to cope with them, i.e, those in poorer, less developed, and more remote areas. These findings have direct policy implications. Social safety nets and adaptation services should account for environmental exposure when identifying the children most at risk. Priority should be targeted to the lowest-income, lowest-HDI, and most remote areas, where the pollution and heat burden is the highest. Without such action, widening exposure gaps may translate into widening gaps in health, development, and long-term outcomes between children in poorer and in more prosperous regions.

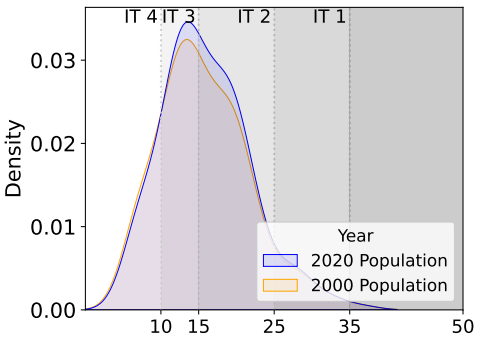
## Figures and Tables

Figure 1: The distribution of air pollution exposure and heat exposure for children

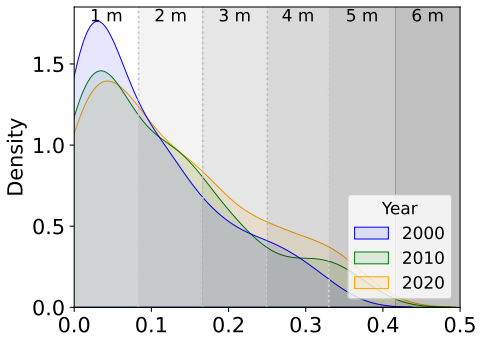
(a) $PM_{2.5}$, population of each year



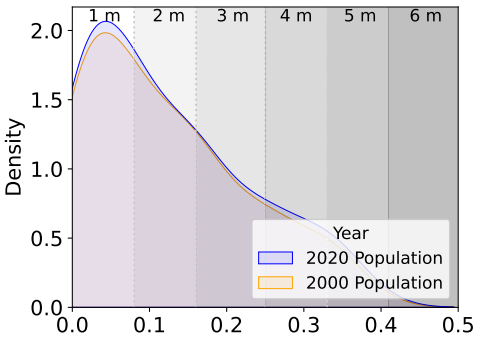

(b) $PM_{2.5}$ in 2020; 2020 vs. 2000 population weights




(c) Share of year above 32°C, population of each year



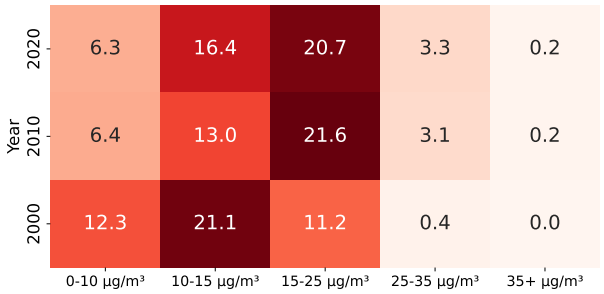

(d) Share of year above 32°C in 2020; 2020 vs. 2000 population weights



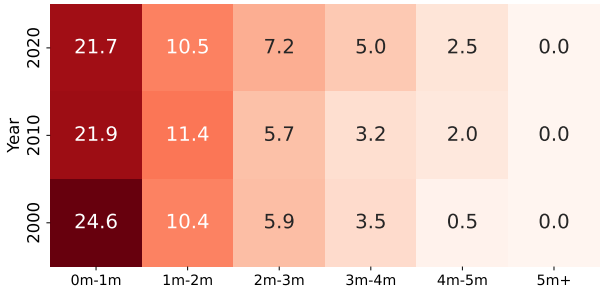

Notes: Kernel density distributions of cell-level $PM_{2.5}$ levels (top row) and share of the year above 32°C (bottom row), weighted by cell-level child population. The left panels show 2000, 2010, and 2020, each weighted by the child population of that year. The right panels show the 2020 distribution weighted by the 2020 child population (blue) and by the 2000 child population (orange), which separates changes in environmental exposure from changes in where children live. Background shading indicates severity ranges; vertical dotted lines mark the WHO interim targets (IT) for $PM_{2.5}$ and one-month steps for heat (m = months above 32°C). Summary means and marginal category shares are reported in Appendix Tables C.1, C.3, and C.5; distribution and counterfactual definitions are in Sections A.1 and A.1.

Figure 2: Children (in millions) exposed to air pollution and heat thresholds in LAC

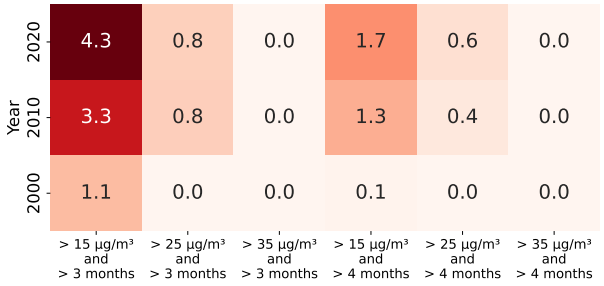
(a) $PM_{2.5}$ Interim Targets




(b) Months above 32°C

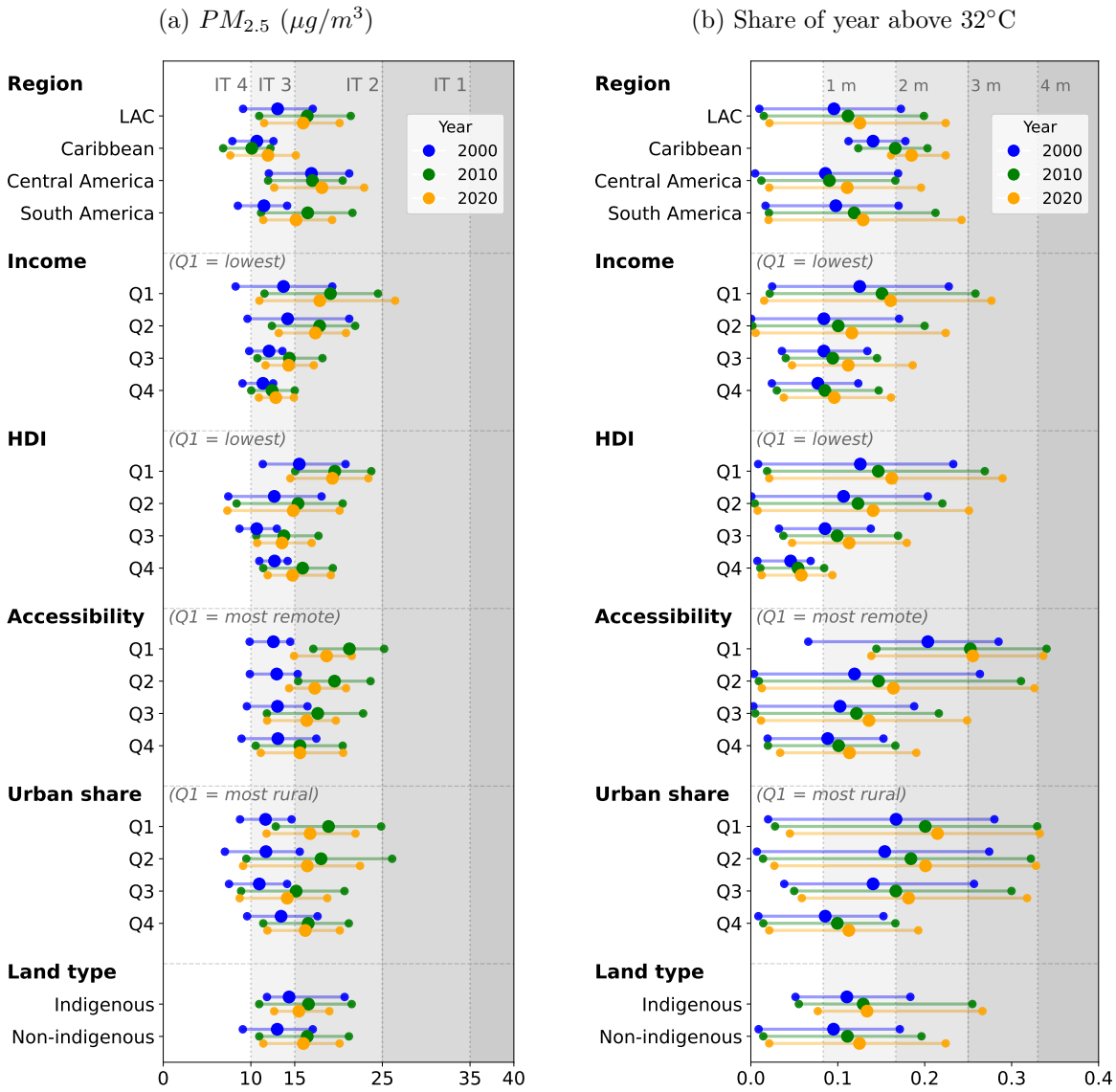



(c) Combined thresholds




Notes: Number of children (in millions) exposed to each threshold in 2000, 2010, and 2020. Panel (a) classifies children by $PM_{2.5}$ concentration ($\mu g/m^3$) following the WHO interim targets: guideline ($< 10$), IT4 (10–15), IT3 (15–25), IT2 (25–35), and IT1 (35+). Panel (b) classifies children by the number of months per year above 32°C. Panel (c) shows children exposed to both $PM_{2.5}$ above 15, 25, or 35 $\mu g/m^3$ and at least 3 or 4 months above 32°C. Values are obtained by overlapping cell-level $PM_{2.5}$, UTCI, and child population data. Appendix Tables C.2, C.4, and C.6 give regional counts for panels a, b, and c; country disaggregations are in Appendix Tables C.7, C.11, and C.12.

Figure 3: Group-specific distributional ranges across all years




Notes: The large dot is the child-population-weighted mean and the small dots mark the 20th and 80th percentiles (joined by a line) of cell-level $PM_{2.5}$ levels (left) and share of the year above 32°C (right) in each group. Within each row, the dots from top to bottom correspond to 2000, 2010, and 2020. Groups are region (all of Latin America and the Caribbean pooled, and its Caribbean, Central America, and South America subregions), quartiles of income (GDP per capita at PPP, 2021 constant prices; Q1 lowest), HDI (Q1 lowest), accessibility (travel time to the nearest city; Q1 most remote, Q4 most accessible), urban share (Q1 most rural), and land type (Indigenous vs. non-Indigenous territories). Group assignments use 2020 values (2015 for accessibility) and are held constant across years. Background shading indicates threshold ranges; vertical dotted lines mark the WHO interim targets (IT) for $PM_{2.5}$ and months above 32°C (m). Appendix Table C.1 reports regional means, and Appendix Tables C.8 and C.9 summarize the same group distributions using marginal category shares. Table 1 reports the plotted means and percentiles; Section A.1 defines the conditional statistics.

Figure 4: Overlapping air pollution and heat exposure by income group




Notes: Share (in percent) of children within each group exposed to both $PM_{2.5}$ and heat above the stated thresholds in 2000, 2010, and 2020 (m = months above 32°C). Columns combine a $PM_{2.5}$ threshold ($> 15$, $> 25$, or $> 35$ $\mu g/m^3$) with a heat threshold ($> 3$ or $> 4$ months). The top row shows the lowest (Q1) and highest (Q4) income quartiles, defined by cell-level GDP per capita at PPP (2021 constant prices) in 2020; the bottom row shows the lowest (Q1) and highest (Q4) HDI quartiles, defined by 2020 subnational HDI. Group assignments are held constant across years. Appendix Table C.13 reports the conditional shares for the tabulated pollution thresholds and all geographic and socioeconomic groups; Section A.1 defines these probabilities.

## Tables

Table 1: Child exposure to PM 2.5 and heat: levels and inequality by group and year

| | | PM 2.5 ($\mu g/m^3$) | | | | Share of year above 32°C | | | |
|---|---|---|---|---|---|---|---|---|---|
| Group | Year | Mean | P20 | P80 | P80−P20 | Mean | P20 | P80 | P80−P20 |
| **Region** | | | | | | | | | |
| LAC | 2000 | 13.05 | 9.11 | 17.06 | 7.95 | 0.096 | 0.010 | 0.173 | 0.163 |
| | 2010 | 16.44 | 10.95 | 21.39 | 10.44 | 0.112 | 0.015 | 0.199 | 0.184 |
| | 2020 | 15.96 | 11.51 | 20.12 | 8.61 | 0.125 | 0.021 | 0.224 | 0.203 |
| Caribbean | 2000 | 10.69 | 7.89 | 12.57 | 4.69 | 0.141 | 0.112 | 0.178 | 0.065 |
| | 2010 | 10.10 | 6.83 | 12.23 | 5.40 | 0.166 | 0.124 | 0.203 | 0.080 |
| | 2020 | 11.94 | 7.65 | 15.12 | 7.48 | 0.185 | 0.161 | 0.224 | 0.063 |
| Central America | 2000 | 16.91 | 12.05 | 21.22 | 9.17 | 0.086 | 0.005 | 0.169 | 0.165 |
| | 2010 | 17.01 | 11.96 | 20.47 | 8.51 | 0.090 | 0.012 | 0.166 | 0.154 |
| | 2020 | 18.10 | 12.66 | 22.92 | 10.26 | 0.111 | 0.021 | 0.196 | 0.175 |
| South America | 2000 | 11.49 | 8.51 | 14.13 | 5.62 | 0.098 | 0.017 | 0.170 | 0.153 |
| | 2010 | 16.48 | 11.16 | 21.59 | 10.42 | 0.119 | 0.021 | 0.212 | 0.192 |
| | 2020 | 15.16 | 11.41 | 19.27 | 7.86 | 0.129 | 0.020 | 0.242 | 0.222 |
| **Land type** | | | | | | | | | |
| Indigenous | 2000 | 14.36 | 11.84 | 20.69 | 8.86 | 0.110 | 0.051 | 0.184 | 0.132 |
| | 2010 | 16.57 | 10.95 | 21.50 | 10.55 | 0.129 | 0.055 | 0.255 | 0.200 |
| | 2020 | 15.47 | 12.66 | 18.95 | 6.29 | 0.134 | 0.077 | 0.266 | 0.189 |
| Non-Indigenous | 2000 | 13.01 | 9.08 | 17.06 | 7.98 | 0.095 | 0.009 | 0.171 | 0.162 |
| | 2010 | 16.43 | 10.96 | 21.18 | 10.23 | 0.111 | 0.014 | 0.196 | 0.182 |
| | 2020 | 15.99 | 11.42 | 20.12 | 8.70 | 0.125 | 0.021 | 0.224 | 0.203 |
| **Accessibility** *(Q1 = most remote)* | | | | | | | | | |
| Q1 | 2000 | 12.57 | 9.86 | 14.49 | 4.63 | 0.204 | 0.066 | 0.285 | 0.219 |
| | 2010 | 21.24 | 17.12 | 25.20 | 8.08 | 0.252 | 0.144 | 0.340 | 0.196 |
| | 2020 | 18.64 | 14.92 | 21.51 | 6.59 | 0.255 | 0.139 | 0.337 | 0.198 |
| Q2 | 2000 | 12.95 | 9.88 | 15.34 | 5.46 | 0.119 | 0.004 | 0.263 | 0.260 |
| | 2010 | 19.54 | 15.39 | 23.65 | 8.26 | 0.147 | 0.009 | 0.311 | 0.302 |
| | 2020 | 17.29 | 14.37 | 20.86 | 6.49 | 0.164 | 0.013 | 0.326 | 0.314 |
| Q3 | 2000 | 13.04 | 9.54 | 16.46 | 6.91 | 0.103 | 0.003 | 0.188 | 0.185 |
| | 2010 | 17.63 | 11.83 | 22.81 | 10.98 | 0.121 | 0.005 | 0.216 | 0.211 |
| | 2020 | 16.37 | 11.85 | 19.69 | 7.84 | 0.136 | 0.012 | 0.249 | 0.237 |
| Q4 | 2000 | 13.08 | 8.93 | 17.46 | 8.53 | 0.088 | 0.019 | 0.153 | 0.133 |
| | 2010 | 15.60 | 10.55 | 20.47 | 9.92 | 0.101 | 0.020 | 0.166 | 0.147 |
| | 2020 | 15.59 | 11.13 | 20.53 | 9.40 | 0.113 | 0.034 | 0.190 | 0.157 |
| **Income** *(Q1 = lowest)* | | | | | | | | | |
| Q1 | 2000 | 13.72 | 8.26 | 19.29 | 11.03 | 0.125 | 0.025 | 0.228 | 0.203 |
| | 2010 | 19.08 | 11.55 | 24.50 | 12.95 | 0.151 | 0.022 | 0.258 | 0.237 |
| | 2020 | 17.86 | 10.97 | 26.46 | 15.48 | 0.161 | 0.015 | 0.277 | 0.261 |
| Q2 | 2000 | 14.19 | 9.61 | 21.22 | 11.61 | 0.084 | 0.000 | 0.171 | 0.170 |
| | 2010 | 17.85 | 12.40 | 21.90 | 9.50 | 0.101 | 0.002 | 0.200 | 0.198 |
| | 2020 | 17.36 | 13.18 | 20.85 | 7.67 | 0.116 | 0.005 | 0.224 | 0.219 |
| Q3 | 2000 | 12.07 | 9.81 | 13.60 | 3.79 | 0.084 | 0.036 | 0.134 | 0.098 |
| | 2010 | 14.39 | 10.75 | 18.17 | 7.42 | 0.094 | 0.040 | 0.145 | 0.105 |
| | 2020 | 14.31 | 11.67 | 17.17 | 5.50 | 0.112 | 0.047 | 0.186 | 0.139 |

*Continued on next page*

Table 1: Child exposure to PM 2.5 and heat: levels and inequality by group and year

| | | PM 2.5 ($\mu g/m^3$) | | | | Share of year above 32°C | | | |
|---|---|---|---|---|---|---|---|---|---|
| Group | Year | Mean | P20 | P80 | P80−P20 | Mean | P20 | P80 | P80−P20 |
| Q4 | 2000 | 11.37 | 9.04 | 12.54 | 3.50 | 0.077 | 0.024 | 0.124 | 0.099 |
| | 2010 | 12.42 | 10.04 | 15.00 | 4.96 | 0.085 | 0.030 | 0.147 | 0.117 |
| | 2020 | 12.84 | 10.92 | 14.91 | 3.99 | 0.096 | 0.038 | 0.161 | 0.123 |
| **HDI** *(Q1 = lowest)* | | | | | | | | | |
| Q1 | 2000 | 15.52 | 11.36 | 20.80 | 9.44 | 0.126 | 0.008 | 0.233 | 0.224 |
| | 2010 | 19.57 | 15.05 | 23.74 | 8.68 | 0.147 | 0.019 | 0.269 | 0.250 |
| | 2020 | 19.30 | 14.51 | 23.40 | 8.90 | 0.162 | 0.021 | 0.290 | 0.268 |
| Q2 | 2000 | 12.67 | 7.42 | 18.06 | 10.64 | 0.107 | 0.001 | 0.204 | 0.203 |
| | 2010 | 15.38 | 8.37 | 20.47 | 12.10 | 0.123 | 0.004 | 0.220 | 0.216 |
| | 2020 | 14.83 | 7.32 | 20.12 | 12.80 | 0.141 | 0.008 | 0.251 | 0.243 |
| Q3 | 2000 | 10.66 | 8.70 | 12.96 | 4.26 | 0.085 | 0.032 | 0.138 | 0.106 |
| | 2010 | 13.77 | 10.61 | 17.71 | 7.10 | 0.099 | 0.037 | 0.169 | 0.132 |
| | 2020 | 13.55 | 10.72 | 16.93 | 6.21 | 0.113 | 0.047 | 0.179 | 0.132 |
| Q4 | 2000 | 12.70 | 10.96 | 14.19 | 3.23 | 0.046 | 0.007 | 0.069 | 0.061 |
| | 2010 | 15.90 | 11.41 | 19.34 | 7.93 | 0.054 | 0.011 | 0.084 | 0.073 |
| | 2020 | 14.75 | 11.92 | 19.12 | 7.20 | 0.058 | 0.012 | 0.094 | 0.081 |
| **Urban share** *(Q1 = most rural)* | | | | | | | | | |
| Q1 | 2000 | 11.67 | 8.76 | 14.65 | 5.89 | 0.167 | 0.020 | 0.280 | 0.260 |
| | 2010 | 18.86 | 12.84 | 24.86 | 12.02 | 0.201 | 0.028 | 0.329 | 0.302 |
| | 2020 | 16.76 | 11.79 | 21.94 | 10.15 | 0.215 | 0.045 | 0.332 | 0.288 |
| Q2 | 2000 | 11.72 | 7.03 | 15.58 | 8.54 | 0.154 | 0.007 | 0.274 | 0.267 |
| | 2010 | 18.01 | 9.48 | 26.12 | 16.64 | 0.184 | 0.014 | 0.322 | 0.308 |
| | 2020 | 16.44 | 9.11 | 22.45 | 13.35 | 0.201 | 0.027 | 0.328 | 0.301 |
| Q3 | 2000 | 10.96 | 7.50 | 14.13 | 6.63 | 0.141 | 0.038 | 0.257 | 0.218 |
| | 2010 | 15.16 | 8.88 | 20.68 | 11.79 | 0.167 | 0.050 | 0.300 | 0.250 |
| | 2020 | 14.14 | 8.72 | 18.72 | 10.00 | 0.181 | 0.059 | 0.318 | 0.259 |
| Q4 | 2000 | 13.45 | 9.57 | 17.60 | 8.03 | 0.086 | 0.009 | 0.153 | 0.144 |
| | 2010 | 16.54 | 11.41 | 21.17 | 9.76 | 0.099 | 0.014 | 0.166 | 0.152 |
| | 2020 | 16.21 | 11.88 | 20.14 | 8.26 | 0.113 | 0.021 | 0.193 | 0.171 |

*Note:* Child-population-weighted mean, 20th percentile (P20), and 80th percentile (P80) of cell-level $PM_{2.5}$ ($\mu g/m^3$) and of the share of the year above 32°C; the group statistics are the ones plotted in Figure 3. Dispersion of exposure within a group is measured by the difference P80−P20, in $\mu g/m^3$ for $PM_{2.5}$ and in share of the year for heat. LAC is Latin America and the Caribbean. Land type distinguishes Indigenous and non-Indigenous territories. Quartiles refer to accessibility (Q1 most remote), income (GDP per capita, Q1 lowest), HDI (Q1 lowest), and urban share (Q1 most rural); group assignments use 2020 values (2015 for accessibility) and are held constant across years.

Table 2: Share of children (%) exposed to compound air pollution and heat by group and year

| Group | Year | $> 15\ \mu g/m^3$ & $> 3$ months | $> 15\ \mu g/m^3$ & $> 4$ months | $> 25\ \mu g/m^3$ & $> 3$ months | $> 25\ \mu g/m^3$ & $> 4$ months |
|---|---|---|---|---|---|
| **Region** | | | | | |
| LAC | 2000 | 2.40 | 0.13 | 0.00 | 0.00 |
| | 2010 | 7.48 | 2.86 | 1.81 | 0.79 |
| | 2020 | 9.22 | 3.55 | 1.66 | 1.19 |
| Caribbean | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| Central America | 2000 | 3.11 | 0.00 | 0.00 | 0.00 |
| | 2010 | 1.55 | 0.00 | 1.32 | 0.00 |
| | 2020 | 7.78 | 1.19 | 2.04 | 1.19 |
| South America | 2000 | 2.21 | 0.20 | 0.00 | 0.00 |
| | 2010 | 10.54 | 4.29 | 2.12 | 1.19 |
| | 2020 | 10.40 | 4.86 | 1.57 | 1.25 |
| **Land type** | | | | | |
| Indigenous | 2000 | 3.06 | 0.00 | 0.00 | 0.00 |
| | 2010 | 15.21 | 6.11 | 4.34 | 2.59 |
| | 2020 | 12.62 | 5.76 | 3.26 | 2.41 |
| Non-Indigenous | 2000 | 2.38 | 0.14 | 0.00 | 0.00 |
| | 2010 | 7.19 | 2.73 | 1.71 | 0.72 |
| | 2020 | 9.03 | 3.42 | 1.58 | 1.12 |
| **Accessibility** *(Q1 = most remote)* | | | | | |
| Q1 | 2000 | 15.16 | 0.00 | 0.00 | 0.00 |
| | 2010 | 59.32 | 34.41 | 18.25 | 13.53 |
| | 2020 | 51.15 | 25.82 | 11.61 | 10.96 |
| Q2 | 2000 | 13.34 | 1.95 | 0.00 | 0.00 |
| | 2010 | 26.27 | 14.31 | 9.11 | 4.86 |
| | 2020 | 29.64 | 16.55 | 7.88 | 5.73 |
| Q3 | 2000 | 2.33 | 0.00 | 0.00 | 0.00 |
| | 2010 | 9.03 | 3.20 | 1.07 | 0.78 |
| | 2020 | 9.22 | 3.18 | 0.74 | 0.59 |
| Q4 | 2000 | 1.01 | 0.00 | 0.00 | 0.00 |
| | 2010 | 3.53 | 0.64 | 0.80 | 0.00 |
| | 2020 | 5.54 | 1.47 | 0.93 | 0.55 |
| **Income** *(Q1 = lowest)* | | | | | |
| Q1 | 2000 | 4.03 | 0.00 | 0.00 | 0.00 |
| | 2010 | 12.44 | 3.19 | 2.69 | 0.43 |
| | 2020 | 12.67 | 4.77 | 2.96 | 1.65 |
| Q2 | 2000 | 2.33 | 0.46 | 0.00 | 0.00 |
| | 2010 | 8.85 | 6.03 | 2.51 | 1.98 |
| | 2020 | 10.53 | 5.94 | 2.28 | 2.22 |
| Q3 | 2000 | 1.37 | 0.00 | 0.00 | 0.00 |
| | 2010 | 3.52 | 0.44 | 0.69 | 0.22 |
| | 2020 | 6.38 | 0.82 | 0.29 | 0.11 |
| Q4 | 2000 | 1.23 | 0.00 | 0.00 | 0.00 |
| | 2010 | 2.23 | 0.29 | 0.69 | 0.20 |

*Continued on next page*

Table 2: Share of children exposed to compound air pollution and heat by group and year

| Group | Year | $> 15\ \mu g/m^3$ & $> 3$ months | $> 15\ \mu g/m^3$ & $> 4$ months | $> 25\ \mu g/m^3$ & $> 3$ months | $> 25\ \mu g/m^3$ & $> 4$ months |
|---|---|---|---|---|---|
| | 2020 | 5.28 | 1.30 | 0.36 | 0.17 |
| **HDI** *(Q1 = lowest)* | | | | | |
| Q1 | 2000 | 3.41 | 0.17 | 0.00 | 0.00 |
| | 2010 | 13.59 | 3.96 | 2.32 | 0.17 |
| | 2020 | 15.67 | 4.85 | 2.30 | 1.20 |
| Q2 | 2000 | 4.87 | 0.34 | 0.00 | 0.00 |
| | 2010 | 8.52 | 5.85 | 3.50 | 2.82 |
| | 2020 | 12.38 | 6.80 | 3.45 | 3.21 |
| Q3 | 2000 | 0.64 | 0.00 | 0.00 | 0.00 |
| | 2010 | 4.24 | 0.73 | 0.89 | 0.26 |
| | 2020 | 4.68 | 1.47 | 0.43 | 0.19 |
| Q4 | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| **Urban share** *(Q1 = most rural)* | | | | | |
| Q1 | 2000 | 9.36 | 0.00 | 0.00 | 0.00 |
| | 2010 | 43.66 | 16.02 | 12.67 | 5.23 |
| | 2020 | 40.46 | 17.44 | 9.90 | 5.78 |
| Q2 | 2000 | 9.90 | 0.00 | 0.00 | 0.00 |
| | 2010 | 30.81 | 15.23 | 12.89 | 7.39 |
| | 2020 | 30.46 | 14.78 | 9.00 | 6.67 |
| Q3 | 2000 | 4.81 | 0.00 | 0.00 | 0.00 |
| | 2010 | 17.07 | 7.07 | 3.05 | 1.25 |
| | 2020 | 16.26 | 6.56 | 2.06 | 1.31 |
| Q4 | 2000 | 1.65 | 0.16 | 0.00 | 0.00 |
| | 2010 | 4.68 | 1.57 | 1.05 | 0.41 |
| | 2020 | 6.90 | 2.46 | 1.20 | 0.89 |

*Note:* Share of children (in percent) exposed to both $PM_{2.5}$ above $15\,\mu g/m^3$ (IT3) or $25\,\mu g/m^3$ (IT2) and at least 3 or 4 months per year above 32°C. We obtain it by overlapping cell-level $PM_{2.5}$, UTCI, and child population data. Rows and group definitions are as in Table 1. Figure 4 plots the lowest and highest income and HDI groups, adding the $35\,\mu g/m^3$ threshold. Appendix Table C.13 reports the same nonregional rows; Appendix Figures D.17 and D.18 plot accessibility and land type.

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# Appendix for

# Increasing pollution, increasing heat, and rapid rise of compound environmental risks for Latin American and Caribbean children

## A Materials and Methods

### A.1 Framework for measuring and decomposing child-population environmental exposure distributions

We combine geographically granular ambient pollution and heat measures with the spatial distribution of children to describe the conditions where children live. The resulting child-population exposure distributions provide counts and shares exposed to each hazard separately and jointly, together with moments and quantiles describing exposure heterogeneity. Conditioning on location-bound socioeconomic and geographic characteristics describes differences across and within groups; counterfactual reweighting distinguishes changes in ambient conditions from changes in children's spatial distribution. These are descriptive analyses of ambient exposure, rather than estimates of its effects on child development.

The definitions below use common child-population weights for marginal and joint distributions, followed by cross-sectional conditional and intertemporal counterfactual decompositions. Section A.2 describes their empirical implementation.

#### Why child-population exposure distributions matter

Distributional analysis identifies which children live under higher ambient exposures, whether pollution and heat overlap, and how these conditions vary within and across places with different socioeconomic resources. Population weighting is important because the locations of children need not coincide with the locations of environmental hazards. Conditional distributions describe this heterogeneity without treating every child in a geographic group as equally exposed, while counterfactual distributions distinguish environmental change from shifts in where children live. These statistics provide context for understanding vulnerability and adaptation needs, without equating ambient exposure with a measured developmental outcome.

#### Child-population distribution of ambient exposure

**Units and exposure measures.** The unit of observation is a geographic cell $c$ in year $t$. Let $w_{ct}$ be the number of children living in the cell, $x_{ct}$ the annual mean $PM_{2.5}$ concentration ($\mu g/m^3$), and $h^{\kappa}_{ct} \in [0,1]$ the share of hours in the year with UTCI above a chosen heat-stress threshold $\kappa$. The share $h^{\kappa}_{ct}$ is the mean of the hourly indicator $\mathbf{1}\{UTCI > \kappa\}$ over all hours of the year, aggregated to cell $c$. The threshold $\kappa$ is held fixed when comparing distributions; its dependence is implicit in the distribution and summary-statistic notation below. A group $g$ is a set of cells: a country, a subregion, the whole region, or a land-type, income, HDI, accessibility or urbanicity group. Cells with missing values of $x_{ct}$, $h^{\kappa}_{ct}$ or $w_{ct}$, or with $w_{ct} = 0$, are excluded from the corresponding statistic. These measures assign cell-level ambient conditions to resident children; they do not measure individual behavior or personal exposure.

**Child-weighted distributions.** For a group $g$ and year $t$, the share of the group's children living in cell $c \in g$ is

$$s_{c,g,t} = \frac{w_{ct}}{\sum_{\hat{c} \in g} w_{\hat{c}t}}, \tag{A.1}$$

so that $\sum_{c \in g} s_{c,g,t} = 1$. For any exposure measure $y_{ct}$ (either $x_{ct}$ or $h^{\kappa}_{ct}$), the child-weighted distribution function, which gives the share of children in $g$ exposed to at most $z$, is

$$F_{gt}(z) = P_{gt}(y \leq z) = \sum_{c \in g} s_{c,g,t} \, \mathbf{1}\{y_{ct} \leq z\}. \tag{A.2}$$

A cell with twice as many children therefore counts twice: $P_{gt}$ describes a child drawn from the population of $g$ in year $t$, with exposure $y$ equal to that child's cell-level value. Figure 1, panels a and c, visualizes these marginal distributions with child-weighted kernel densities, rather than plotting the empirical CDFs themselves.

**Mean exposure.** The mean of the child-population distribution is

$$\bar{y}_{gt} = \sum_{c \in g} s_{c,g,t} \, y_{ct}. \tag{A.3}$$

With $y = h^{\kappa}$, Equation (A.3) is the Share of Time for the Average Child (STAC) of Feng et al. (2026): the share of the year that the average child in $g$ spends above $\kappa$. Weighting by children makes $\bar{y}_{gt}$ describe the conditions of the average child, not of the average cell. Table 1 reports these means, with regional summaries in Table C.1 and country means in Figure D.2.

**Exposure percentiles and dispersion.** We define the $\tau$-th percentile, for $0 < \tau \leq 100$, as the smallest exposure level at which at least $\tau\%$ of children in $g$ are covered:

$$Q^{\tau}_{gt} = \min \left\{ z : F_{gt}(z) \geq \tau/100 \right\}. \tag{A.4}$$

For chosen lower and upper percentiles $0 < \underline{\tau} < \overline{\tau} \leq 100$, we measure the dispersion of exposure within a group as their difference,

$$D_{gt} = Q^{\overline{\tau}}_{gt} - Q^{\underline{\tau}}_{gt}, \tag{A.5}$$

in $\mu g/m^3$ for $PM_{2.5}$ and in share of the year for heat. The empirical analysis uses the 20th and 80th percentiles. Figure 3 plots the means and these two percentiles; Table 1 also reports their difference. Figure D.1 gives the corresponding country-specific ranges.

**Threshold counts and shares.** For an interval $[a, b)$ of exposure values, the number of children in group $g$ is

$$N_{gt}[a, b) = \sum_{c \in g} w_{ct} \, \mathbf{1}\{a \leq y_{ct} < b\}, \tag{A.6}$$

and their share is

$$S_{gt}[a, b) = \sum_{c \in g} s_{c,g,t} \, \mathbf{1}\{a \leq y_{ct} < b\} = \frac{N_{gt}[a, b)}{\sum_{c \in g} w_{ct}}. \tag{A.7}$$

These are counts and probability masses of the same child-population distribution; taking $b = \infty$ gives the count and share at or above $a$. Interval boundaries can be chosen to reflect pollution guidelines or heat durations expressed as shares of the year. Figure 2, panels a and b, reports

marginal category counts. Tables C.2 and C.3 report pollution counts and shares, and Tables C.4 and C.5 report their heat counterparts.

### Joint distribution and compound exposure

**Joint child-population distribution.** Combining pollution and heat values in the same cells gives the bivariate distribution

$$F_{gt}^{joint}(z_x, z_h) = P_{gt}(x \le z_x, h^\kappa \le z_h) = \sum_{c \in g} s_{c,g,t}\, \mathbf{1}\{x_{ct} \le z_x\}\, \mathbf{1}\{h_{ct}^\kappa \le z_h\}. \tag{A.8}$$

Its marginals recover Equation (A.2) for pollution and heat, respectively, on the same set of cells. The joint distribution records which children face both ambient conditions. Due to the spatial correlation in pollution and heat exposures, the probability of compound exposure is generally not the product of the marginal exceedance probabilities. Because pollution is an annual mean and heat is an annual duration, this is co-exposure at the cell-year level, not a measure of simultaneous hourly exceedances.

**Compound-exposure counts and probabilities.** The number of children in the joint upper-tail region defined by pollution above $\pi$ and heat duration at least $m$ is

$$J_{gt}(\pi, m) = \sum_{c \in g} w_{ct}\, \mathbf{1}\{x_{ct} > \pi\}\, \mathbf{1}\{h_{ct}^\kappa \ge m\}, \tag{A.9}$$

and the corresponding probability under Equation (A.8) is

$$P_{gt}(x > \pi, h^\kappa \ge m) = \frac{J_{gt}(\pi, m)}{\sum_{c \in g} w_{ct}} = \sum_{c \in g} s_{c,g,t}\, \mathbf{1}\{x_{ct} > \pi\}\, \mathbf{1}\{h_{ct}^\kappa \ge m\}. \tag{A.10}$$

Here $\pi$ is a chosen pollution threshold and $m \in [0, 1]$ a chosen heat-duration threshold. Figure 2, panel c, reports joint threshold counts, with regional counts in Table C.6. Table 2 reports compound probabilities as percentages.

### Cross-sectional decomposition into conditional distributions

Our first decomposition examines exposure heterogeneity at a given moment in time. We partition the child-population ambient exposure distribution by geographically observable characteristics and compare the moments and quantiles of the resulting conditional distributions. This distinguishes differences across groups from dispersion within groups associated with location-bound socioeconomic and geographic conditions.

**Conditioning on spatial partitions.** For a given population $g$, let $R$ be a categorical spatial characteristic taking values $r_1, \ldots, r_K$, with cell assignment $R_c$. Each choice of $R$ defines a mutually exclusive and exhaustive partition of $g$: for example, Indigenous and non-Indigenous land types, or the four income groups. HDI, accessibility, urbanicity, and administrative geography provide other partitions, considered separately rather than pooling overlapping categories from different layers. Assume each partition group has a positive child-population share. The

probability that a child belongs to group $k$ is $p_{gtk} = P_{gt}(R = r_k) = \sum_{c \in g} s_{c,g,t}\, \mathbf{1}\{R_c = r_k\}$, with $\sum_{k=1}^{K} p_{gtk} = 1$. These are shares of children.

The ambient exposure distribution conditional on membership in group $k$ is

$$F_{gt}(z \mid R = r_k) = \frac{\sum_{c \in g} s_{c,g,t}\, \mathbf{1}\{R_c = r_k\}\, \mathbf{1}\{y_{ct} \leq z\}}{p_{gtk}}. \tag{A.11}$$

Within a partition group, the weights are renormalized to sum to one. The overall distribution therefore decomposes exactly as the mixture

$$F_{gt}(z) = \sum_{k=1}^{K} p_{gtk}\, F_{gt}(z \mid R = r_k). \tag{A.12}$$

At each exposure cutoff, $p_{gtk} F_{gt}(z \mid R = r_k)$ is group $k$'s contribution to the overall share of children at or below that cutoff.

**Conditional means, quantiles, and dispersion.** Applying Equations (A.3)–(A.5) to $F_{gt}(z \mid R = r_k)$ gives the same statistics with an additional group subscript $k$: $\bar{y}_{gtk}$, $Q^{\tau}_{gtk}$, and $D_{gtk}$. Conditional means and quantiles describe differences across groups; $D_{gtk}$ measures within-group dispersion. Unlike means, overall quantiles are not generally weighted averages of conditional quantiles. These summaries underlie Figure 3 and Table 1. Figures D.8 and D.9 visualize the full conditional distributions for income and HDI, and Tables C.8 and C.9 report their marginal threshold probabilities.

**Conditional joint distributions.** Conditioning Equation (A.8) on $R = r_k$ likewise restricts its sum to cells in group $k$ and divides by $p_{gtk}$. Thus $F^{joint}_{gt}(z_x, z_h)$ is a mixture of conditional joint distributions with the same child-population shares. In particular,

$$P_{gt}(x > \pi, h^{\kappa} \geq m) = \sum_{k=1}^{K} p_{gtk}\, P_{gt}(x > \pi, h^{\kappa} \geq m \mid R = r_k). \tag{A.13}$$

The conditional probability uses the children within group $k$ as its denominator, rather than all children in $g$. Figure 4 and Table 2 report these probabilities for socioeconomic and geographic groups, with accessibility and land-type comparisons in Figures D.17 and D.18.

**Fixed spatial characteristics.** Group assignments $R_c$ are defined using a chosen reference year for each spatial characteristic and held fixed across the years compared. Changes in conditional distributions therefore reflect changes in ambient conditions and child-population weights within fixed sets of cells, not reclassification. The mixture shares $p_{gtk}$ can nevertheless change as the distribution of children across those sets changes.

### Intertemporal decomposition using counterfactual distributions

Our second decomposition separates changes in child pollution and heat exposure into a population-driven component and a component driven by ambient environmental conditions. We construct counterfactual distributions by replacing later-period child-population weights with earlier-period weights while holding later-period ambient conditions fixed. Comparing the counter-

factual with the baseline distribution isolates environmental change; comparing it with the later-period observed distribution isolates changes in where children live.

**Environmental change versus changes in where children live.** Let $\underline{t} < \bar{t}$ denote an earlier and a later year. On a fixed set of cells $g$, use the earlier-year child shares and later-year ambient conditions to define the counterfactual exposure distribution

$$F^{CF}_{g,\bar{t}}(z) = \sum_{c \in g} s_{c,g,\underline{t}} \, \mathbf{1}\{y_{c,\bar{t}} \leq z\}. \tag{A.14}$$

The baseline year $\underline{t}$ is implicit in the counterfactual notation. At every cutoff $z$, the observed change decomposes exactly as

$$F_{g,\bar{t}}(z) - F_{g,\underline{t}}(z) = \underbrace{\left(F^{CF}_{g,\bar{t}}(z) - F_{g,\underline{t}}(z)\right)}_{\text{environmental change}} + \underbrace{\left(F_{g,\bar{t}}(z) - F^{CF}_{g,\bar{t}}(z)\right)}_{\text{change in where children live}}. \tag{A.15}$$

The first term changes ambient conditions while fixing the earlier-year child weights; the second changes those weights while fixing later-year ambient conditions. Figure 1, panels b and d, compares the observed and counterfactual later-year distributions through their kernel-density visualizations.

**Counterfactual summaries.** Applying the earlier definitions to $F^{CF}_{g,\bar{t}}$ gives the counterfactual mean $\bar{y}^{CF}_{g,\bar{t}}$, percentiles $Q^{\tau,CF}_{g,\bar{t}}$, and dispersion $D^{CF}_{g,\bar{t}}$. The change in the mean between $\underline{t}$ and $\bar{t}$ then splits exactly into

$$\bar{y}_{g,\bar{t}} - \bar{y}_{g,\underline{t}} = \underbrace{\left(\bar{y}^{CF}_{g,\bar{t}} - \bar{y}_{g,\underline{t}}\right)}_{\text{environmental change}} + \underbrace{\left(\bar{y}_{g,\bar{t}} - \bar{y}^{CF}_{g,\bar{t}}\right)}_{\text{change in where children live}}. \tag{A.16}$$

This mean identity and Equation (A.15) have no residual. Their ordering evaluates environmental change with baseline weights and redistribution with later-period ambient conditions. They are descriptive accounting decompositions, not estimates of causal effects or evidence that the same children moved between cells. Counterfactual quantiles and dispersion summarize the reweighted distribution but do not inherit the linear mixture identity for means.

**Joint counterfactuals and population totals.** The same construction applies to the joint distribution:

$$F^{joint,CF}_{g,\bar{t}}(z_x, z_h) = \sum_{c \in g} s_{c,g,\underline{t}} \, \mathbf{1}\{x_{c,\bar{t}} \leq z_x\} \, \mathbf{1}\{h^{\kappa}_{c,\bar{t}} \leq z_h\}. \tag{A.17}$$

Its compound-exposure probability is obtained by using $s_{c,g,\underline{t}}$ with later-year exposures in Equation (A.10). Within a fixed partition group, the conditional counterfactual instead uses the baseline weights renormalized within that group. These normalized distributions isolate changes in children's relative spatial weights. Absolute counts additionally depend on the total number of children: multiplying a counterfactual probability by the earlier-year population holds that total fixed, whereas multiplying it by the later-year population allows population growth or decline. Similar observed and counterfactual distributions therefore indicate a limited contribution of spatial redistribution to exposure shares and distributional summaries, not the absence of a population-total contribution to changes in exposed counts.

## A.2 Data

In this section, we describe our sources for population, air pollution, and extreme heat data. For a particular subnational, national and supranational region, we compute the child-population-weighted pollution or heat exposure measures using the distribution of children jointly with the distribution of climatic burdens at the finest level of geography where data are available. Our statistics inform policy makers where a specific population group, in this case children under five, faces different magnitudes of climatic exposures. Given that the spatial distribution of children in Latin America and the Caribbean might not overlap with the spatial distribution of climatic burdens, it is essential to combine the population distribution and climatic distribution data jointly to identify locations where children are most at risk.

### Children population

We generate cell-specific global population estimates based on the WorldPop population estimates dataset for 2000, 2010, and 2020 (Tatem 2017). The WorldPop gridded data provides total population estimates at 30 arc-second grids ($\sim$ 1km at the equator), and it is aggregated based on up to administrative level 6 population data from global economies. Population estimates are provided by age and gender categories, which we restrict to ages 0–4 to calculate child population.

### Air pollution

Atmospheric pollution by aerosols is important to human health and well-being, especially when there are higher concentrations of $PM_{2.5}$ particles that are smaller than 2.5 micrometers (Jacobson 2002; Bishop, Ketcham, and Kuminoff 2023; Currie et al. 2015; Currie and Walker 2011). We use Van Donkelaar et al. (2021) estimates available at 0.01° as our measure of air pollution. We chose particulate matter (PM 2.5) as a pollution measure following research indicating that these particles are small enough to enter human bodies and cause health complications (Odo et al. 2023; Fisher et al. 2021; Fu, Viard, and Zhang 2021). We aggregated the data into $1° \times 1°$ grids, calculating the annual average $PM_{2.5}$ level for each grid based on the cells within it. We aggregate these measures to the child population mean exposure for higher geographic groups using cell-level information.

To have benchmarks for air-pollution severity, we use the recommended annual level of PM 2.5 concentration before 2021 — 10 $\mu g/m^3$ — and interim targets (ITs) of the World Health Organization (WHO) as a threshold analysis (*WHO global air quality guidelines* 2021). These thresholds represent staged concentration limits that provide a framework for progressive air quality improvement. Each successive target (IT1 through IT4) corresponds to a significant decrease in air pollution concentrations, allowing for a gradual approach to air quality policies in regions where immediate adherence to the Air Quality Guideline may be technically or economically infeasible. The $PM_{2.5}$ concentrations corresponding to Interim Targets (IT) 4, 3, 2, and 1 are 10, 15, 25, and 35 $\mu g/m^3$, respectively. We use these targets as discrete measures of health-risk exposure to calculate the number of children exposed to pollution levels above and within these thresholds. For example, using population and air pollution data, we compute

the population exposed to pollution levels above IT 1, 35 $\mu g/m^3$, and between IT2 and IT 1, 25-35 $\mu g/m^3$.

### Temperature

We use the fifth generation of the European Centre for Medium-Range Weather Forecasts (ECMWF) atmospheric reanalyses of global climate: the ERA5-HEAT dataset (Napoli 2020). ERA5-HEAT offers hourly data on numerous climatic variables with a spatial resolution of 0.25 degrees. We utilize the Universal Thermal Climate Index (UTCI), an integrative measure of the human-perceived equivalent temperature, taking into account factors like air temperature, humidity, wind speed, and radiant heat (Bröde et al. 2012; Jendritzky, De Dear, and Havenith 2012; Jendritzky and Höppe 2017).

Following Feng et al. (2026), within a particular span of time at a particular gridded location, we compute Share of Time for the Average Child (STAC) statistics. STAC statistics measuring the share of time exposed to extreme heat for the average child, considering multiple temperature thresholds that capture different magnitudes of heat stress. Our main results focus on 32° as a threshold, but we present findings using 26° and 38° in our appendix. We choose to focus on 32°, the lower bound temperature associated with the category of strong heat stress using UTCI (Bröde et al. 2012). To make the share of time easier to interpret, we transform the share of time into months of exposure per year. We check how many children are exposed to 0 to 1, 1 to 2, 2 to 3, 3 to 4, 4 to 5, and 5+ months of extreme heat. We focus our discussion of results on the number of months above 32°, but we also compute these numbers using 26° and 38° of exposure as temperature thresholds.

### GDP per capita

Gross domestic product (GDP) per capita is obtained from the Becker Friedman Institute for Economics Data Studio (Rossi-Hansberg and Zhang 2026). The data provide cell-level annual estimates of GDP per capita for 2012 to 2022, expressed in purchasing power parity (PPP) terms in constant 2021 US dollars. Because estimates are not available before 2012, we use 2020 as the reference year, as it is the only year in the series that coincides with one of our analysis years (2000, 2010, and 2020). We assign cells to income quartiles using their 2020 values and hold these assignments constant across all years.

### Human Development Index (HDI)

Human development is measured using the Subnational Human Development Index (SHDI) from the Global Data Lab (Smits and Permanyer 2019). The SHDI measures development on a scale from 0 to 1 incorporating measures in health, education, and standard of living. The dataset provides annual HDI estimates at the national and subnational levels. We use observations corresponding to 2000, 2010, and 2020, matching the years available to our master dataset

We assign HDI values to our $1° \times 1°$ geographic grid using the GDL's subnational boundary shapefiles. Each grid coordinate is spatially matched to a GDL subnational polygon and

the subnational HDI of the polygon is then attached to the interior grid point. Where a subnational HDI value is unavailable or a coordinate cannot be matched to a subnational region, we substitute the corresponding country's national HDI for the corresponding year though these cases were largely relevant for small nations under which the entire country could be considered a subnational unit. Of the 6,540 grid-year observations, 77.4% received subnational HDI estimates, 20.5% received national estimates, and 2.1% remained unmatched.

### Accessibility

Accessibility is measured using the 2015 Accessibility to Cities map (Weiss et al. 2018), distributed by the Malaria Atlas Project. The map provides, at around 1 km resolution, the travel time in minutes to the nearest city of at least 50,000 inhabitants. We average travel times over the pixels within each $1° \times 1°$ cell and divide cells into quartiles of this mean travel time across Latin America and the Caribbean. Q1 contains the most remote cells, i.e., longest travel times, and Q4 the most accessible. The map provide a single year 2015 information, which implies that we hold remoteness quartiles constant across 2000, 2010, and 2020.

### Land type

Land type is defined using the LandMark database of formally recognized Indigenous and community land rights (LandMark 2026). We classify each $1° \times 1°$ cell as Indigenous if its centroid falls within an Indigenous territory, and as non-Indigenous otherwise, yielding two groups. LandMark boundaries are treated as time-invariant, so each cell keeps the same classification in 2000, 2010, and 2020.

### Rural-Urban Classification

Rural-urban characteristics are measured using the Global Human Settlement Layer Settlement Model (GHS-SMOD R2023A), developed by the European Commission's Joint Research Centre (Pesaresi et al. 2024). The dataset provides global settlement classifications at approximately 1km resolution using a combination of gridded population estimates and built-up surface information. We use the 2000, 2010, and 2020 settlement layers and aggregate the original, highly detailed classifications into the individual classifications of urban, rural, and water.

To integrate these data into our $1° \times 1°$ geographic grid, we calculate the area of intersection between each grid cell and the underlying 1km raster values that fall within a grid cell. These areas are aggregated by settlement category to obtain three variables representing the proportion of each grid cell classified as urban, rural, or water. All three measures are calculated separately for 2000, 2010, and 2020.

### Capturing overlapping pollution and temperature exposures

To capture overlapping exposures to environmental hazards, we merge our cell-level data on air pollution, temperature, and child population. Using the merged dataset, we calculate the number of children exposed to both poor air quality and months with extreme heat. We combine Interim Targets (ITs) of the World Health Organization and months exposed to hot

temperatures to define groups. For example, we calculate the number of children exposed to $PM_{2.5}$ above IT2 — 25 $\mu g/m^3$ — and 3 months above 32° in LAC, its subregions, and countries. Our approach reveals hotspots where children are more vulnerable to both high pollution and high temperatures.

## B Additional results

This section complements the main results using the four components of the framework in Section A.1: marginal exposure distributions, joint exposure, intratemporal conditioning, and intertemporal counterfactual decomposition. Appendix C reports the underlying statistics, and Appendix D provides additional visualizations.

### B.1 Child-population distributions of pollution and heat exposure

The distributions and threshold statistics defined in Section A.1 describe ambient conditions among children, rather than among equally weighted cells. Figure 1, panels a and c, visualizes the pollution and heat distributions for 2000, 2010, and 2020. Table C.1 reports means, and Tables C.2–C.5 report marginal category counts and shares.

**Air pollution.** Mean child-population-weighted $PM_{2.5}$ exposure in LAC increased from 13.05 to 15.96 $\mu g/m^3$ between 2000 and 2020. In 2020, 40.6 million children (87% of the child population) lived in cells above the 10 $\mu g/m^3$ guideline used in the analysis, 7.9 million more than in 2000. Exposure also shifted toward more severe categories: approximately 200,000 children were exposed to concentrations above 35 $\mu g/m^3$ in 2020, whereas no children occupied that category in 2000 (Tables C.2 and C.3; Figure 2a).

**Heat stress.** The average child's exposure to UTCI above 32°C increased from approximately one month per year to 12.5% of the year. The number of children exposed for at least three months nearly doubled, reaching 7.53 million in 2020, approximately 3.5 million more than in 2000. Children facing three to five months of heat stress accounted for about 16% of the child population in 2020 (Tables C.4 and C.5; Figure 2b).

### B.2 Joint distribution of pollution and heat exposure

Section A.1 defines compound exposure as probability mass in the joint pollution–heat distribution. Table C.6 reports the regional counts visualized in Figure 2c, and Table C.12 disaggregates them by country.

**Regional compound exposure.** In 2020, 4.3 million children (9.2%) faced both $PM_{2.5}$ above 15 $\mu g/m^3$ and at least three months of heat stress, approximately 3.2 million more than in 2000. At the four-month heat threshold, the count rose from 0.06 to 1.66 million. For pollution above 25 $\mu g/m^3$ and the four-month heat threshold, it rose from zero to 0.56 million. These combinations illustrate the growing overlap of ambient hazards; they do not establish corresponding health effects.

**Subregional and country patterns.** Compound exposure is concentrated in Central America and South America. For pollution above 15 $\mu g/m^3$ and the four-month heat threshold, South America's count increased from 60,000 to 1.49 million children between 2000 and 2020; Central America's count reached 170,000 in 2020 (Table C.6). Country disaggregations show increases of approximately 850,000 children in Brazil and 280,000 in Colombia for this combination (Table C.12). Conditional shares are reported in Table 2 and Table C.13.

### B.3 Intratemporal decomposition by geographic and socioeconomic groups

The conditional distributions in Section A.1 distinguish differences across groups from dispersion within them. Countries and subregions provide geographic partitions; income, HDI, accessibility, land type, and urban share provide additional partitions. Figure 3 summarizes conditional means and percentiles, while Figures D.8 and D.9 show the full income- and HDI-specific distributions.

**Subregional pollution differences.** Mean pollution exposure increased in all three subregions, but not by the same amount. South America experienced the largest increase, approximately 3.7 $\mu g/m^3$, while Central America retained the highest mean in 2020, 18.1 $\mu g/m^3$ (Table C.1). Central America was the only subregion with children above 25 $\mu g/m^3$ in 2000 and above 35 $\mu g/m^3$ in 2020. By 2020, more than 15% of its children faced concentrations above 25 $\mu g/m^3$ (Tables C.2 and C.3).

The Caribbean illustrates why means alone are insufficient. The share of children below 10 $\mu g/m^3$ increased from approximately 33% in 2000 to 40% in 2020, while the share above 15 $\mu g/m^3$ increased from zero to approximately 38% (Table C.3). The conditional distribution thus changed at both its lower and upper exposure ranges.

**Subregional heat differences.** Mean heat duration increased in all three subregions. The Caribbean experienced the largest increase, approximately four percentage points of the year, or 15 additional days (Table C.1). South America had the largest share of children in the four-to-five-month category in 2020: 2.32 million children, or 7.59%, compared with approximately 470,000 in 2000 (Tables C.4 and C.5). These are distinct comparisons of average heat duration and long-duration exposure.

**Country-level distributions.** Figures D.3 and D.4 provide mapped spatial context for pollution and heat, with category cutoffs given in each year's legend; they should not be interpreted as reproducing the child-weighted country means. Table C.10 reports those means, and Figure D.2 plots them by subregion. Figure D.1 adds the 20th and 80th percentiles, distinguishing changes in mean exposure from changes in within-country dispersion. Means shifted toward higher exposure in many countries, and percentile ranges widened in some countries.

Country threshold counts provide a complementary view. In Brazil, approximately 400,000 children faced pollution above 25 $\mu g/m^3$ in 2020 (Table C.7). Other countries, including Argentina, Bolivia, Honduras, and Peru, also experienced shifts across pollution categories. Table C.11 reports the corresponding heat-duration counts, including categories occupied in 2020 but not in 2000. Changes in category counts describe the distribution of exposure, but do not by themselves establish a change in dispersion.

**Socioeconomic and geographic conditioning.** Tables C.8 and C.9 report marginal category shares within each group, complementing the means and percentile ranges in Figure 3 and Table 1. Table C.13 reports conditional compound-exposure shares. Figure 4 visualizes the income and HDI contrasts, with accessibility and land-type comparisons in Figures D.17 and D.18. Together, these exhibits distinguish high average exposure, broad within-group dispersion, and joint exceedance, rather than treating them as interchangeable measures.

Figures D.10–D.12 visualize the pollution-category shares for income, HDI, and accessibility; Figures D.13–D.15 give the corresponding heat-duration shares. Figure D.16 presents both marginal comparisons for land type. Urban-share comparisons are tabulated rather than separately plotted.

### B.4 Intertemporal decomposition of environmental change and population shifts

We apply the counterfactual construction in Section A.1, combining the ambient conditions of 2010 and 2020 with the 2000 spatial shares of the child population. This holds children's relative spatial distribution fixed while allowing pollution and heat to change. Comparing the counterfactual with the baseline distribution describes environmental change; comparing it with the observed later-year distribution describes the contribution of spatial redistribution.

Figure 1, panels b and d, shows that the observed and counterfactual 2020 distributions overlap closely for both hazards. The substantial exposure increases would therefore have occurred even without changes in where children live. This finding points to worsening ambient conditions, rather than children moving to or being born in more exposed places, as the principal source of the distributional shift. The policy implication is to reduce pollution and heat stress where children reside. These comparisons concern normalized exposure distributions; changes in total child population can still affect absolute exposed counts.

## C Tables

Table C.1: Mean PM 2.5 and time above temperature thresholds by region

| Region | Year | Mean PM 2.5 | time > 26° | time > 32° | time > 38° |
|---|---|---|---|---|---|
| LAC | 2000 | 13.05 | 0.28 | 0.10 | 0.01 |
| LAC | 2010 | 16.44 | 0.31 | 0.11 | 0.01 |
| LAC | 2020 | 15.96 | 0.34 | 0.13 | 0.02 |
| Caribbean | 2000 | 10.69 | 0.46 | 0.14 | 0.01 |
| Caribbean | 2010 | 10.10 | 0.52 | 0.17 | 0.01 |
| Caribbean | 2020 | 11.94 | 0.56 | 0.18 | 0.01 |
| Central America | 2000 | 16.91 | 0.28 | 0.09 | 0.01 |
| Central America | 2010 | 17.01 | 0.29 | 0.09 | 0.01 |
| Central America | 2020 | 18.10 | 0.32 | 0.11 | 0.01 |
| South America | 2000 | 11.49 | 0.28 | 0.10 | 0.01 |
| South America | 2010 | 16.48 | 0.32 | 0.12 | 0.01 |
| South America | 2020 | 15.16 | 0.33 | 0.13 | 0.02 |

*Note:* Child-population-weighted mean $PM_{2.5}$ concentration and shares of the year above the indicated UTCI thresholds, for LAC and its subregions. Section A.1 defines the mean. The pollution and 32°C heat means appear in the regional rows of main Table 1 and Figure 3, supporting the main Results discussion of subregional exposure.

Table C.2: Number of children exposure to PM 2.5 interim targets in millions by subregion

| Region | Year | WHO Guideline | WHO IT4 | WHO IT3 | WHO IT2 | WHO IT1 |
|---|---|---|---|---|---|---|
| LAC | 2000 | 12.28 | 21.09 | 11.16 | 0.45 | 0.00 |
| LAC | 2010 | 6.43 | 12.97 | 21.58 | 3.09 | 0.18 |
| LAC | 2020 | 6.26 | 16.44 | 20.74 | 3.27 | 0.21 |
| Caribbean | 2000 | 0.49 | 0.99 | 0.00 | 0.00 | 0.00 |
| Caribbean | 2010 | 0.78 | 0.61 | 0.03 | 0.00 | 0.00 |
| Caribbean | 2020 | 0.66 | 0.36 | 0.63 | 0.00 | 0.00 |
| Central America | 2000 | 1.18 | 4.10 | 7.44 | 0.45 | 0.00 |
| Central America | 2010 | 1.57 | 3.73 | 6.65 | 1.44 | 0.00 |
| Central America | 2020 | 1.36 | 4.42 | 6.49 | 2.18 | 0.21 |
| South America | 2000 | 10.61 | 16.00 | 3.72 | 0.00 | 0.00 |
| South America | 2010 | 4.07 | 8.63 | 14.90 | 1.65 | 0.18 |
| South America | 2020 | 4.25 | 11.67 | 13.62 | 1.09 | 0.00 |

*Note:* Children in millions within each pollution category, for LAC and its subregions. Categories follow the WHO guideline and interim targets; counts are defined in Section A.1. Figure 2a visualizes the LAC row used in the main air-pollution results. Table C.7 disaggregates these counts by country.

Table C.3: Share of children exposed to PM 2.5 interim targets by subregion

| Region | Year | WHO Guideline | WHO IT-4 | WHO IT-3 | WHO IT-2 | WHO IT-1 |
|---|---|---|---|---|---|---|
| LAC | 2000 | 27.30 | 46.89 | 24.81 | 1.00 | 0.00 |
| LAC | 2010 | 14.53 | 29.30 | 48.76 | 6.99 | 0.41 |
| LAC | 2020 | 13.35 | 35.04 | 44.21 | 6.97 | 0.44 |
| Caribbean | 2000 | 32.95 | 67.05 | 0.00 | 0.00 | 0.00 |
| Caribbean | 2010 | 54.90 | 42.70 | 2.40 | 0.00 | 0.00 |
| Caribbean | 2020 | 39.88 | 21.66 | 38.46 | 0.00 | 0.00 |
| Central America | 2000 | 8.99 | 31.11 | 56.49 | 3.41 | 0.00 |
| Central America | 2010 | 11.76 | 27.83 | 49.65 | 10.76 | 0.00 |
| Central America | 2020 | 9.27 | 30.18 | 44.28 | 14.87 | 1.40 |
| South America | 2000 | 34.98 | 52.76 | 12.26 | 0.00 | 0.00 |
| South America | 2010 | 13.84 | 29.32 | 50.61 | 5.62 | 0.62 |
| South America | 2020 | 13.87 | 38.08 | 44.48 | 3.57 | 0.00 |

*Note:* Percent of children within each pollution category, using the child population of each region and year as the denominator. Figure 2a visualizes the corresponding LAC counts, and Figure 1a shows the underlying pollution distribution.

Table C.4: Number of children exposed to temperature > 32 ° in millions

| Region | Year | 0-1 months | 1-2 months | 2-3 months | 3-4 months | 4-5 months | 5+ months |
|---|---|---|---|---|---|---|---|
| LAC | 2000 | 24.63 | 10.40 | 5.95 | 3.54 | 0.47 | 0.00 |
| LAC | 2010 | 21.92 | 11.38 | 5.72 | 3.22 | 2.02 | 0.00 |
| LAC | 2020 | 21.69 | 10.49 | 7.21 | 5.03 | 2.50 | 0.00 |
| Caribbean | 2000 | 0.25 | 0.60 | 0.63 | 0.00 | 0.00 | 0.00 |
| Caribbean | 2010 | 0.02 | 0.80 | 0.61 | 0.00 | 0.00 | 0.00 |
| Caribbean | 2020 | 0.02 | 0.85 | 0.68 | 0.11 | 0.00 | 0.00 |
| Central America | 2000 | 8.07 | 2.36 | 2.23 | 0.52 | 0.00 | 0.00 |
| Central America | 2010 | 7.55 | 3.24 | 2.20 | 0.40 | 0.00 | 0.00 |
| Central America | 2020 | 7.76 | 2.82 | 2.41 | 1.48 | 0.17 | 0.00 |
| South America | 2000 | 16.31 | 7.45 | 3.09 | 3.01 | 0.47 | 0.00 |
| South America | 2010 | 14.35 | 7.34 | 2.91 | 2.82 | 2.02 | 0.00 |
| South America | 2020 | 13.92 | 6.82 | 4.12 | 3.45 | 2.32 | 0.00 |

*Note:* Children in millions within each duration category of strong heat stress. Duration is months per year with UTCI above 32°C. Figure 2b visualizes the LAC counts used in the main heat-stress results. Table C.11 disaggregates them in thousands; Section A.1 defines the statistic.

Table C.5: Share of children exposed to temperature > 32 ° by subregion

| Region | Year | 0-1 months | 1-2 months | 2-3 months | 3-4 months | 4-5 months | 5+ months |
|---|---|---|---|---|---|---|---|
| LAC | 2000 | 54.75 | 23.12 | 13.22 | 7.87 | 1.04 | 0.00 |
| LAC | 2010 | 49.53 | 25.71 | 12.92 | 7.28 | 4.56 | 0.00 |
| LAC | 2020 | 46.23 | 22.35 | 15.37 | 10.72 | 5.32 | 0.00 |
| Caribbean | 2000 | 17.02 | 40.24 | 42.74 | 0.00 | 0.00 | 0.00 |
| Caribbean | 2010 | 1.47 | 55.69 | 42.84 | 0.00 | 0.00 | 0.00 |
| Caribbean | 2020 | 1.02 | 51.37 | 41.21 | 6.40 | 0.00 | 0.00 |
| Central America | 2000 | 61.24 | 17.90 | 16.89 | 3.98 | 0.00 | 0.00 |
| Central America | 2010 | 56.38 | 24.21 | 16.41 | 2.99 | 0.00 | 0.00 |
| Central America | 2020 | 52.99 | 19.24 | 16.46 | 10.12 | 1.19 | 0.00 |
| South America | 2000 | 53.77 | 24.56 | 10.19 | 9.94 | 1.54 | 0.00 |
| South America | 2010 | 48.74 | 24.94 | 9.88 | 9.58 | 6.85 | 0.00 |
| South America | 2020 | 45.43 | 22.27 | 13.46 | 11.25 | 7.59 | 0.00 |

*Note:* Percent of children within each heat-duration category, using each region's child population as the denominator. Figure 2b visualizes the corresponding LAC counts, and Figure 1c shows the underlying heat-duration distribution.

Table C.6: Number of children exposed to combined air pollution and temperature thresholds by region

| Region | Year | > IT3 & > 3 months | > IT3 & > 4 months | > IT2 & > 3 months | > IT2 & > 4 months |
|---|---|---|---|---|---|
| LAC | 2000 | 1.08 | 0.06 | 0.00 | 0.00 |
| LAC | 2010 | 3.31 | 1.26 | 0.80 | 0.35 |
| LAC | 2020 | 4.33 | 1.66 | 0.78 | 0.56 |
| Caribbean | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| Caribbean | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| Caribbean | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| Central America | 2000 | 0.41 | 0.00 | 0.00 | 0.00 |
| Central America | 2010 | 0.21 | 0.00 | 0.18 | 0.00 |
| Central America | 2020 | 1.14 | 0.17 | 0.30 | 0.17 |
| South America | 2000 | 0.67 | 0.06 | 0.00 | 0.00 |
| South America | 2010 | 3.10 | 1.26 | 0.62 | 0.35 |
| South America | 2020 | 3.19 | 1.49 | 0.48 | 0.38 |

*Note:* Children in millions jointly exceeding the indicated pollution and heat-duration thresholds. Figure 2c visualizes the LAC counts used in the main joint-exposure results. Table C.12 disaggregates them in thousands. Section A.1 defines the statistics.

Table C.7: Number of children exposure to PM 2.5 interim targets in thousands by country

| Country | Year | WHO Guideline | WHO IT-4 | WHO IT-3 | WHO IT-2 | WHO IT-1 |
|---|---|---|---|---|---|---|
| ARG | 2000 | 463.13 | 1276.28 | 859.99 | 0.00 | 0.00 |
| ARG | 2010 | 451.51 | 738.06 | 1384.37 | 39.92 | 0.00 |
| ARG | 2020 | 302.47 | 721.04 | 1732.51 | 6.28 | 0.00 |
| BLZ | 2000 | 0.00 | 122.53 | 0.00 | 0.00 | 0.00 |
| BLZ | 2010 | 0.00 | 47.73 | 84.67 | 0.00 | 0.00 |
| BLZ | 2020 | 0.00 | 58.95 | 94.61 | 0.00 | 0.00 |
| BOL | 2000 | 0.00 | 284.76 | 956.53 | 0.00 | 0.00 |
| BOL | 2010 | 0.00 | 0.00 | 216.26 | 873.99 | 181.71 |
| BOL | 2020 | 0.00 | 47.21 | 663.50 | 660.87 | 0.00 |
| BRA | 2000 | 8889.33 | 5168.84 | 538.03 | 0.00 | 0.00 |
| BRA | 2010 | 3529.20 | 5610.68 | 3452.99 | 514.64 | 0.00 |
| BRA | 2020 | 3691.13 | 7246.86 | 1603.29 | 401.19 | 0.00 |
| CHL | 2000 | 130.03 | 332.82 | 96.66 | 0.00 | 0.00 |
| CHL | 2010 | 8.79 | 225.06 | 294.91 | 0.00 | 0.00 |
| CHL | 2020 | 36.22 | 270.39 | 272.47 | 0.00 | 0.00 |
| COL | 2000 | 409.74 | 3017.76 | 93.20 | 0.00 | 0.00 |
| COL | 2010 | 0.00 | 85.39 | 3664.38 | 0.47 | 0.00 |
| COL | 2020 | 0.00 | 634.75 | 3544.21 | 0.00 | 0.00 |
| CRI | 2000 | 0.00 | 139.08 | 0.00 | 0.00 | 0.00 |
| CRI | 2010 | 0.00 | 0.00 | 124.89 | 0.00 | 0.00 |
| CRI | 2020 | 0.00 | 0.00 | 116.04 | 0.00 | 0.00 |
| CUB | 2000 | 127.97 | 157.46 | 0.00 | 0.00 | 0.00 |
| CUB | 2010 | 232.40 | 0.00 | 0.00 | 0.00 | 0.00 |
| CUB | 2020 | 231.59 | 0.00 | 0.00 | 0.00 | 0.00 |
| DOM | 2000 | 334.76 | 791.63 | 0.00 | 0.00 | 0.00 |
| DOM | 2010 | 531.39 | 609.62 | 0.00 | 0.00 | 0.00 |
| DOM | 2020 | 408.99 | 356.98 | 600.62 | 0.00 | 0.00 |
| ECU | 2000 | 348.81 | 811.02 | 228.11 | 0.00 | 0.00 |
| ECU | 2010 | 0.00 | 0.68 | 1468.06 | 0.00 | 0.00 |
| ECU | 2020 | 0.00 | 334.57 | 1237.72 | 0.00 | 0.00 |
| FLK | 2000 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| FLK | 2010 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| FLK | 2020 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| GTM | 2000 | 0.00 | 0.00 | 1574.74 | 39.75 | 0.00 |
| GTM | 2010 | 0.00 | 39.26 | 1037.12 | 608.03 | 0.00 |
| GTM | 2020 | 0.00 | 108.17 | 565.48 | 1196.09 | 0.00 |
| GUF | 2000 | 1.65 | 2.62 | 0.00 | 0.00 | 0.00 |
| GUF | 2010 | 0.00 | 5.26 | 0.24 | 0.00 | 0.00 |
| GUF | 2020 | 0.00 | 7.83 | 0.00 | 0.00 | 0.00 |
| GUY | 2000 | 45.00 | 5.00 | 0.00 | 0.00 | 0.00 |
| GUY | 2010 | 0.00 | 5.75 | 52.10 | 0.00 | 0.00 |
| GUY | 2020 | 0.00 | 27.15 | 53.06 | 0.00 | 0.00 |
| HND | 2000 | 24.14 | 5.80 | 1058.21 | 390.17 | 0.00 |
| HND | 2010 | 0.00 | 7.11 | 517.71 | 832.48 | 0.00 |
| HND | 2020 | 0.00 | 9.08 | 196.80 | 981.66 | 205.26 |
| JAM | 2000 | 0.00 | 42.72 | 0.00 | 0.00 | 0.00 |
| JAM | 2010 | 0.00 | 0.00 | 34.29 | 0.00 | 0.00 |
| JAM | 2020 | 0.00 | 0.00 | 33.39 | 0.00 | 0.00 |
| MEX | 2000 | 1101.20 | 3406.32 | 4564.65 | 18.77 | 0.00 |
| MEX | 2010 | 1574.02 | 3462.01 | 4322.97 | 0.00 | 0.00 |
| MEX | 2020 | 1357.54 | 3773.27 | 5244.57 | 0.00 | 0.00 |
| MTQ | 2000 | 12.45 | 0.00 | 0.00 | 0.00 | 0.00 |
| MTQ | 2010 | 9.72 | 0.00 | 0.00 | 0.00 | 0.00 |
| MTQ | 2020 | 8.49 | 0.00 | 0.00 | 0.00 | 0.00 |
| NIC | 2000 | 59.49 | 315.60 | 245.55 | 0.00 | 0.00 |
| NIC | 2010 | 0.00 | 93.50 | 518.46 | 0.00 | 0.00 |
| NIC | 2020 | 0.00 | 345.38 | 267.59 | 0.00 | 0.00 |
| PAN | 2000 | 0.00 | 110.11 | 0.00 | 0.00 | 0.00 |
| PAN | 2010 | 0.00 | 76.40 | 40.78 | 0.00 | 0.00 |
| PAN | 2020 | 0.00 | 124.53 | 0.00 | 0.00 | 0.00 |
| PER | 2000 | 2.55 | 1717.74 | 789.99 | 0.00 | 0.00 |
| PER | 2010 | 0.00 | 101.26 | 2419.14 | 58.14 | 0.00 |
| PER | 2020 | 13.32 | 297.81 | 2516.80 | 0.00 | 0.00 |
| PRI | 2000 | 12.26 | 0.00 | 0.00 | 0.00 | 0.00 |
| PRI | 2010 | 10.37 | 0.00 | 0.00 | 0.00 | 0.00 |
| PRI | 2020 | 8.36 | 0.00 | 0.00 | 0.00 | 0.00 |
| PRY | 2000 | 75.07 | 567.40 | 4.42 | 0.00 | 0.00 |
| PRY | 2010 | 0.00 | 304.30 | 300.34 | 0.00 | 0.00 |
| PRY | 2020 | 0.00 | 0.00 | 611.76 | 23.83 | 0.00 |
| SUR | 2000 | 2.67 | 2.02 | 0.00 | 0.00 | 0.00 |
| SUR | 2010 | 0.00 | 2.27 | 2.11 | 0.00 | 0.00 |
| SUR | 2020 | 0.00 | 4.58 | 0.00 | 0.00 | 0.00 |
| URY | 2000 | 37.13 | 1297.06 | 0.00 | 0.00 | 0.00 |
| URY | 2010 | 84.70 | 1399.34 | 0.00 | 0.00 | 0.00 |
| URY | 2020 | 83.45 | 1483.28 | 0.00 | 0.00 | 0.00 |
| USA | 2000 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| USA | 2010 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| USA | 2020 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| VEN | 2000 | 202.59 | 1517.86 | 152.60 | 0.00 | 0.00 |
| VEN | 2010 | 0.00 | 156.64 | 1646.74 | 166.64 | 0.00 |
| VEN | 2020 | 122.38 | 590.29 | 1389.16 | 0.00 | 0.00 |

*Note:* Children in thousands within each pollution category by country. These country-level counts disaggregate the regional results summarized in Figure 2a; the country distributions are summarized in Figure D.1. Country codes are as supplied in the source data.

Table C.8: PM 2.5 threshold exposure (shares) by group and year

| Group | Year | WHO Guideline | WHO IT-4 | WHO IT-3 | WHO IT-2 | WHO IT-1 |
|---|---|---|---|---|---|---|
| **Land type** | | | | | | |
| Indigenous | 2000 | 7.29 | 55.90 | 36.81 | 0.00 | 0.00 |
| | 2010 | 1.43 | 46.50 | 43.21 | 8.86 | 0.00 |
| | 2020 | 0.99 | 59.30 | 36.13 | 3.59 | 0.00 |
| Non-Indigenous | 2000 | 27.98 | 46.58 | 24.40 | 1.03 | 0.00 |
| | 2010 | 15.03 | 28.65 | 48.97 | 6.92 | 0.43 |
| | 2020 | 14.03 | 33.70 | 44.65 | 7.15 | 0.46 |
| **Accessibility** *(Q1 = most remote)* | | | | | | |
| Q1 | 2000 | 21.10 | 61.40 | 17.51 | 0.00 | 0.00 |
| | 2010 | 0.49 | 7.39 | 70.04 | 22.08 | 0.00 |
| | 2020 | 0.56 | 21.59 | 66.18 | 11.67 | 0.00 |
| Q2 | 2000 | 20.08 | 55.25 | 23.39 | 1.28 | 0.00 |
| | 2010 | 6.07 | 11.32 | 68.53 | 14.09 | 0.00 |
| | 2020 | 6.28 | 24.39 | 59.91 | 9.42 | 0.00 |
| Q3 | 2000 | 25.17 | 46.77 | 28.06 | 0.00 | 0.00 |
| | 2010 | 14.45 | 18.45 | 59.11 | 7.62 | 0.38 |
| | 2020 | 13.18 | 23.54 | 57.79 | 5.50 | 0.00 |
| Q4 | 2000 | 28.80 | 45.73 | 24.18 | 1.29 | 0.00 |
| | 2010 | 15.85 | 35.22 | 42.84 | 5.61 | 0.48 |
| | 2020 | 14.62 | 40.37 | 37.36 | 7.01 | 0.64 |
| **Income** *(Q1 = lowest)* | | | | | | |
| Q1 | 2000 | 29.43 | 32.20 | 35.26 | 3.10 | 0.00 |
| | 2010 | 16.63 | 10.46 | 53.56 | 18.02 | 1.32 |
| | 2020 | 18.53 | 16.82 | 43.86 | 20.78 | 0.00 |
| Q2 | 2000 | 25.72 | 39.36 | 34.91 | 0.00 | 0.00 |
| | 2010 | 8.11 | 20.21 | 67.18 | 4.42 | 0.07 |
| | 2020 | 8.90 | 18.67 | 68.51 | 2.42 | 1.50 |
| Q3 | 2000 | 21.61 | 66.82 | 11.58 | 0.00 | 0.00 |
| | 2010 | 16.41 | 41.26 | 41.56 | 0.76 | 0.00 |
| | 2020 | 9.64 | 55.77 | 34.18 | 0.40 | 0.00 |
| Q4 | 2000 | 34.99 | 55.54 | 8.89 | 0.58 | 0.00 |
| | 2010 | 16.65 | 63.31 | 18.77 | 1.27 | 0.00 |
| | 2020 | 14.84 | 66.26 | 18.47 | 0.42 | 0.00 |
| **HDI** *(Q1 = lowest)* | | | | | | |
| Q1 | 2000 | 14.83 | 37.60 | 44.39 | 3.18 | 0.00 |
| | 2010 | 4.02 | 15.15 | 63.80 | 15.82 | 1.20 |
| | 2020 | 3.93 | 20.95 | 57.72 | 16.08 | 1.33 |
| Q2 | 2000 | 37.48 | 31.90 | 30.62 | 0.00 | 0.00 |
| | 2010 | 33.54 | 14.84 | 45.04 | 6.50 | 0.09 |
| | 2020 | 34.61 | 13.56 | 45.16 | 6.67 | 0.00 |
| Q3 | 2000 | 42.49 | 55.55 | 1.96 | 0.00 | 0.00 |
| | 2010 | 13.97 | 58.82 | 26.30 | 0.92 | 0.00 |
| | 2020 | 12.27 | 58.63 | 28.67 | 0.43 | 0.00 |
| Q4 | 2000 | 14.34 | 69.63 | 16.03 | 0.00 | 0.00 |
| | 2010 | 9.20 | 33.13 | 57.19 | 0.48 | 0.00 |
| | 2020 | 4.86 | 54.75 | 40.31 | 0.07 | 0.00 |
| **Urban share** *(Q1 = most rural)* | | | | | | |
| Q1 | 2000 | 33.77 | 47.82 | 18.41 | 0.00 | 0.00 |
| | 2010 | 11.95 | 18.71 | 50.04 | 19.30 | 0.00 |
| | 2020 | 14.13 | 28.02 | 45.77 | 12.08 | 0.00 |
| Q2 | 2000 | 37.97 | 39.41 | 22.62 | 0.00 | 0.00 |
| | 2010 | 23.63 | 18.00 | 36.22 | 19.73 | 2.42 |
| | 2020 | 22.06 | 21.82 | 42.39 | 13.74 | 0.00 |
| Q3 | 2000 | 46.51 | 41.30 | 11.18 | 1.01 | 0.00 |
| | 2010 | 26.67 | 27.87 | 39.51 | 5.95 | 0.00 |
| | 2020 | 26.15 | 31.29 | 38.94 | 3.62 | 0.00 |
| Q4 | 2000 | 23.82 | 48.05 | 27.08 | 1.05 | 0.00 |
| | 2010 | 12.34 | 30.10 | 50.67 | 6.50 | 0.40 |
| | 2020 | 11.04 | 36.24 | 45.06 | 7.14 | 0.53 |

*Note:* Percent of children within each pollution category, conditional on land type, accessibility, income, HDI, or urban share. Assignments use fixed 2020 values (2015 for accessibility). Main Figure 3a and Table 1 summarize these distributions with means and percentiles. Figures D.10–D.12 and D.16 visualize the tabulated category shares, supporting the main intratemporal comparisons. See Section A.1.

Table C.9: Temperature threshold exposure (t32, shares) by group and year

| Group | Year | 0–1 m | 1–2 m | 2–3 m | 3–4 m | 4–5 m | 5+ m |
|---|---|---|---|---|---|---|---|
| **Land type** | | | | | | | |
| Indigenous | 2000 | 55.01 | 22.44 | 10.91 | 11.57 | 0.08 | 0.00 |
| | 2010 | 51.77 | 21.53 | 4.62 | 15.03 | 7.05 | 0.00 |
| | 2020 | 60.06 | 15.90 | 2.78 | 15.38 | 5.88 | 0.00 |
| Non-Indigenous | 2000 | 54.74 | 23.15 | 13.30 | 7.74 | 1.07 | 0.00 |
| | 2010 | 49.44 | 25.87 | 13.24 | 6.98 | 4.46 | 0.00 |
| | 2020 | 45.47 | 22.70 | 16.07 | 10.47 | 5.29 | 0.00 |
| **Accessibility** *(Q1 = most remote)* | | | | | | | |
| Q1 | 2000 | 21.00 | 10.05 | 15.30 | 53.66 | 0.00 | 0.00 |
| | 2010 | 17.95 | 9.36 | 7.14 | 27.91 | 37.64 | 0.00 |
| | 2020 | 8.94 | 14.19 | 10.63 | 38.10 | 28.14 | 0.00 |
| Q2 | 2000 | 53.97 | 7.19 | 14.00 | 22.72 | 2.12 | 0.00 |
| | 2010 | 47.04 | 9.77 | 10.78 | 16.46 | 15.96 | 0.00 |
| | 2020 | 43.08 | 11.00 | 8.07 | 19.28 | 18.57 | 0.00 |
| Q3 | 2000 | 48.87 | 24.68 | 16.43 | 9.28 | 0.75 | 0.00 |
| | 2010 | 41.62 | 28.13 | 16.26 | 8.29 | 5.70 | 0.00 |
| | 2020 | 39.00 | 20.73 | 21.40 | 13.00 | 5.87 | 0.00 |
| Q4 | 2000 | 57.47 | 24.56 | 12.14 | 4.79 | 1.05 | 0.00 |
| | 2010 | 53.18 | 27.10 | 12.27 | 5.41 | 2.05 | 0.00 |
| | 2020 | 50.13 | 24.41 | 14.41 | 8.13 | 2.91 | 0.00 |
| **Income** *(Q1 = lowest)* | | | | | | | |
| Q1 | 2000 | 42.16 | 25.05 | 17.58 | 13.36 | 1.86 | 0.00 |
| | 2010 | 33.03 | 27.79 | 17.59 | 13.92 | 7.67 | 0.00 |
| | 2020 | 29.12 | 21.27 | 24.66 | 15.33 | 9.62 | 0.00 |
| Q2 | 2000 | 59.78 | 17.14 | 13.48 | 8.03 | 1.58 | 0.00 |
| | 2010 | 54.76 | 21.67 | 10.76 | 5.46 | 7.35 | 0.00 |
| | 2020 | 51.95 | 20.21 | 11.35 | 9.52 | 6.96 | 0.00 |
| Q3 | 2000 | 59.92 | 26.07 | 9.39 | 4.55 | 0.08 | 0.00 |
| | 2010 | 56.54 | 28.71 | 9.69 | 4.64 | 0.44 | 0.00 |
| | 2020 | 49.84 | 27.56 | 12.68 | 8.90 | 1.02 | 0.00 |
| Q4 | 2000 | 62.96 | 24.95 | 9.06 | 2.92 | 0.10 | 0.00 |
| | 2010 | 61.21 | 23.93 | 11.91 | 2.66 | 0.29 | 0.00 |
| | 2020 | 62.42 | 20.44 | 9.41 | 6.36 | 1.37 | 0.00 |
| **HDI** *(Q1 = lowest)* | | | | | | | |
| Q1 | 2000 | 45.09 | 15.91 | 22.74 | 13.97 | 2.28 | 0.00 |
| | 2010 | 37.37 | 23.96 | 16.17 | 13.56 | 8.94 | 0.00 |
| | 2020 | 33.79 | 18.49 | 18.18 | 19.71 | 9.84 | 0.00 |
| Q2 | 2000 | 44.44 | 32.33 | 11.06 | 10.83 | 1.35 | 0.00 |
| | 2010 | 40.69 | 30.48 | 14.73 | 7.99 | 6.11 | 0.00 |
| | 2020 | 35.26 | 27.11 | 16.51 | 13.80 | 7.31 | 0.00 |
| Q3 | 2000 | 56.10 | 27.80 | 12.64 | 3.47 | 0.00 | 0.00 |
| | 2010 | 50.73 | 29.09 | 15.40 | 3.95 | 0.84 | 0.00 |
| | 2020 | 50.26 | 26.24 | 17.91 | 4.06 | 1.53 | 0.00 |
| Q4 | 2000 | 81.79 | 17.17 | 1.04 | 0.00 | 0.00 | 0.00 |
| | 2010 | 79.72 | 18.40 | 1.89 | 0.00 | 0.00 | 0.00 |
| | 2020 | 76.57 | 17.99 | 5.44 | 0.00 | 0.00 | 0.00 |
| **Urban share** *(Q1 = most rural)* | | | | | | | |
| Q1 | 2000 | 31.56 | 11.36 | 22.20 | 34.36 | 0.52 | 0.00 |
| | 2010 | 28.43 | 6.87 | 15.29 | 31.39 | 18.02 | 0.00 |
| | 2020 | 24.89 | 7.73 | 14.20 | 31.57 | 21.61 | 0.00 |
| Q2 | 2000 | 35.39 | 14.65 | 19.58 | 29.15 | 1.24 | 0.00 |
| | 2010 | 30.47 | 13.15 | 16.19 | 23.04 | 17.16 | 0.00 |
| | 2020 | 26.74 | 11.18 | 15.33 | 27.75 | 18.99 | 0.00 |
| Q3 | 2000 | 35.87 | 25.99 | 15.70 | 22.16 | 0.28 | 0.00 |
| | 2010 | 29.39 | 25.76 | 16.43 | 15.18 | 13.23 | 0.00 |
| | 2020 | 25.40 | 22.14 | 19.38 | 18.82 | 14.26 | 0.00 |
| Q4 | 2000 | 58.70 | 23.14 | 12.49 | 4.51 | 1.15 | 0.00 |
| | 2010 | 53.61 | 26.41 | 12.23 | 5.17 | 2.58 | 0.00 |
| | 2020 | 50.45 | 23.02 | 14.78 | 8.54 | 3.21 | 0.00 |

*Note:* Percent of children within each heat-duration category, conditional on the listed groups. Assignments use fixed 2020 values (2015 for accessibility). Main Figure 3b and Table 1 summarize these distributions with means and percentiles. Figures D.13–D.15 and D.16 visualize the tabulated heat shares, supporting the main intratemporal comparisons. See Section A.1.

Table C.10: Mean PM 2.5 and time above temperature thresholds by country

| Country | Year | Mean PM 2.5 | time > 26° | time > 32° | time > 38° |
|---|---|---|---|---|---|
| ARG | 2000 | 13.64 | 0.14 | 0.04 | 0.00 |
| ARG | 2010 | 16.27 | 0.15 | 0.05 | 0.01 |
| ARG | 2020 | 16.16 | 0.17 | 0.06 | 0.01 |
| BLZ | 2000 | 11.82 | 0.43 | 0.16 | 0.00 |
| BLZ | 2010 | 16.00 | 0.50 | 0.20 | 0.02 |
| BLZ | 2020 | 18.55 | 0.51 | 0.24 | 0.03 |
| BOL | 2000 | 17.31 | 0.14 | 0.05 | 0.00 |
| BOL | 2010 | 30.00 | 0.17 | 0.06 | 0.01 |
| BOL | 2020 | 24.53 | 0.20 | 0.08 | 0.01 |
| BRA | 2000 | 9.42 | 0.34 | 0.13 | 0.01 |
| BRA | 2010 | 13.45 | 0.39 | 0.16 | 0.02 |
| BRA | 2020 | 12.17 | 0.41 | 0.17 | 0.02 |
| CHL | 2000 | 12.19 | 0.08 | 0.00 | 0.00 |
| CHL | 2010 | 15.52 | 0.08 | 0.01 | 0.00 |
| CHL | 2020 | 13.90 | 0.11 | 0.02 | 0.00 |
| COL | 2000 | 12.77 | 0.28 | 0.08 | 0.01 |
| COL | 2010 | 19.48 | 0.34 | 0.10 | 0.01 |
| COL | 2020 | 17.92 | 0.38 | 0.13 | 0.02 |
| CRI | 2000 | 13.19 | 0.62 | 0.14 | 0.00 |
| CRI | 2010 | 17.44 | 0.63 | 0.17 | 0.00 |
| CRI | 2020 | 16.51 | 0.69 | 0.19 | 0.00 |
| CUB | 2000 | 10.25 | 0.45 | 0.18 | 0.02 |
| CUB | 2010 | 8.26 | 0.48 | 0.18 | 0.02 |
| CUB | 2020 | 8.77 | 0.55 | 0.22 | 0.03 |
| DOM | 2000 | 10.78 | 0.46 | 0.14 | 0.00 |
| DOM | 2010 | 10.37 | 0.53 | 0.17 | 0.00 |
| DOM | 2020 | 12.44 | 0.56 | 0.18 | 0.01 |
| ECU | 2000 | 12.91 | 0.22 | 0.05 | 0.00 |
| ECU | 2010 | 19.62 | 0.27 | 0.07 | 0.00 |
| ECU | 2020 | 16.85 | 0.28 | 0.09 | 0.00 |
| FLK | 2000 | 2.71 | 0.00 | 0.00 | 0.00 |
| FLK | 2010 | 3.66 | 0.00 | 0.00 | 0.00 |
| FLK | 2020 | 3.89 | 0.00 | 0.00 | 0.00 |
| GTM | 2000 | 20.62 | 0.30 | 0.09 | 0.00 |
| GTM | 2010 | 23.00 | 0.33 | 0.10 | 0.01 |
| GTM | 2020 | 24.05 | 0.36 | 0.12 | 0.01 |
| GUF | 2000 | 10.38 | 0.45 | 0.21 | 0.00 |
| GUF | 2010 | 13.63 | 0.55 | 0.26 | 0.01 |
| GUF | 2020 | 12.17 | 0.52 | 0.24 | 0.01 |
| GUY | 2000 | 9.51 | 0.50 | 0.23 | 0.01 |
| GUY | 2010 | 19.13 | 0.60 | 0.28 | 0.02 |
| GUY | 2020 | 14.67 | 0.59 | 0.28 | 0.02 |
| HND | 2000 | 21.39 | 0.40 | 0.14 | 0.01 |
| HND | 2010 | 26.31 | 0.42 | 0.16 | 0.01 |
| HND | 2020 | 30.29 | 0.45 | 0.18 | 0.01 |
| JAM | 2000 | 13.47 | 0.45 | 0.08 | 0.00 |
| JAM | 2010 | 15.74 | 0.55 | 0.11 | 0.00 |
| JAM | 2020 | 16.10 | 0.57 | 0.12 | 0.00 |
| MEX | 2000 | 15.90 | 0.24 | 0.07 | 0.01 |
| MEX | 2010 | 14.58 | 0.23 | 0.07 | 0.01 |
| MEX | 2020 | 15.59 | 0.28 | 0.09 | 0.01 |
| MTQ | 2000 | 6.04 | 0.34 | 0.01 | 0.00 |
| MTQ | 2010 | 6.90 | 0.51 | 0.04 | 0.00 |
| MTQ | 2020 | 6.44 | 0.46 | 0.04 | 0.00 |
| NIC | 2000 | 13.77 | 0.43 | 0.14 | 0.01 |
| NIC | 2010 | 17.70 | 0.52 | 0.18 | 0.02 |
| NIC | 2020 | 15.93 | 0.52 | 0.19 | 0.02 |
| PAN | 2000 | 13.52 | 0.51 | 0.12 | 0.00 |
| PAN | 2010 | 14.18 | 0.59 | 0.16 | 0.00 |
| PAN | 2020 | 13.67 | 0.63 | 0.18 | 0.00 |
| PER | 2000 | 14.25 | 0.12 | 0.03 | 0.00 |
| PER | 2010 | 20.32 | 0.15 | 0.04 | 0.00 |
| PER | 2020 | 18.30 | 0.16 | 0.04 | 0.00 |
| PRI | 2000 | 6.75 | 0.33 | 0.03 | 0.00 |
| PRI | 2010 | 6.21 | 0.47 | 0.06 | 0.00 |
| PRI | 2020 | 6.78 | 0.44 | 0.05 | 0.00 |
| PRY | 2000 | 13.03 | 0.32 | 0.16 | 0.03 |
| PRY | 2010 | 13.89 | 0.32 | 0.14 | 0.03 |
| PRY | 2020 | 21.77 | 0.37 | 0.19 | 0.04 |
| SUR | 2000 | 8.25 | 0.47 | 0.22 | 0.00 |
| SUR | 2010 | 15.28 | 0.57 | 0.27 | 0.01 |
| SUR | 2020 | 12.53 | 0.55 | 0.26 | 0.01 |
| URY | 2000 | 12.24 | 0.12 | 0.03 | 0.00 |
| URY | 2010 | 11.17 | 0.13 | 0.03 | 0.00 |
| URY | 2020 | 11.54 | 0.14 | 0.04 | 0.00 |
| USA | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| USA | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| USA | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| VEN | 2000 | 12.47 | 0.45 | 0.18 | 0.02 |
| VEN | 2010 | 20.09 | 0.56 | 0.23 | 0.03 |
| VEN | 2020 | 16.18 | 0.55 | 0.24 | 0.05 |

*Note:* Child-population-weighted country means, defined in Section A.1. Figures D.1 and D.2 plot pollution and 32°C heat means. These country summaries complement the subregional comparisons in main Figure 3 and the marginal-exposure results. Country codes follow the source data; maps are separate spatial illustrations.

Table C.11: Number of children exposed to temperature > 32 ° in thousands by country

| Country | Year | 0-1 months | 1-2 months | 2-3 months | 3-4 months | 4-5 months | 5+ months |
|---|---|---|---|---|---|---|---|
| ARG | 2000 | 2219.10 | 380.29 | 0.00 | 0.00 | 0.00 | 0.00 |
| ARG | 2010 | 2032.46 | 576.62 | 4.78 | 0.00 | 0.00 | 0.00 |
| ARG | 2020 | 1983.82 | 719.04 | 59.43 | 0.00 | 0.00 | 0.00 |
| BLZ | 2000 | 0.00 | 40.16 | 82.37 | 0.00 | 0.00 | 0.00 |
| BLZ | 2010 | 0.00 | 0.00 | 132.40 | 0.00 | 0.00 | 0.00 |
| BLZ | 2020 | 0.00 | 0.00 | 153.56 | 0.00 | 0.00 | 0.00 |
| BOL | 2000 | 893.64 | 260.40 | 53.74 | 33.52 | 0.00 | 0.00 |
| BOL | 2010 | 823.62 | 287.67 | 98.92 | 61.74 | 0.00 | 0.00 |
| BOL | 2020 | 855.72 | 62.56 | 332.74 | 108.99 | 11.58 | 0.00 |
| BRA | 2000 | 5156.82 | 5575.35 | 1462.22 | 2107.41 | 294.39 | 0.00 |
| BRA | 2010 | 3830.00 | 4588.24 | 1769.60 | 1299.55 | 1620.13 | 0.00 |
| BRA | 2020 | 3346.99 | 4196.38 | 2228.27 | 1546.39 | 1624.42 | 0.00 |
| CHL | 2000 | 559.51 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| CHL | 2010 | 528.76 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| CHL | 2020 | 579.08 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| COL | 2000 | 2383.99 | 297.11 | 494.56 | 174.99 | 170.06 | 0.00 |
| COL | 2010 | 2284.51 | 417.95 | 390.03 | 462.70 | 195.05 | 0.00 |
| COL | 2020 | 2341.95 | 556.93 | 138.37 | 837.08 | 304.62 | 0.00 |
| CRI | 2000 | 0.00 | 139.08 | 0.00 | 0.00 | 0.00 | 0.00 |
| CRI | 2010 | 0.00 | 124.89 | 0.00 | 0.00 | 0.00 | 0.00 |
| CRI | 2020 | 0.00 | 0.00 | 116.04 | 0.00 | 0.00 | 0.00 |
| CUB | 2000 | 1.14 | 126.17 | 158.12 | 0.00 | 0.00 | 0.00 |
| CUB | 2010 | 0.86 | 103.09 | 128.45 | 0.00 | 0.00 | 0.00 |
| CUB | 2020 | 0.00 | 47.41 | 78.66 | 105.52 | 0.00 | 0.00 |
| DOM | 2000 | 183.24 | 469.08 | 474.07 | 0.00 | 0.00 | 0.00 |
| DOM | 2010 | 0.00 | 657.77 | 483.24 | 0.00 | 0.00 | 0.00 |
| DOM | 2020 | 0.00 | 765.97 | 600.62 | 0.00 | 0.00 | 0.00 |
| ECU | 2000 | 1084.26 | 303.22 | 0.47 | 0.00 | 0.00 | 0.00 |
| ECU | 2010 | 925.37 | 444.39 | 98.30 | 0.68 | 0.00 | 0.00 |
| ECU | 2020 | 594.27 | 723.49 | 253.63 | 0.91 | 0.00 | 0.00 |
| FLK | 2000 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| FLK | 2010 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| FLK | 2020 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| GTM | 2000 | 911.75 | 569.29 | 23.91 | 109.54 | 0.00 | 0.00 |
| GTM | 2010 | 878.01 | 652.38 | 114.76 | 39.26 | 0.00 | 0.00 |
| GTM | 2020 | 944.27 | 56.97 | 692.70 | 175.79 | 0.00 | 0.00 |
| GUF | 2000 | 0.00 | 0.00 | 4.27 | 0.00 | 0.00 | 0.00 |
| GUF | 2010 | 0.00 | 0.00 | 1.82 | 3.68 | 0.00 | 0.00 |
| GUF | 2020 | 0.00 | 0.00 | 6.38 | 1.45 | 0.00 | 0.00 |
| GUY | 2000 | 2.48 | 2.60 | 14.26 | 30.66 | 0.00 | 0.00 |
| GUY | 2010 | 1.25 | 4.90 | 1.84 | 49.85 | 0.00 | 0.00 |
| GUY | 2020 | 3.26 | 9.08 | 1.86 | 21.14 | 44.87 | 0.00 |
| HND | 2000 | 720.84 | 24.14 | 521.82 | 211.51 | 0.00 | 0.00 |
| HND | 2010 | 215.34 | 514.29 | 451.34 | 176.33 | 0.00 | 0.00 |
| HND | 2020 | 0.00 | 731.06 | 487.14 | 0.00 | 174.60 | 0.00 |
| JAM | 2000 | 42.72 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| JAM | 2010 | 0.00 | 34.29 | 0.00 | 0.00 | 0.00 | 0.00 |
| JAM | 2020 | 0.00 | 33.39 | 0.00 | 0.00 | 0.00 | 0.00 |
| MEX | 2000 | 6175.89 | 1378.03 | 1333.46 | 203.55 | 0.00 | 0.00 |
| MEX | 2010 | 6454.53 | 1520.26 | 1229.65 | 154.57 | 0.00 | 0.00 |
| MEX | 2020 | 6816.17 | 1640.09 | 829.08 | 1090.04 | 0.00 | 0.00 |
| MTQ | 2000 | 12.45 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| MTQ | 2010 | 9.72 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| MTQ | 2020 | 8.49 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| NIC | 2000 | 259.98 | 101.84 | 258.81 | 0.00 | 0.00 | 0.00 |
| NIC | 2010 | 0.00 | 317.47 | 263.79 | 30.70 | 0.00 | 0.00 |
| NIC | 2020 | 0.00 | 347.51 | 49.93 | 215.54 | 0.00 | 0.00 |
| PAN | 2000 | 0.00 | 105.47 | 4.64 | 0.00 | 0.00 | 0.00 |
| PAN | 2010 | 0.00 | 111.94 | 5.24 | 0.00 | 0.00 | 0.00 |
| PAN | 2020 | 0.00 | 42.04 | 82.48 | 0.00 | 0.00 | 0.00 |
| PER | 2000 | 2273.85 | 46.27 | 70.19 | 119.97 | 0.00 | 0.00 |
| PER | 2010 | 2268.76 | 83.01 | 81.30 | 60.41 | 85.06 | 0.00 |
| PER | 2020 | 2471.09 | 93.15 | 98.90 | 136.05 | 28.74 | 0.00 |
| PRI | 2000 | 12.26 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| PRI | 2010 | 10.37 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| PRI | 2020 | 8.36 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| PRY | 2000 | 0.00 | 247.31 | 399.58 | 0.00 | 0.00 | 0.00 |
| PRY | 2010 | 0.00 | 559.23 | 45.41 | 0.00 | 0.00 | 0.00 |
| PRY | 2020 | 0.00 | 143.11 | 487.45 | 5.03 | 0.00 | 0.00 |
| SUR | 2000 | 0.00 | 0.00 | 4.69 | 0.00 | 0.00 | 0.00 |
| SUR | 2010 | 0.00 | 0.00 | 0.54 | 3.84 | 0.00 | 0.00 |
| SUR | 2020 | 0.00 | 0.00 | 0.43 | 4.15 | 0.00 | 0.00 |
| URY | 2000 | 1334.20 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| URY | 2010 | 1484.03 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| URY | 2020 | 1566.73 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| USA | 2000 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| USA | 2010 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| USA | 2020 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| VEN | 2000 | 400.44 | 335.17 | 587.10 | 547.11 | 3.23 | 0.00 |
| VEN | 2010 | 173.67 | 381.02 | 418.15 | 879.18 | 118.00 | 0.00 |
| VEN | 2020 | 174.07 | 318.58 | 515.24 | 784.46 | 309.49 | 0.00 |

*Note:* Children in thousands within each heat-duration category by country. Summing countries and dividing by 1,000 gives the LAC counts in Table C.4 and main Figure 2b. These counts support the main heat-stress results; Figure D.1 instead shows country means and percentiles. Country codes follow the source data.

Table C.12: Number of children exposed to combined air pollution and temperature thresholds by country

| Country | Year | > IT3 & > 3 months | > IT3 & > 4 month | > IT2 & > 3 month | > IT2 & > 4 month |
|---|---|---|---|---|---|
| ARG | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| ARG | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| ARG | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| BLZ | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| BLZ | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| BLZ | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| BOL | 2000 | 25.42 | 0.00 | 0.00 | 0.00 |
| BOL | 2010 | 61.74 | 0.00 | 52.56 | 0.00 |
| BOL | 2020 | 120.57 | 11.58 | 75.47 | 11.58 |
| BRA | 2000 | 531.36 | 36.66 | 0.00 | 0.00 |
| BRA | 2010 | 1296.41 | 866.09 | 511.66 | 335.84 |
| BRA | 2020 | 1256.27 | 884.46 | 401.19 | 370.75 |
| CHL | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| CHL | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| CHL | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| COL | 2000 | 23.96 | 23.96 | 0.00 | 0.00 |
| COL | 2010 | 572.36 | 195.05 | 0.47 | 0.47 |
| COL | 2020 | 916.43 | 304.62 | 0.00 | 0.00 |
| CRI | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| CRI | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| CRI | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| CUB | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| CUB | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| CUB | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| DOM | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| DOM | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| DOM | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| ECU | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| ECU | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| ECU | 2020 | 0.91 | 0.00 | 0.00 | 0.00 |
| FLK | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| FLK | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| FLK | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| GTM | 2000 | 109.54 | 0.00 | 0.00 | 0.00 |
| GTM | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| GTM | 2020 | 124.59 | 0.00 | 124.59 | 0.00 |
| GUF | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| GUF | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| GUF | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| GUY | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| GUY | 2010 | 44.11 | 0.00 | 0.00 | 0.00 |
| GUY | 2020 | 47.11 | 43.38 | 0.00 | 0.00 |
| HND | 2000 | 211.51 | 0.00 | 0.00 | 0.00 |
| HND | 2010 | 176.33 | 0.00 | 176.33 | 0.00 |
| HND | 2020 | 174.60 | 174.60 | 174.60 | 174.60 |
| JAM | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| JAM | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| JAM | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| MEX | 2000 | 88.50 | 0.00 | 0.00 | 0.00 |
| MEX | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| MEX | 2020 | 624.95 | 0.00 | 0.00 | 0.00 |
| MTQ | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| MTQ | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| MTQ | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| NIC | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| NIC | 2010 | 30.70 | 0.00 | 0.00 | 0.00 |
| NIC | 2020 | 215.54 | 0.00 | 0.00 | 0.00 |
| PAN | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| PAN | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| PAN | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| PER | 2000 | 5.20 | 0.00 | 0.00 | 0.00 |
| PER | 2010 | 143.00 | 85.06 | 0.83 | 0.00 |
| PER | 2020 | 156.77 | 28.74 | 0.00 | 0.00 |
| PRI | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| PRI | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| PRI | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| PRY | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| PRY | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| PRY | 2020 | 5.03 | 0.00 | 5.03 | 0.00 |
| SUR | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| SUR | 2010 | 1.57 | 0.00 | 0.00 | 0.00 |
| SUR | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| URY | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| URY | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| URY | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| USA | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| USA | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| USA | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| VEN | 2000 | 84.14 | 0.00 | 0.00 | 0.00 |
| VEN | 2010 | 985.31 | 118.00 | 57.68 | 13.77 |
| VEN | 2020 | 683.27 | 216.70 | 0.00 | 0.00 |

*Note:* Children in thousands jointly above the indicated thresholds. Summing countries and dividing by 1,000 gives the LAC counts in Table C.6 and main Figure 2c, supporting the joint-exposure results. Main Figure 4 instead shows conditional group shares. See Section A.1; country codes follow the source data.

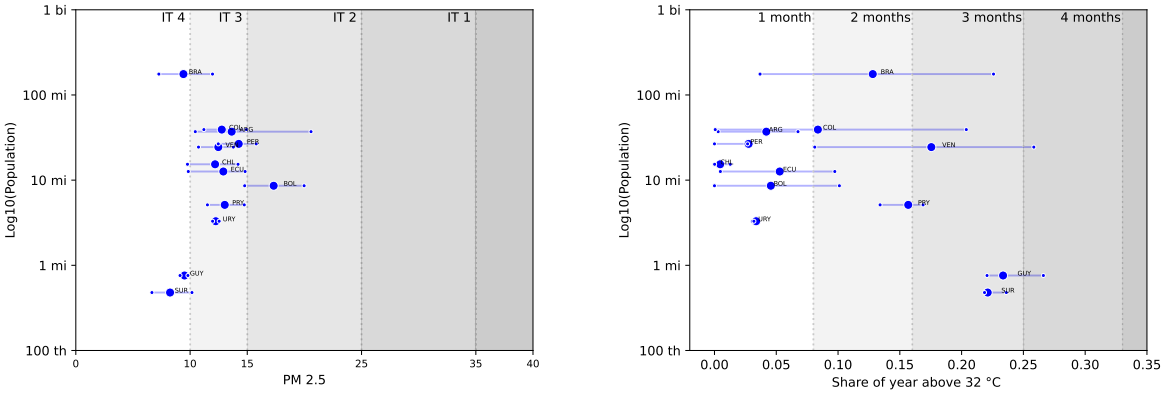
Table C.13: Combined PM 2.5 and heat threshold exposure (shares) by group and year

| Group | Year | > IT3 & > 3 months | > IT3 & > 4 months | > IT2 & > 3 months | > IT2 & > 4 months |
|---|---|---|---|---|---|
| **Land type** | | | | | |
| Indigenous | 2000 | 3.06 | 0.00 | 0.00 | 0.00 |
| | 2010 | 15.21 | 6.11 | 4.34 | 2.59 |
| | 2020 | 12.62 | 5.76 | 3.26 | 2.41 |
| Non-Indigenous | 2000 | 2.38 | 0.14 | 0.00 | 0.00 |
| | 2010 | 7.19 | 2.73 | 1.71 | 0.72 |
| | 2020 | 9.03 | 3.42 | 1.58 | 1.12 |
| **Accessibility** *(Q1 = most remote)* | | | | | |
| Q1 | 2000 | 15.16 | 0.00 | 0.00 | 0.00 |
| | 2010 | 59.32 | 34.41 | 18.25 | 13.53 |
| | 2020 | 51.15 | 25.82 | 11.61 | 10.96 |
| Q2 | 2000 | 13.34 | 1.95 | 0.00 | 0.00 |
| | 2010 | 26.27 | 14.31 | 9.11 | 4.86 |
| | 2020 | 29.64 | 16.55 | 7.88 | 5.73 |
| Q3 | 2000 | 2.33 | 0.00 | 0.00 | 0.00 |
| | 2010 | 9.03 | 3.20 | 1.07 | 0.78 |
| | 2020 | 9.22 | 3.18 | 0.74 | 0.59 |
| Q4 | 2000 | 1.01 | 0.00 | 0.00 | 0.00 |
| | 2010 | 3.53 | 0.64 | 0.80 | 0.00 |
| | 2020 | 5.54 | 1.47 | 0.93 | 0.55 |
| **Income** *(Q1 = lowest)* | | | | | |
| Q1 | 2000 | 4.03 | 0.00 | 0.00 | 0.00 |
| | 2010 | 12.44 | 3.19 | 2.69 | 0.43 |
| | 2020 | 12.67 | 4.77 | 2.96 | 1.65 |
| Q2 | 2000 | 2.33 | 0.46 | 0.00 | 0.00 |
| | 2010 | 8.85 | 6.03 | 2.51 | 1.98 |
| | 2020 | 10.53 | 5.94 | 2.28 | 2.22 |
| Q3 | 2000 | 1.37 | 0.00 | 0.00 | 0.00 |
| | 2010 | 3.52 | 0.44 | 0.69 | 0.22 |
| | 2020 | 6.38 | 0.82 | 0.29 | 0.11 |
| Q4 | 2000 | 1.23 | 0.00 | 0.00 | 0.00 |
| | 2010 | 2.23 | 0.29 | 0.69 | 0.20 |
| | 2020 | 5.28 | 1.30 | 0.36 | 0.17 |
| **HDI** *(Q1 = lowest)* | | | | | |
| Q1 | 2000 | 3.41 | 0.17 | 0.00 | 0.00 |
| | 2010 | 13.59 | 3.96 | 2.32 | 0.17 |
| | 2020 | 15.67 | 4.85 | 2.30 | 1.20 |
| Q2 | 2000 | 4.87 | 0.34 | 0.00 | 0.00 |
| | 2010 | 8.52 | 5.85 | 3.50 | 2.82 |
| | 2020 | 12.38 | 6.80 | 3.45 | 3.21 |
| Q3 | 2000 | 0.64 | 0.00 | 0.00 | 0.00 |
| | 2010 | 4.24 | 0.73 | 0.89 | 0.26 |
| | 2020 | 4.68 | 1.47 | 0.43 | 0.19 |
| Q4 | 2000 | 0.00 | 0.00 | 0.00 | 0.00 |
| | 2010 | 0.00 | 0.00 | 0.00 | 0.00 |
| | 2020 | 0.00 | 0.00 | 0.00 | 0.00 |
| **Urban share** *(Q1 = most rural)* | | | | | |
| Q1 | 2000 | 9.36 | 0.00 | 0.00 | 0.00 |
| | 2010 | 43.66 | 16.02 | 12.67 | 5.23 |
| | 2020 | 40.46 | 17.44 | 9.90 | 5.78 |
| Q2 | 2000 | 9.90 | 0.00 | 0.00 | 0.00 |
| | 2010 | 30.81 | 15.23 | 12.89 | 7.39 |
| | 2020 | 30.46 | 14.78 | 9.00 | 6.67 |
| Q3 | 2000 | 4.81 | 0.00 | 0.00 | 0.00 |
| | 2010 | 17.07 | 7.07 | 3.05 | 1.25 |
| | 2020 | 16.26 | 6.56 | 2.06 | 1.31 |
| Q4 | 2000 | 1.65 | 0.16 | 0.00 | 0.00 |
| | 2010 | 4.68 | 1.57 | 1.05 | 0.41 |
| | 2020 | 6.90 | 2.46 | 1.20 | 0.89 |

*Note:* Percent of children jointly above the thresholds within each group, also reported in main Table 2. Main Figure 4 visualizes the lowest and highest income and HDI groups, adding a higher pollution threshold. Figures D.17 and D.18 show accessibility and land-type comparisons. These support the main intratemporal results; see Section A.1.

# D Figures

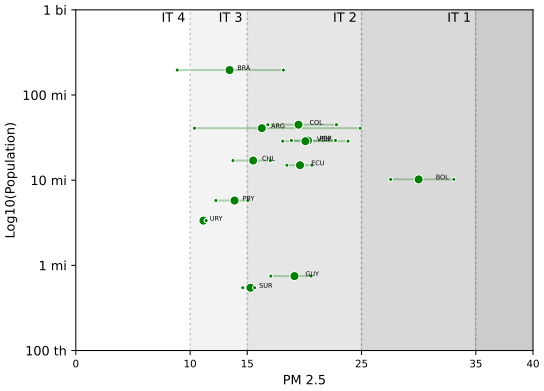
Figure D.1: Country exposure distributions



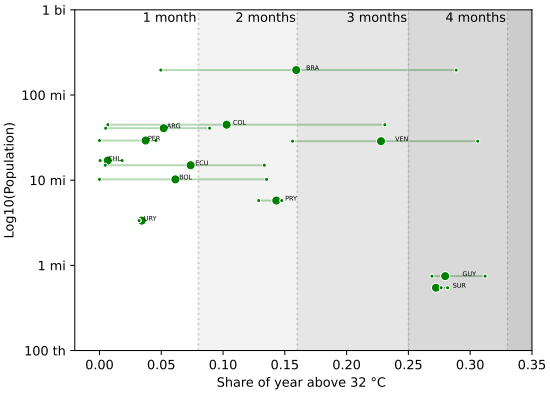

(a) Pollution, 2000



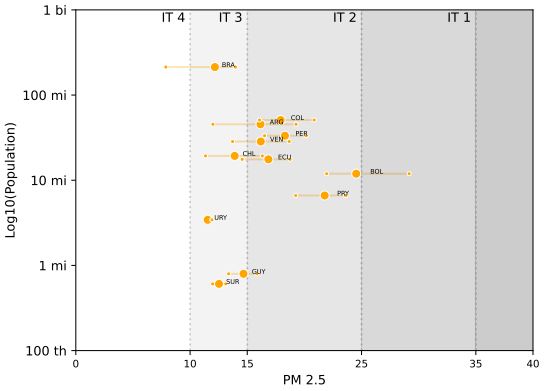

(b) Heat, 2000



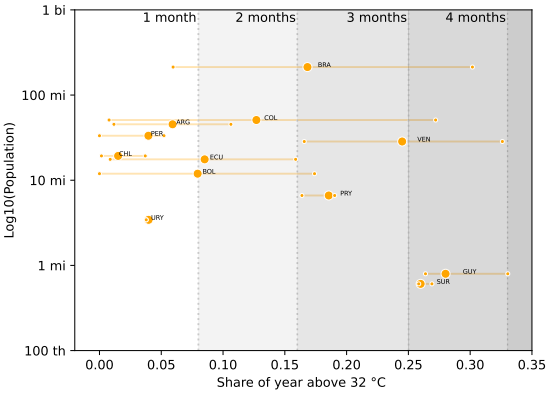

(c) Pollution, 2010




(d) Heat, 2010



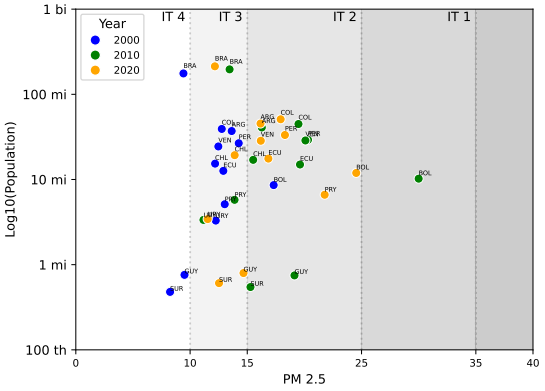

(e) Pollution, 2020



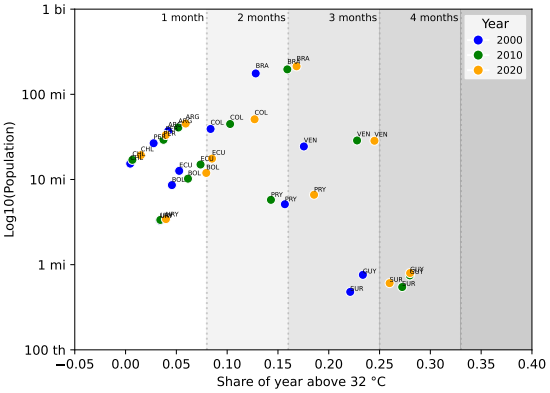

(f) Heat, 2020

*Notes:* Endpoints and central dots mark P20, P80, and the child-weighted country mean. Pollution is in $\mu g/m^3$; heat is the share of the year with UTCI above 32°C. Shading marks severity categories; population is on a logarithmic scale. Table C.10 reports means; Tables C.7 and C.11 report category counts. These country comparisons complement main Figure 3 and the subregional results.

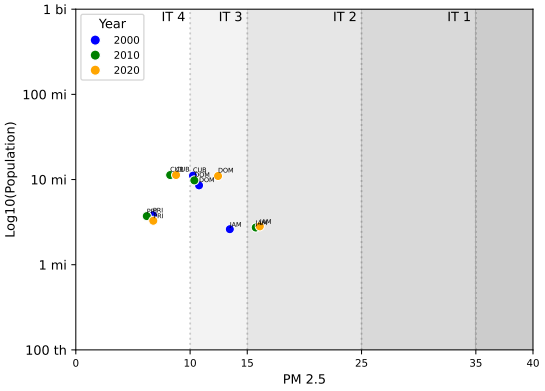
Figure D.2: Regional exposure distributions over time



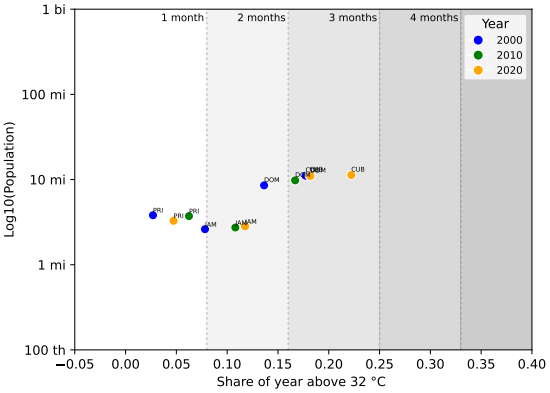

(a) Pollution, South America



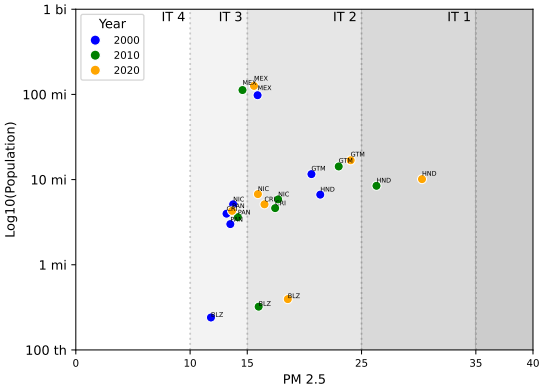

(b) Heat, South America



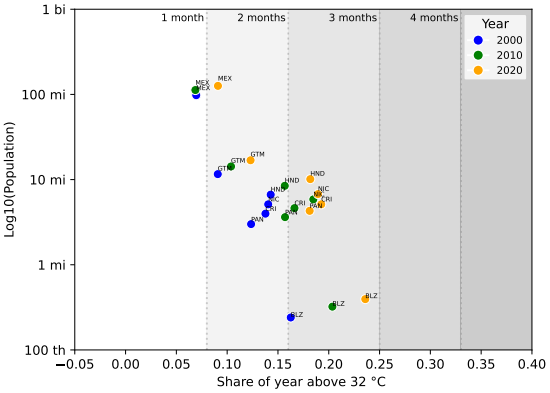

(c) Pollution, Caribbean




(d) Heat, Caribbean



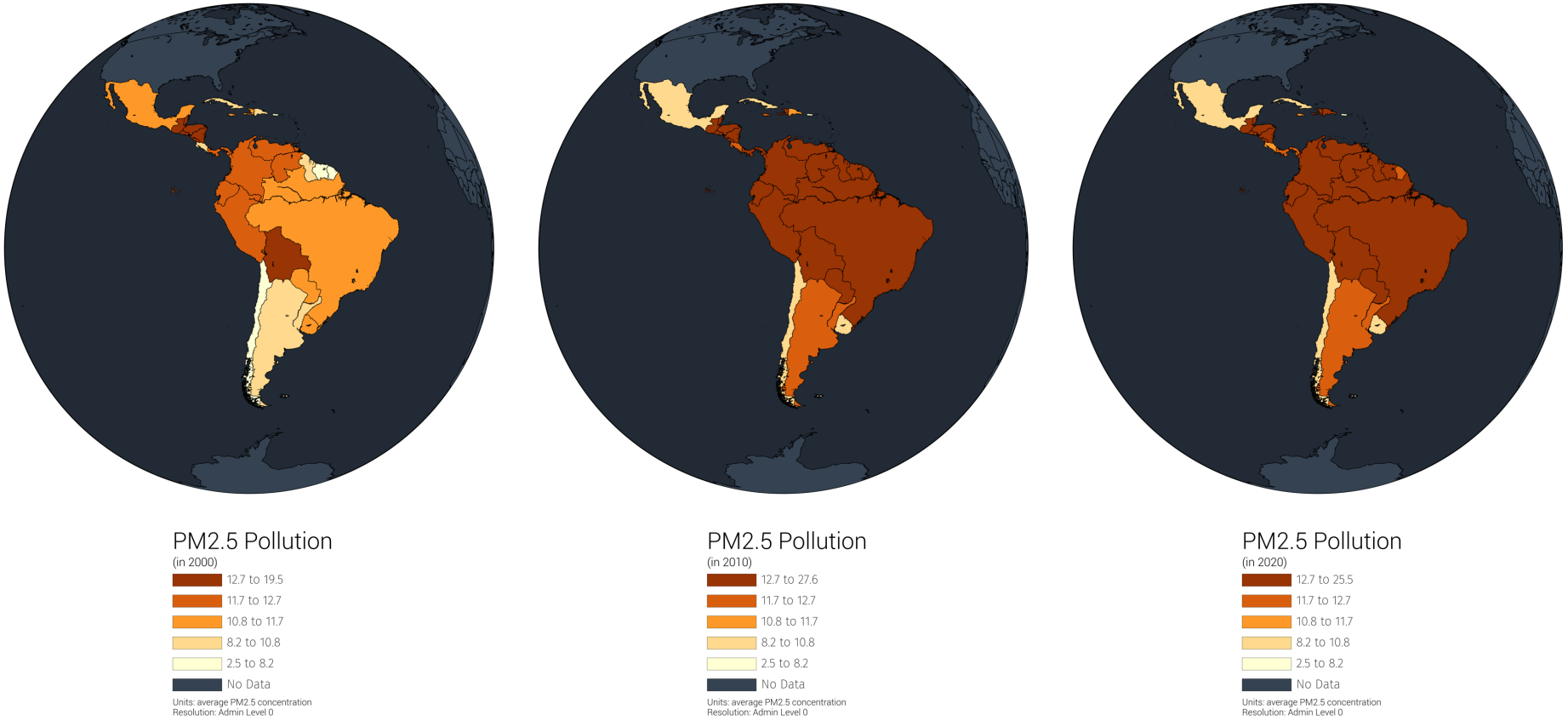

(e) Pollution, Central America




(f) Heat, Central America

*Notes:* Child-weighted country means from Table C.10, grouped by subregion. Pollution is in $\mu g/m^3$; heat is the share of the year with UTCI above 32°C. Colors distinguish years; population is on a logarithmic scale. These country trajectories complement the subregional means in Table C.1 and main Figure 3, discussed in the main marginal-exposure results.

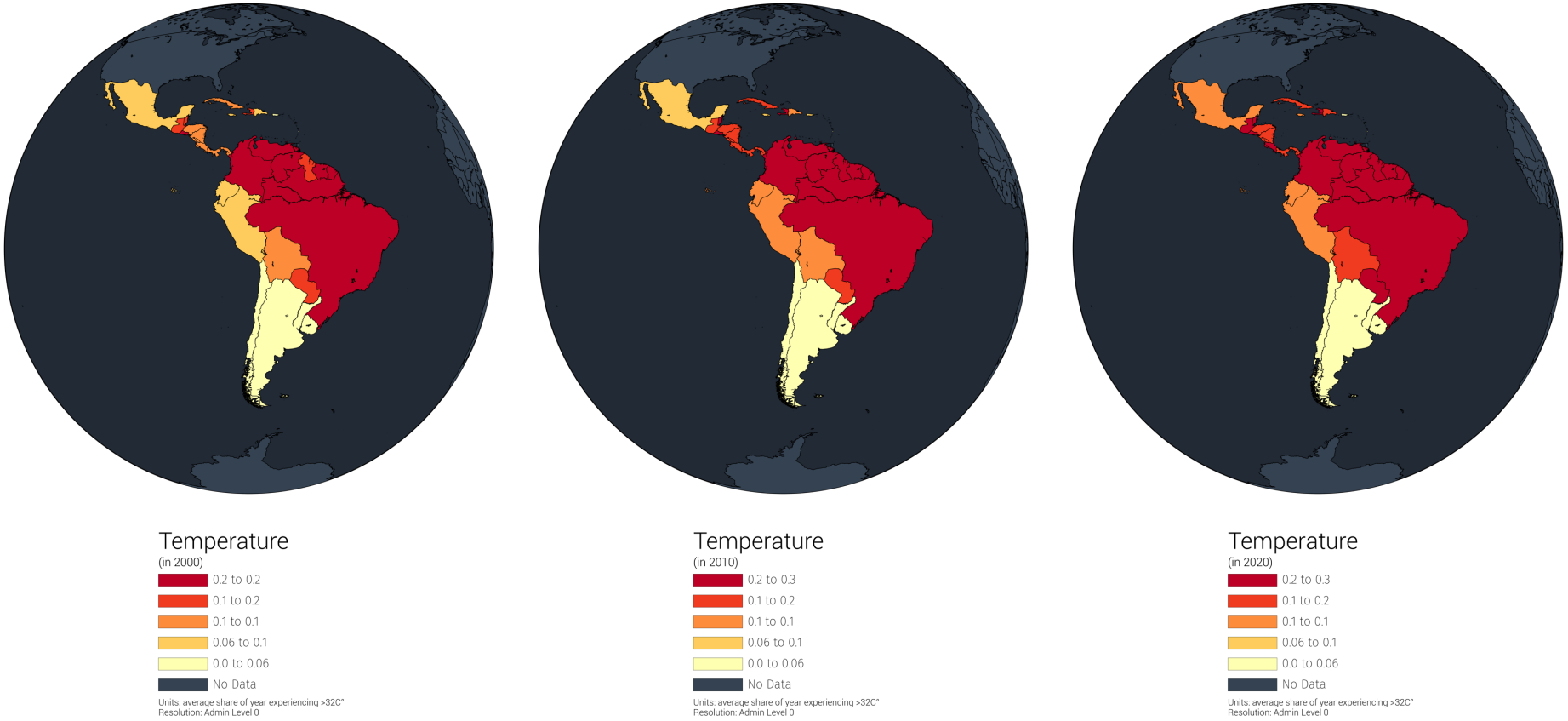
Figure D.3: Country-level pollution exposure




(a) 2000 (b) 2010 (c) 2020

*Notes:* Mapped country-level $PM_{2.5}$ from Van Donkelaar et al. (2021). Darker colors indicate higher pollution; dark gray denotes no data. Each year's legend gives its cutoffs. These maps provide spatial context for the main air-pollution results and Figures 1a and 3a; they should not be read as reproducing the child-weighted country means in Table C.10.

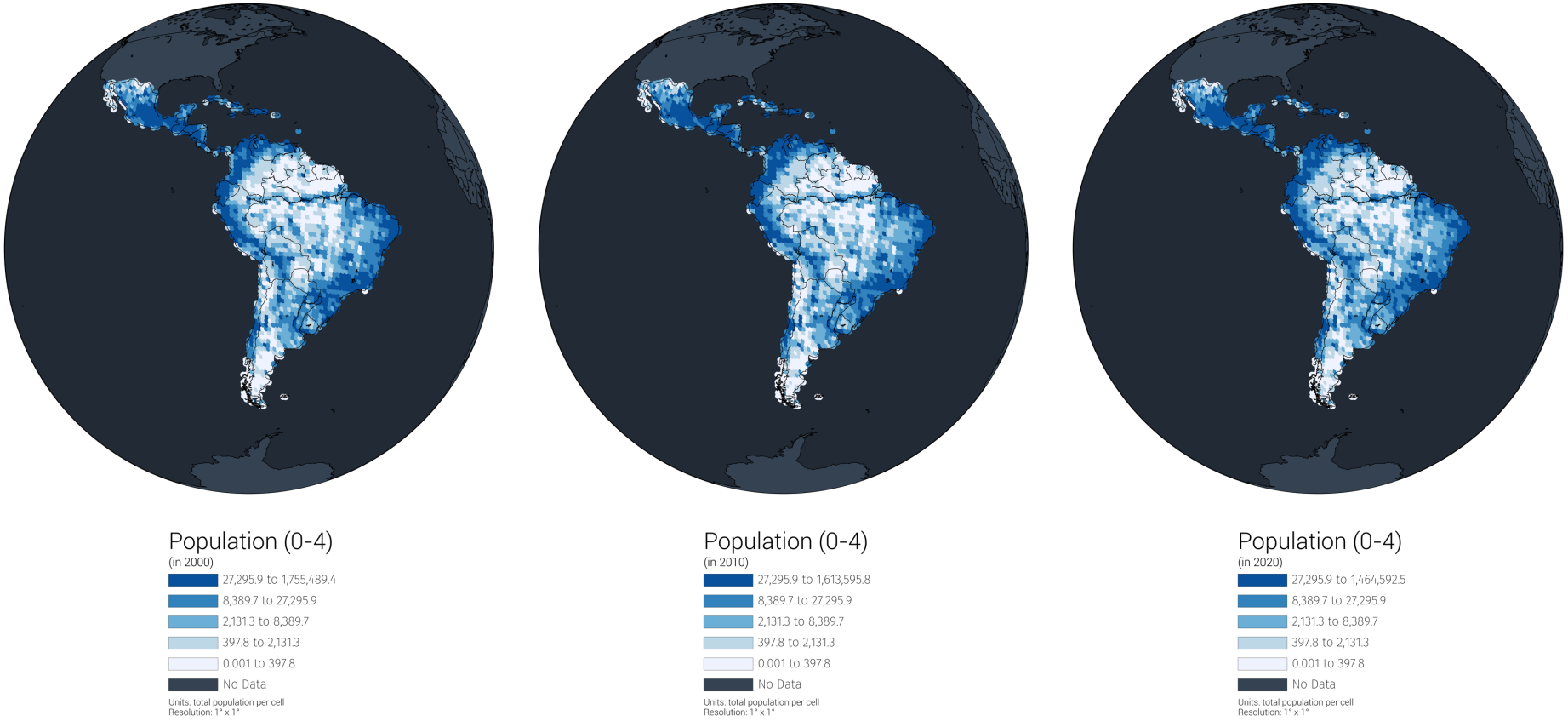
Figure D.4: Country-level heat exposure




(a) 2000 (b) 2010 (c) 2020

*Notes:* Mapped country-level share of the year with UTCI above 32°C. Darker colors indicate longer heat exposure; dark gray denotes no data. Each year's legend gives its cutoffs. These maps provide spatial context for the main heat-stress results and Figures 1c and 3b; they should not be read as reproducing the child-weighted country means in Table C.10.

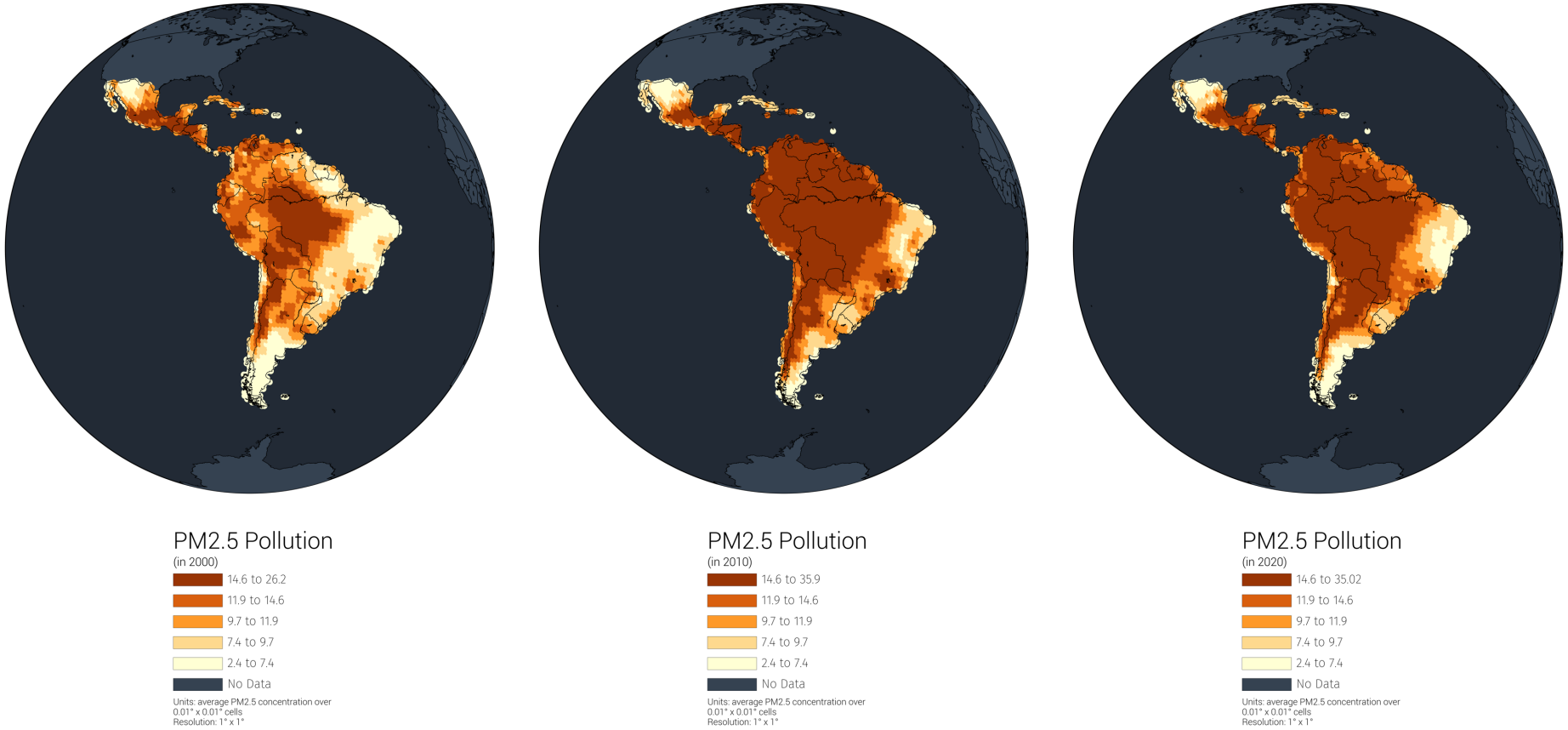
Figure D.5: Child population by grid cell




(a) 2000 (b) 2010 (c) 2020

*Notes:* WorldPop estimates of children aged 0–4 on $1^\circ \times 1^\circ$ cells. Darker blue indicates larger populations; dark gray denotes no data or locations outside LAC. Each year's legend gives its cutoffs. These maps illustrate the spatial population input (Section A.2) underlying main Figure 1. They provide context for the main intertemporal discussion, not estimates of the counterfactual or redistribution effect.

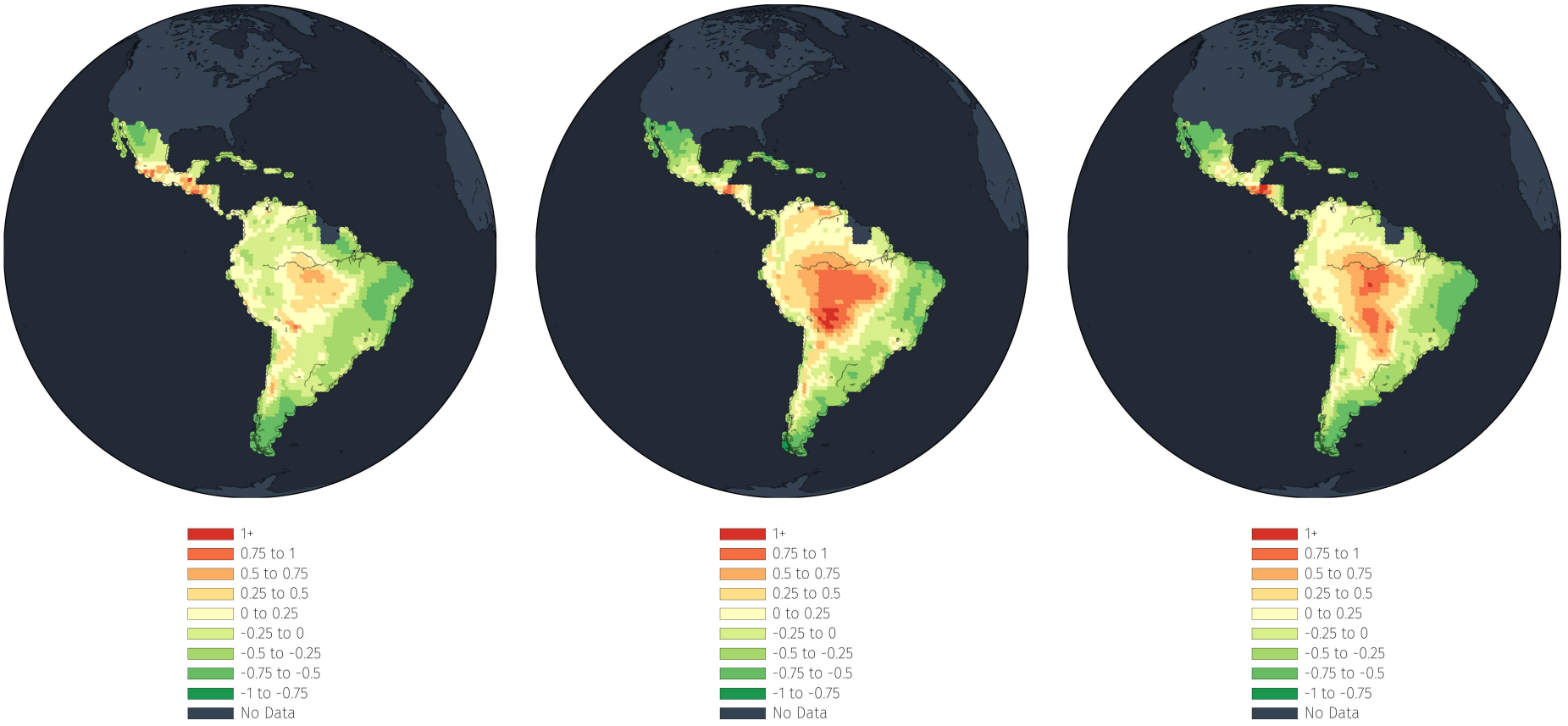
Figure D.6: Pollution exposure by grid cell




(a) 2000 (b) 2010 (c) 2020

*Notes:* $PM_{2.5}$ estimates from Van Donkelaar et al. (2021) on $1^\circ \times 1^\circ$ cells. Darker colors indicate higher pollution; dark gray denotes no data or locations outside LAC. Each year's legend gives its cutoffs. These maps provide spatial context for main Figure 1a and the air-pollution results. Unlike the distribution in that figure, the map does not weight locations by child population.

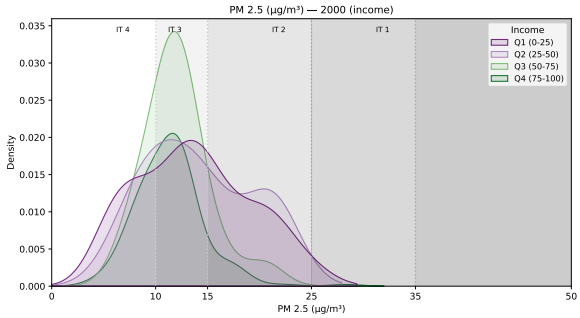
Figure D.7: Excess pollution burden by grid cell



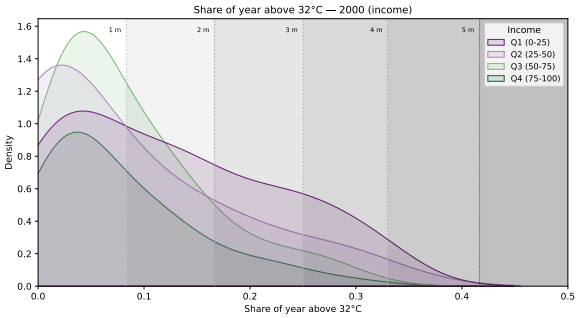

(a) 2000 (b) 2010 (c) 2020

*Notes:* Spatial burden diagnostic combining WorldPop child-population data and $PM_{2.5}$ estimates from Van Donkelaar et al. (2021). Red and green indicate higher and lower relative burdens; legends retain the source scale. Dark gray denotes no data or locations outside LAC. This provides spatial context for the main pollution results, not the concentration distribution in Figure 1a or a counterfactual redistribution effect.

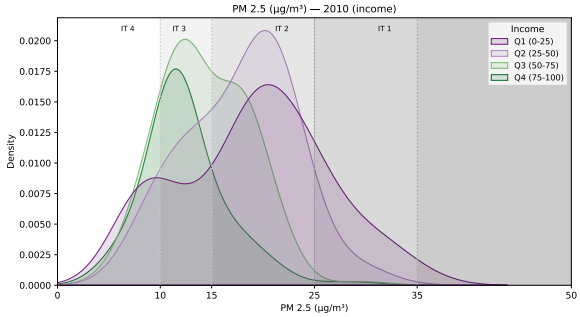
Figure D.8: Exposure distributions by income group



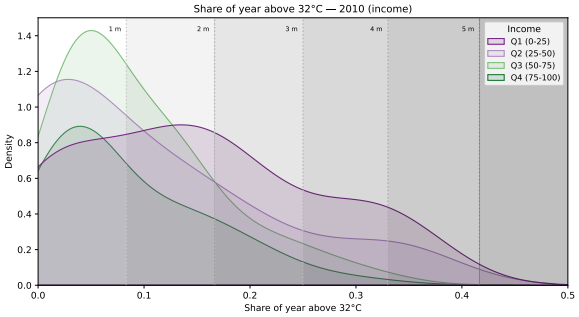

(a) Pollution, 2000



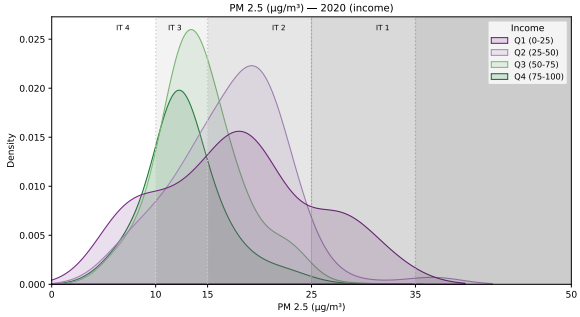

(b) Heat, 2000



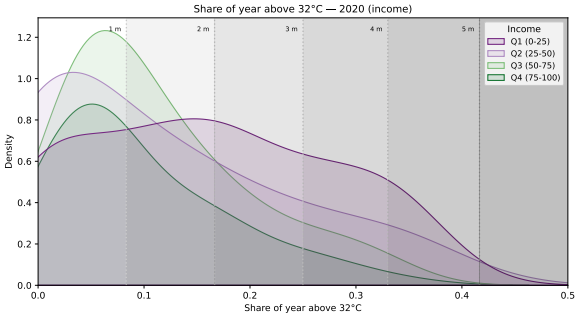

(c) Pollution, 2010




(d) Heat, 2010



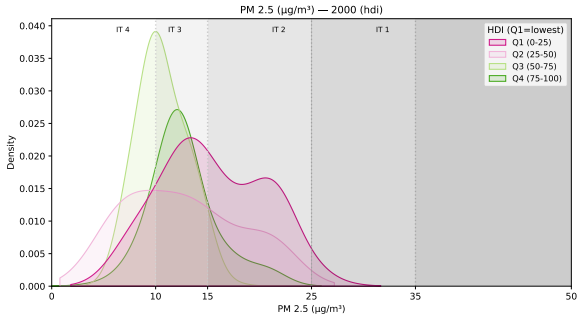

(e) Pollution, 2020



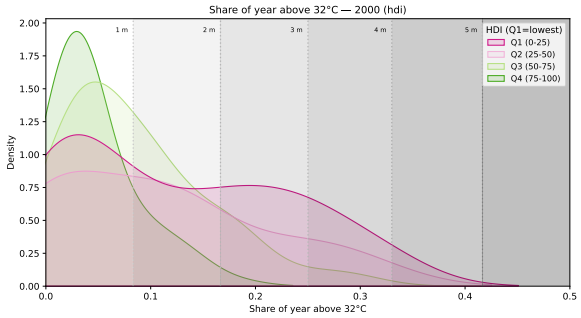

(f) Heat, 2020

*Notes:* Child-weighted kernel densities by fixed 2020 income quartile (GDP per capita at PPP, 2021 prices). Heat is the share of the year with UTCI above 32°C. These distributions complement the income means and percentiles in main Figure 3 and Table 1, supporting the main socioeconomic comparisons. Tables C.8 and C.9 report marginal category shares; main Figure 4 instead shows joint exceedances.

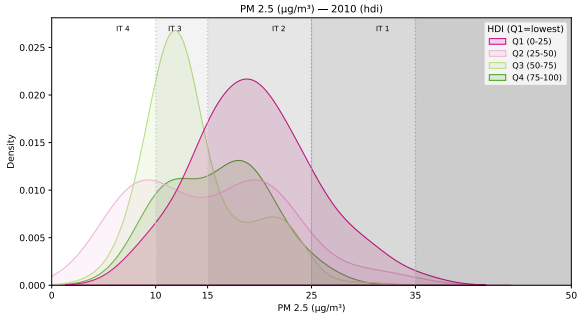
Figure D.9: Exposure distributions by HDI group



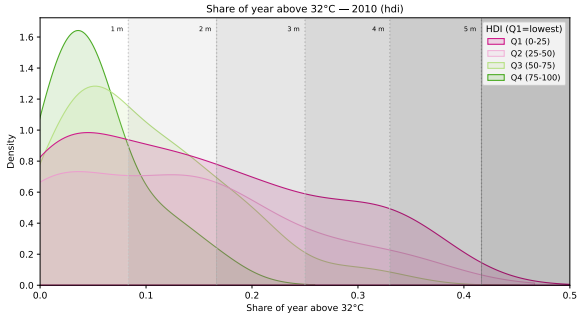

(a) Pollution, 2000



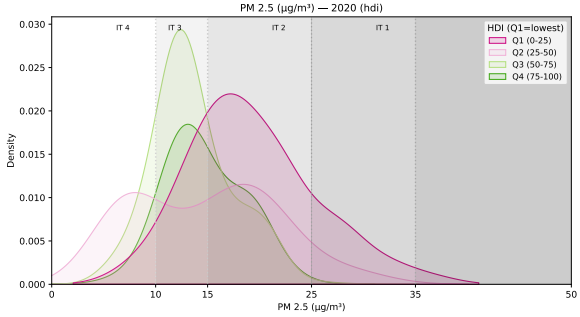

(b) Heat, 2000



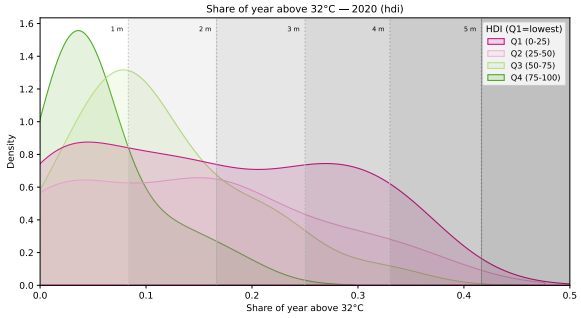

(c) Pollution, 2010




(d) Heat, 2010



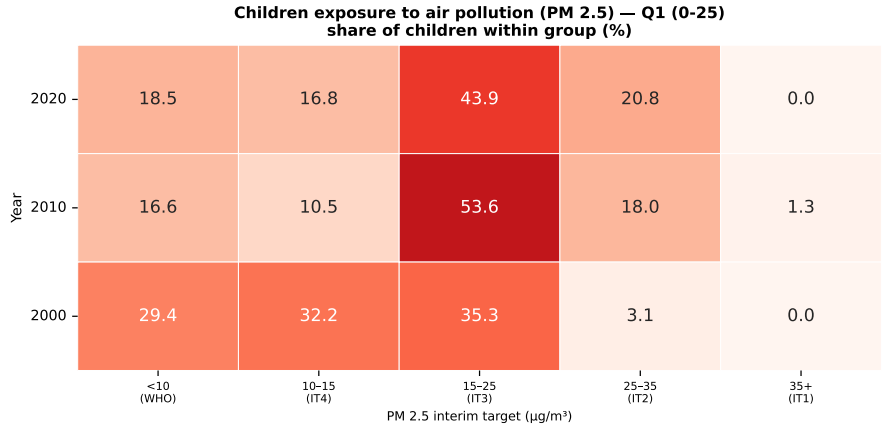

(e) Pollution, 2020



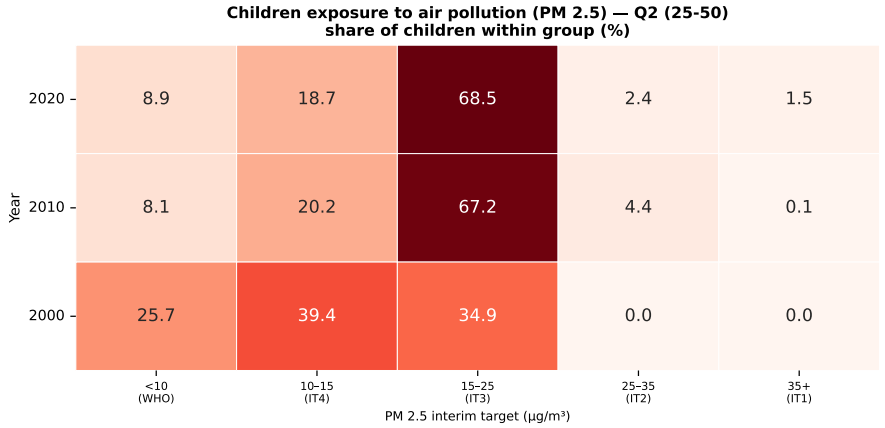

(f) Heat, 2020

*Notes:* Child-weighted kernel densities by fixed 2020 HDI quartile, from Q1 (lowest) to Q4 (highest). Heat is the share of the year with UTCI above 32°C. These distributions complement the HDI means and percentiles in main Figure 3 and Table 1, supporting the main socioeconomic comparisons. Tables C.8 and C.9 report marginal category shares; main Figure 4 instead shows joint exceedances.

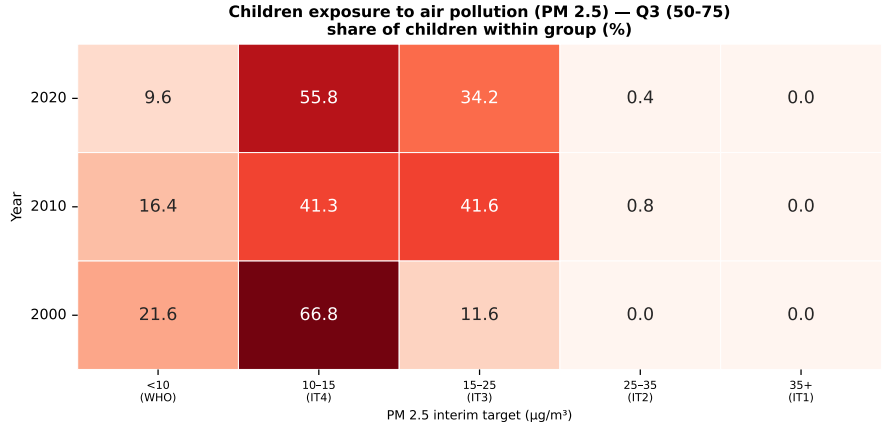
Figure D.10: Children's air pollution exposure by income group



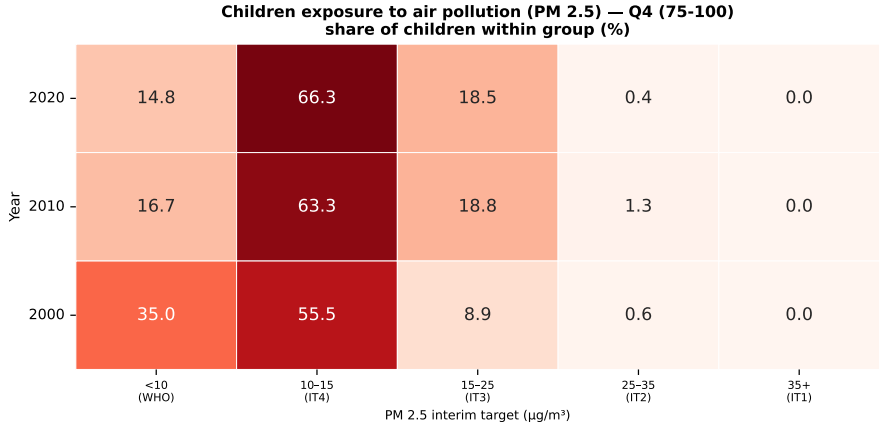

(a) Q1




(b) Q2



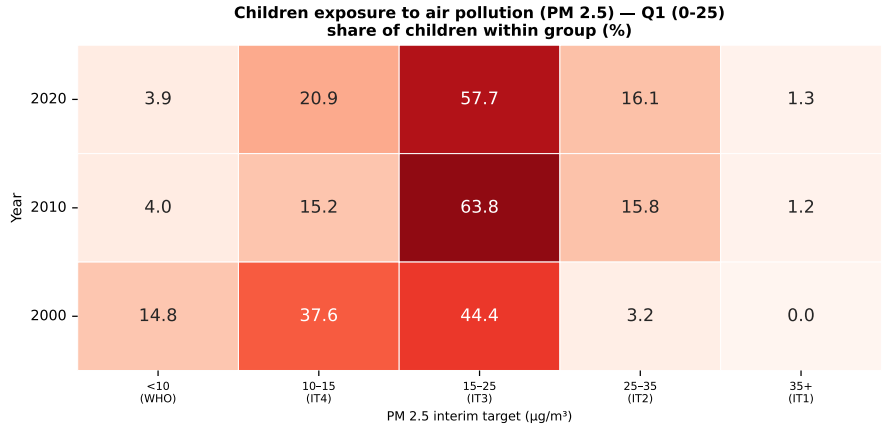

(c) Q3



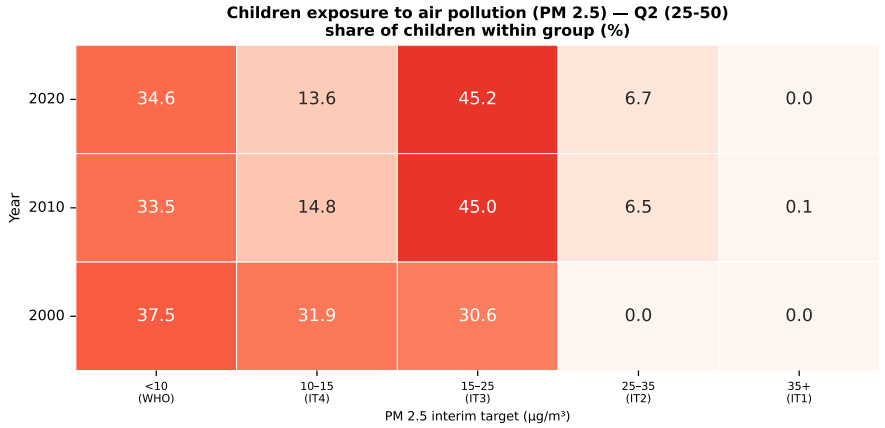

(d) Q4

*Notes:* Pollution-category percentages within fixed 2020 income quartiles (Q1 lowest, Q4 highest), tabulated in Table C.8. These support the main socioeconomic comparisons and complement the income means and percentiles in Figure 3a. Main Figure 4a,b instead reports joint pollution–heat exceedances.

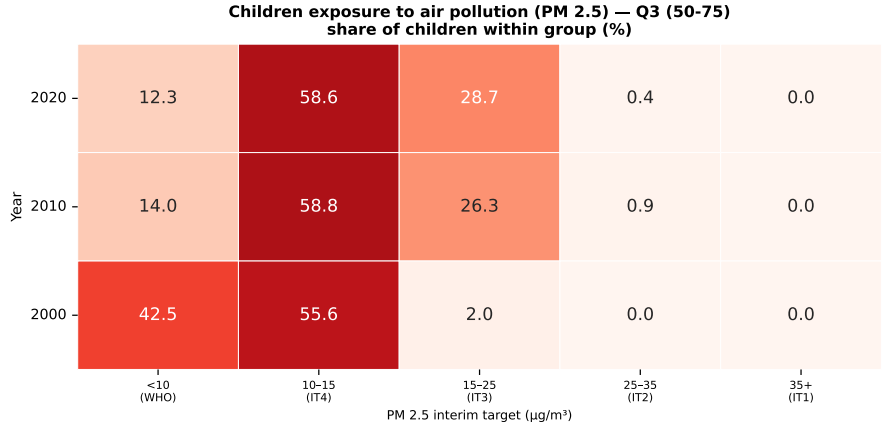
Figure D.11: Children's air pollution exposure by HDI group



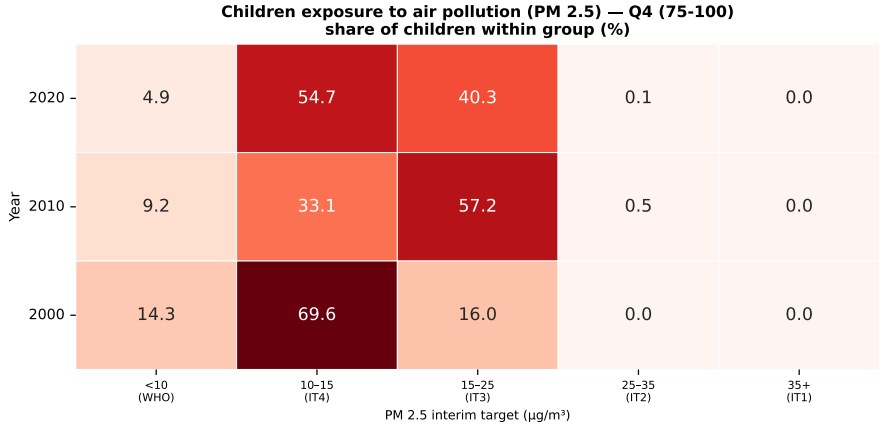

(a) Q1




(b) Q2



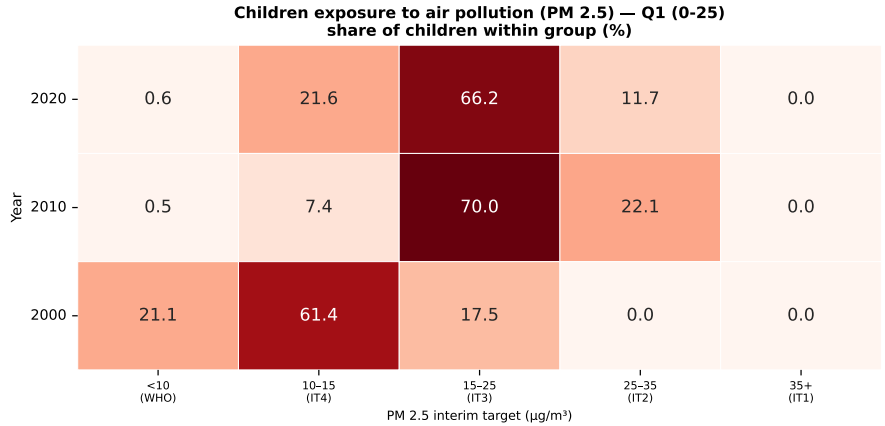

(c) Q3



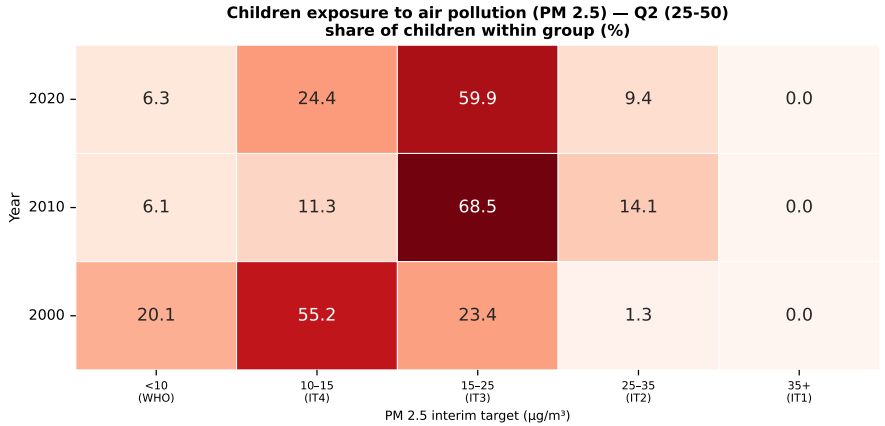

(d) Q4

*Notes:* Pollution-category percentages within fixed 2020 HDI quartiles (Q1 lowest, Q4 highest), tabulated in Table C.8. These support the main socioeconomic comparisons and complement the HDI means and percentiles in Figure 3a. Main Figure 4c,d instead reports joint pollution–heat exceedances.

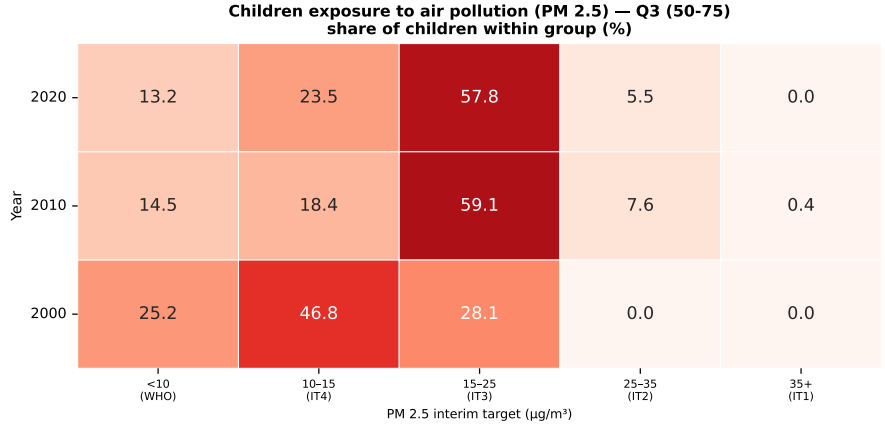
Figure D.12: Children's air pollution exposure by accessibility group



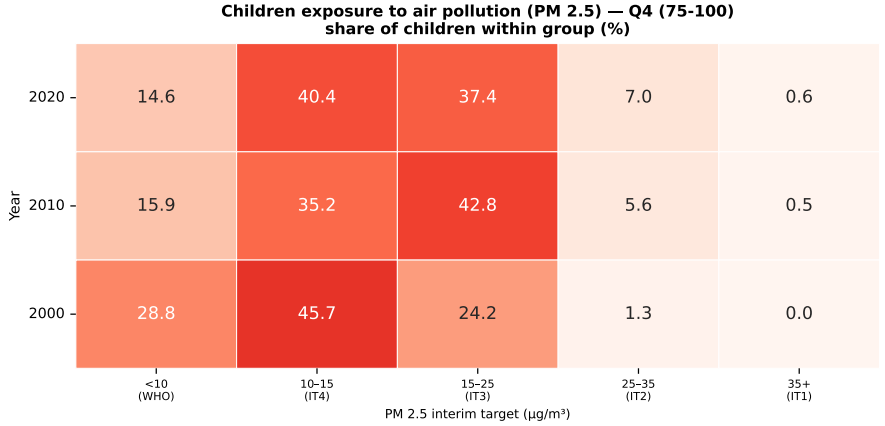

(a) Q1




(b) Q2



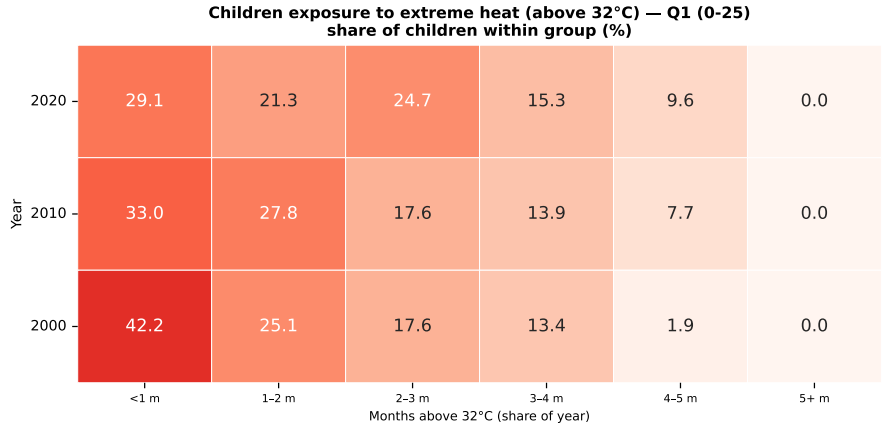

(c) Q3



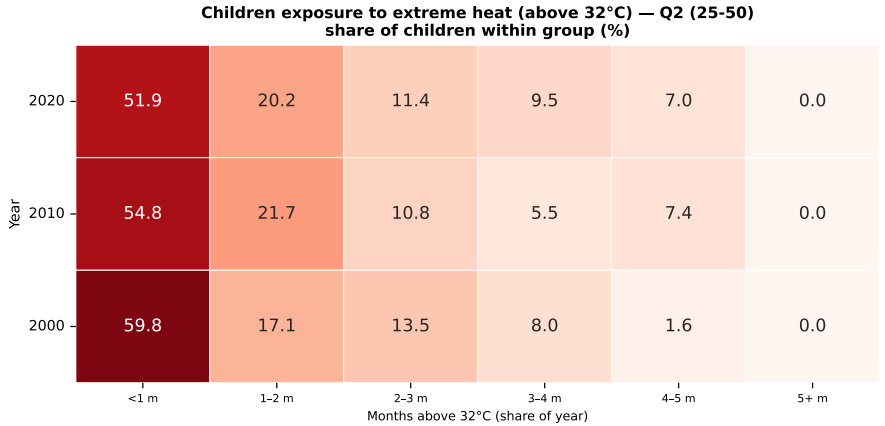

(d) Q4

*Notes:* Pollution-category percentages within fixed 2015 accessibility quartiles (Q1 most remote, Q4 most accessible), tabulated in Table C.8. These complement the accessibility means and percentiles in main Figure 3a and the geographic-group discussion. Figure D.17 instead shows joint pollution–heat exceedances.

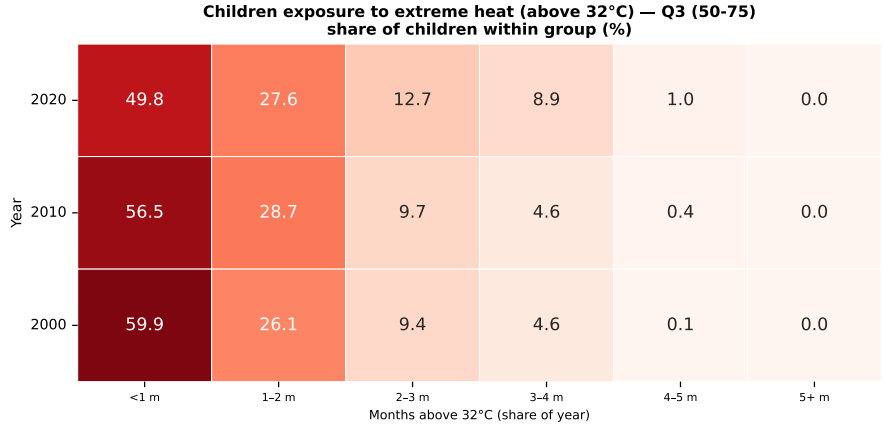
Figure D.13: Children's heat exposure by income group



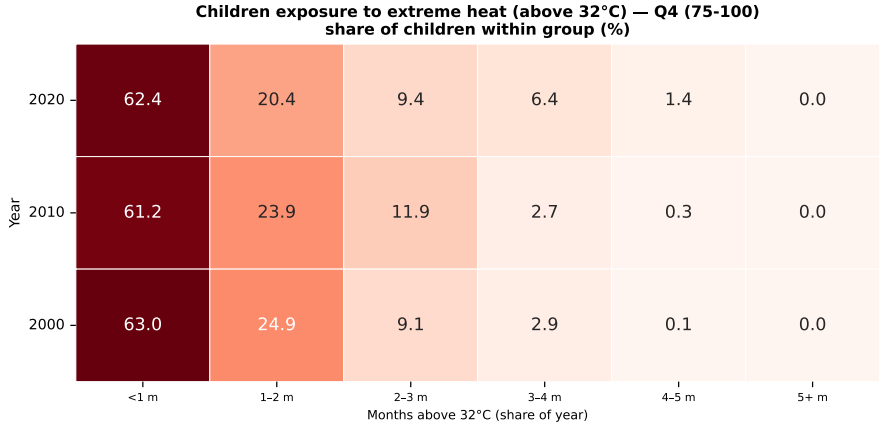

(a) Q1




(b) Q2



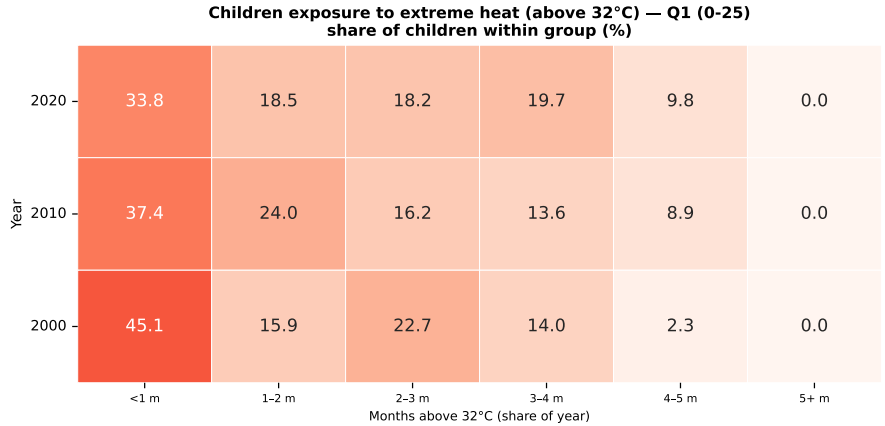

(c) Q3



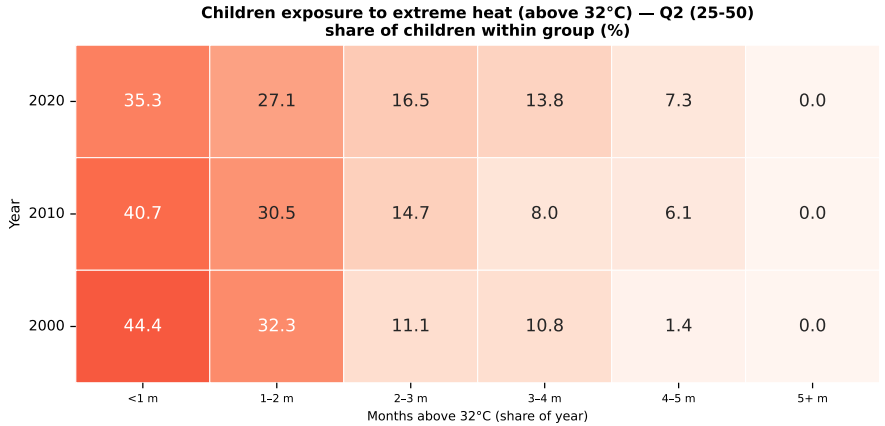

(d) Q4

*Notes:* Heat-duration category percentages within fixed 2020 income quartiles (Q1 lowest, Q4 highest), tabulated in Table C.9. Heat duration is months with UTCI above 32°C. These complement the income means and percentiles in main Figure 3b and the socioeconomic discussion. Main Figure 4a,b instead reports joint exceedances.

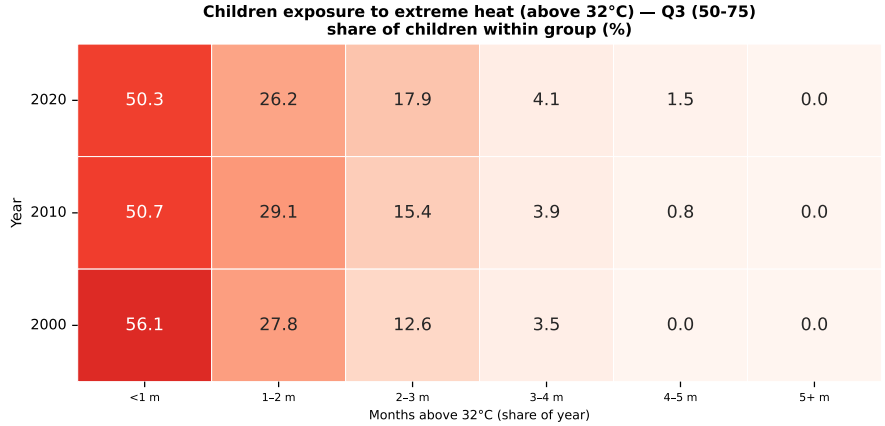
Figure D.14: Children's heat exposure by HDI group



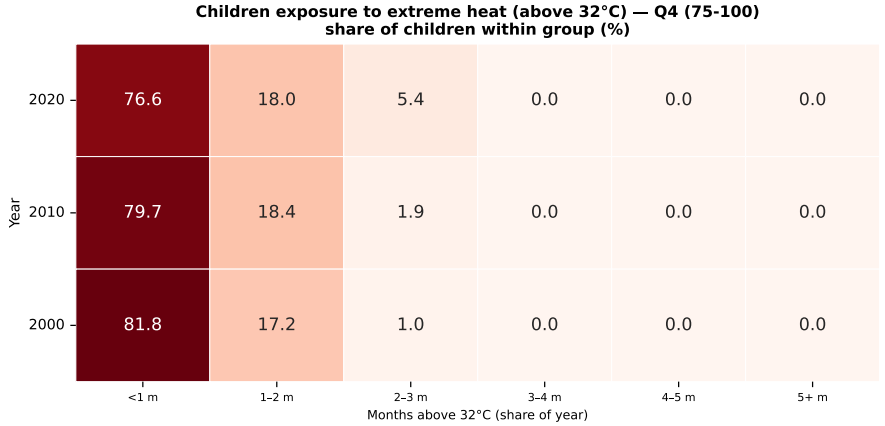

(a) Q1




(b) Q2



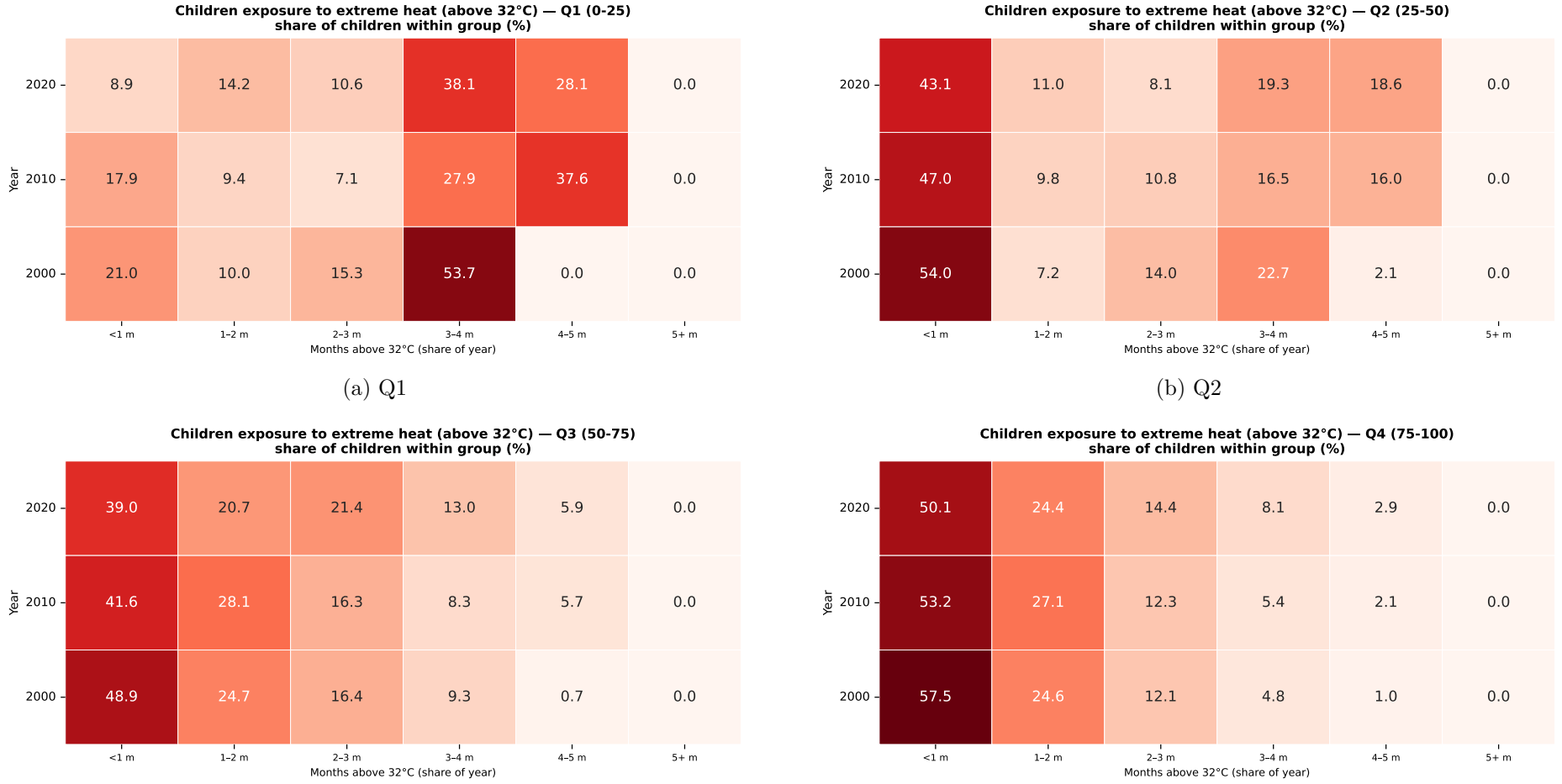

(c) Q3




(d) Q4

*Notes:* Heat-duration category percentages within fixed 2020 HDI quartiles (Q1 lowest, Q4 highest), tabulated in Table C.9. Heat duration is months with UTCI above 32°C. These complement the HDI means and percentiles in main Figure 3b and the socioeconomic discussion. Main Figure 4c,d instead reports joint exceedances.

Figure D.15: Children's heat exposure by accessibility group




(a) Q1

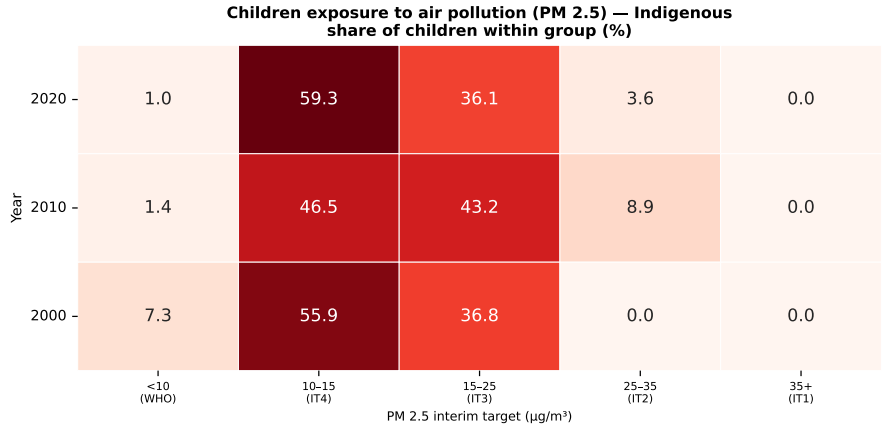
(b) Q2

(c) Q3

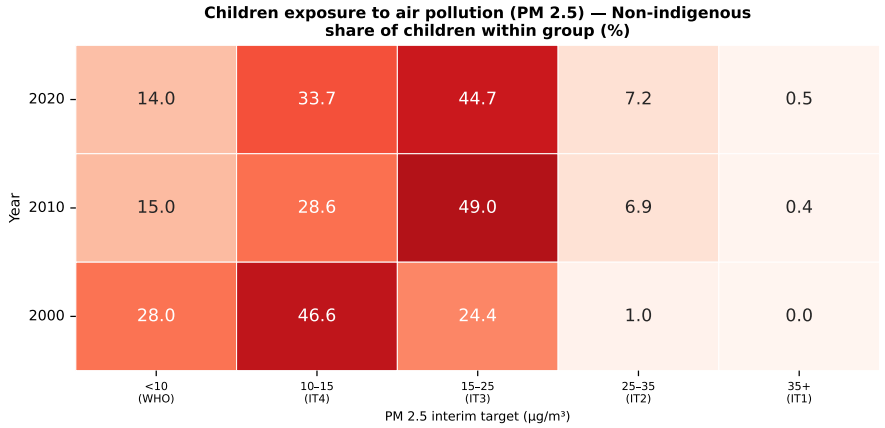
(d) Q4

*Notes:* Heat-duration category percentages within fixed 2015 accessibility quartiles (Q1 most remote, Q4 most accessible), tabulated in Table C.9. Heat duration is months with UTCI above 32°C. These complement the accessibility means and percentiles in main Figure 3b and the geographic-group discussion. Figure D.17 instead shows joint exceedances.

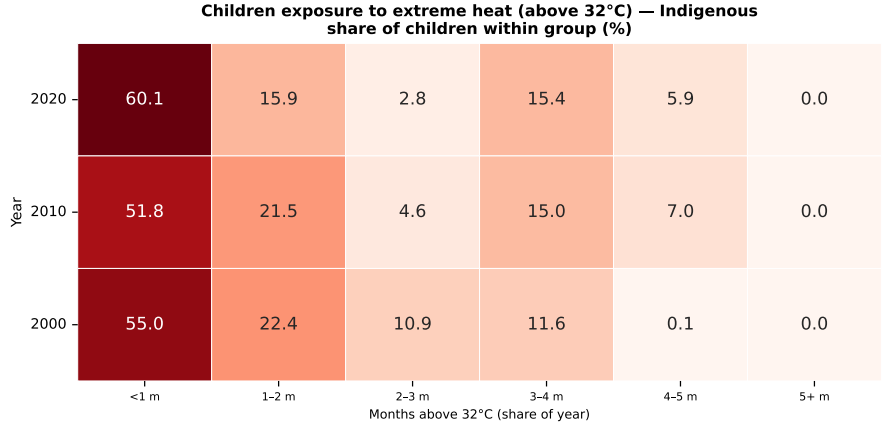
Figure D.16: Children's air pollution and heat exposure by land type



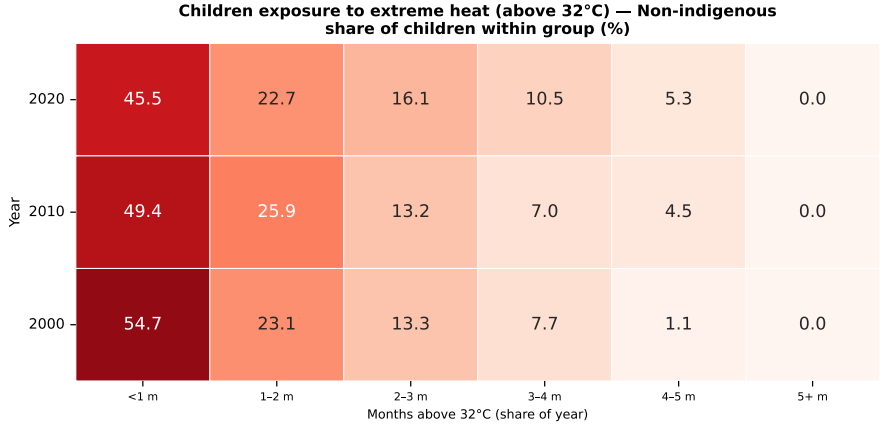

(a) Pollution, Indigenous




(b) Pollution, non-Indigenous



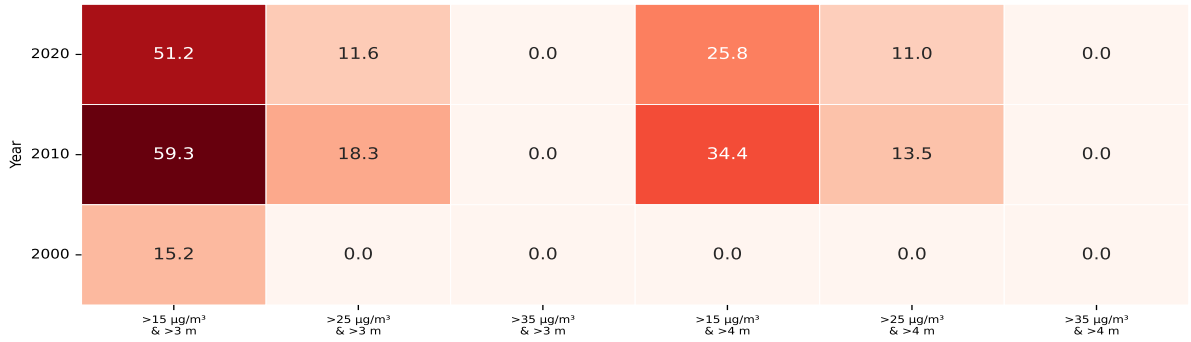

(c) Heat, Indigenous



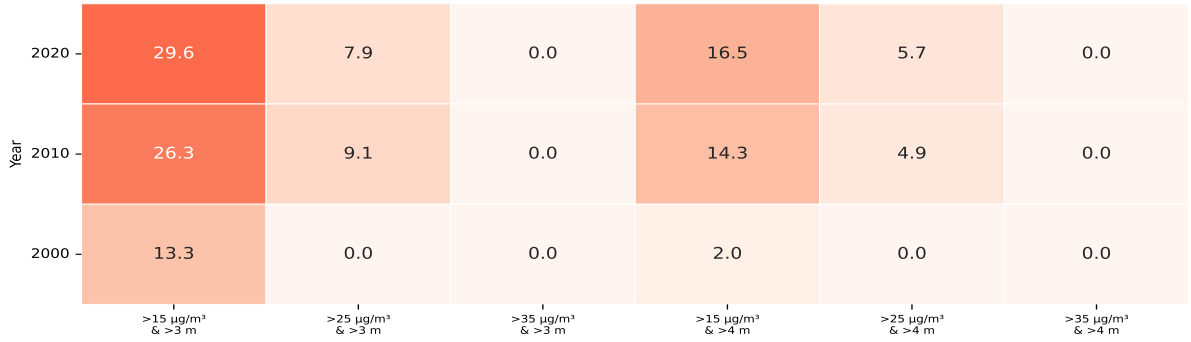

(d) Heat, non-Indigenous

*Notes:* Marginal pollution and heat-duration category percentages within Indigenous and non-Indigenous territories, tabulated in Tables C.8 and C.9. These complement the land-type means and percentiles in main Figure 3 and the geographic-group discussion. Figure D.18 instead shows joint exceedances.

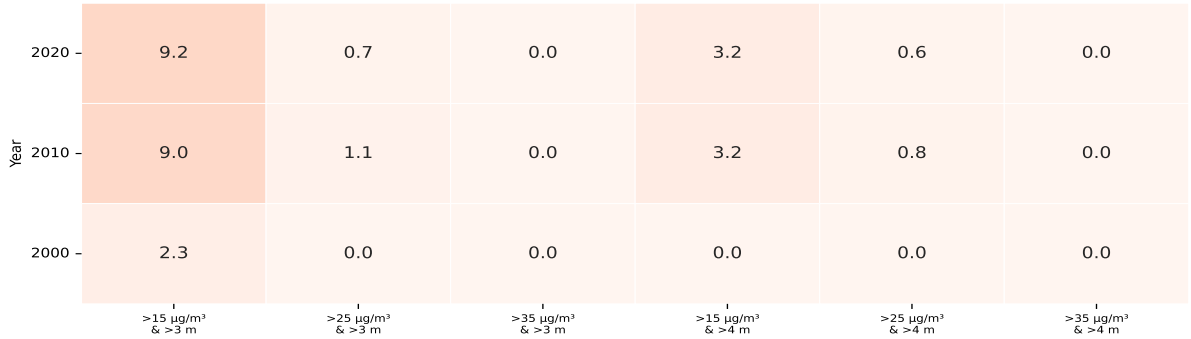
Figure D.17: Overlapping air pollution and heat exposure by accessibility group



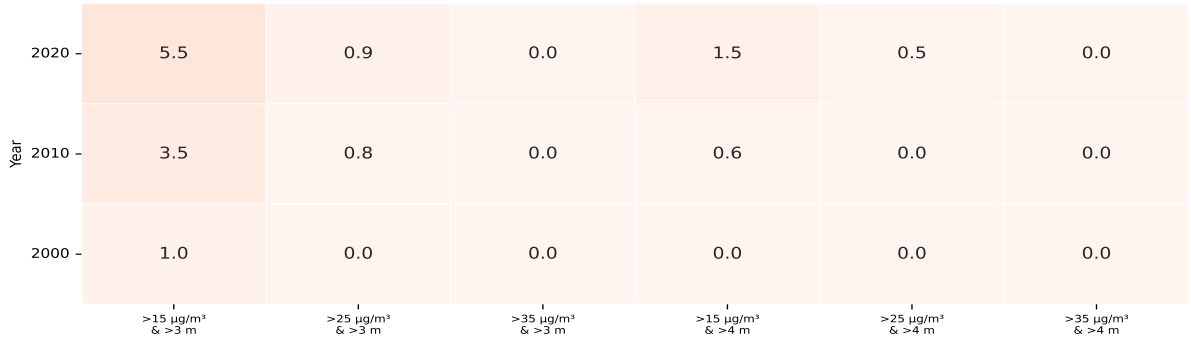

(a) Q1: most remote




(b) Q2



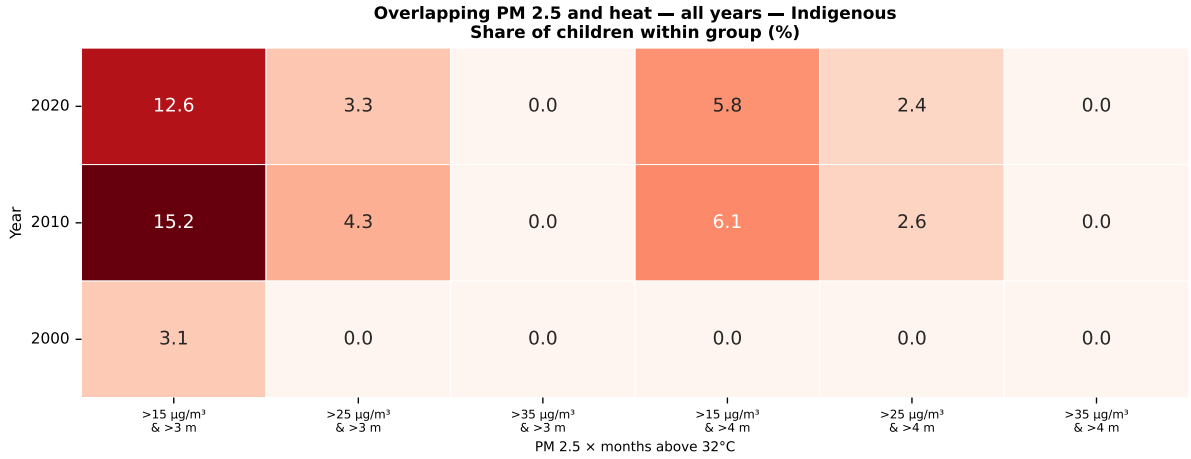

(c) Q3



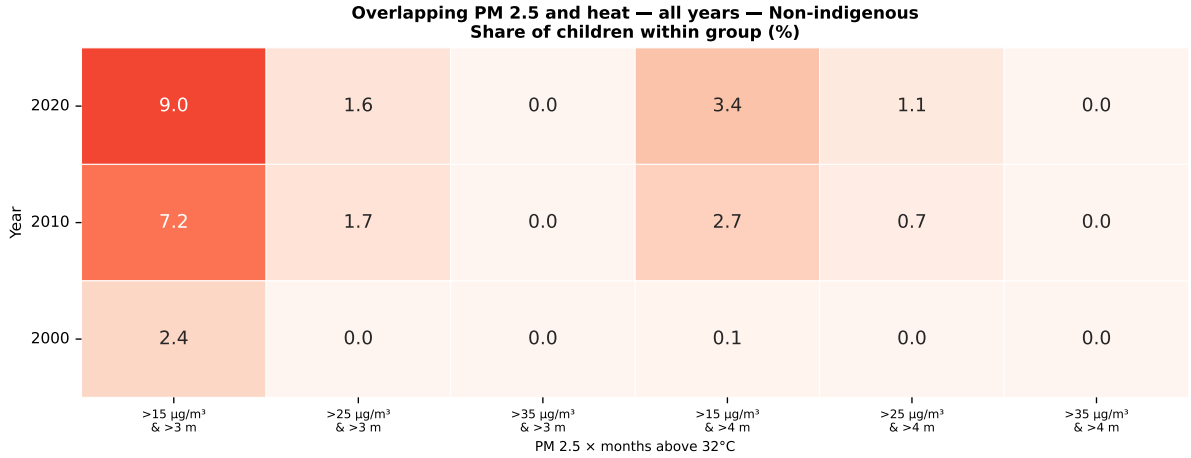

(d) Q4: most accessible

*Notes:* Joint exceedance percentages by accessibility (Q1 most remote, Q4 most accessible). Heat is months with UTCI above 32°C. Main Table 2 and Table C.13 report four of these six combinations. These extend the group contrasts in main Figure 4 to the geographic groups discussed in the main Results.

Figure D.18: Overlapping air pollution and heat exposure by land type




(a) Indigenous




(b) Non-Indigenous

*Notes:* Joint pollution–heat exceedance percentages within Indigenous and non-Indigenous territories. Heat is months with UTCI above 32°C. Main Table 2 and Table C.13 report four of these six combinations. These extend the group contrasts in main Figure 4 to the land-type comparison in the main Results; Figure D.16 instead shows marginal categories.